FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

Jakub Vokoun

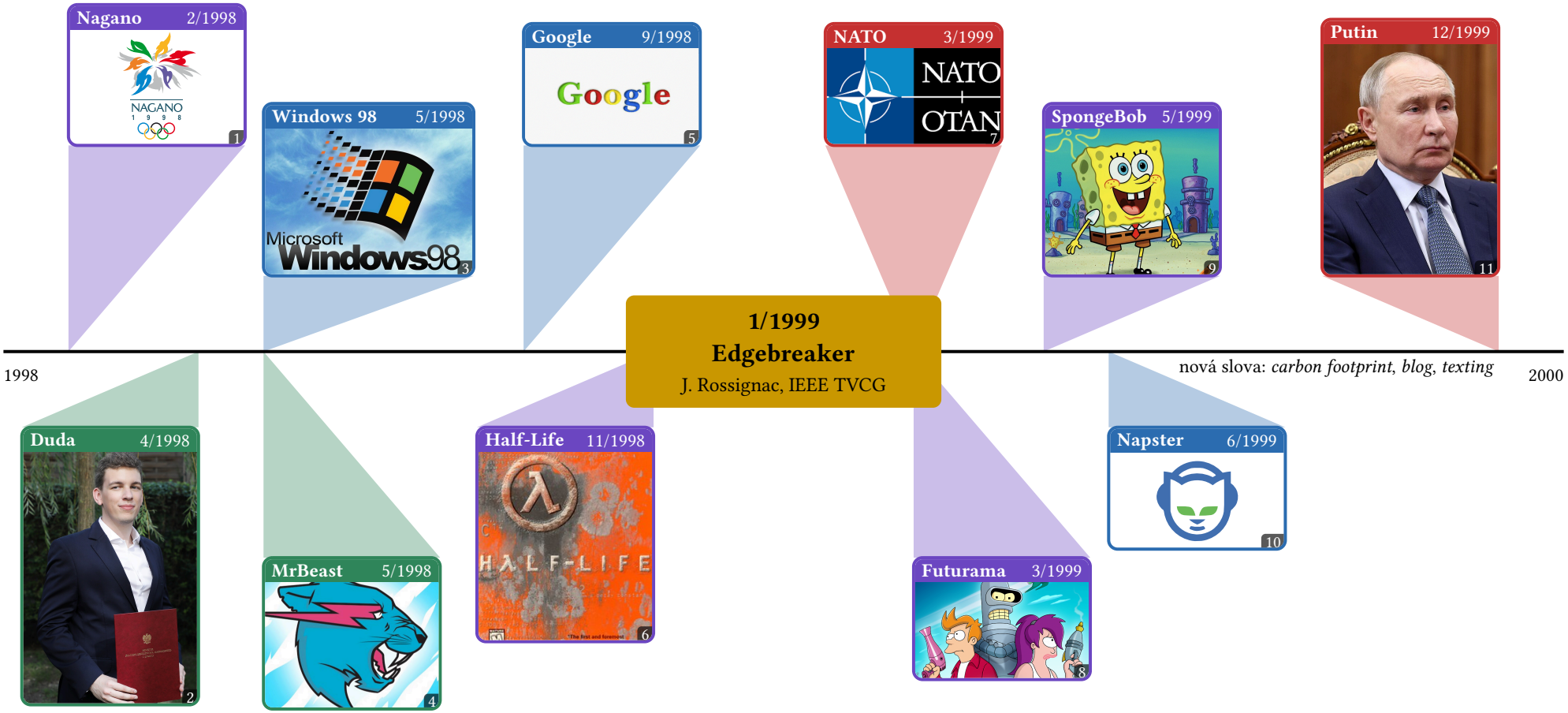
Fakulta aplikovaných věd ZČU

javok@students.zcu.cz

5.11.2026

1. Motivace

1.1 Dobový kontext



Obrázek 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext

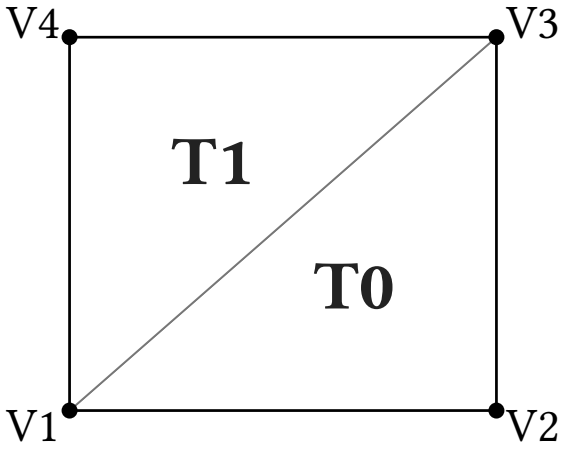


Obrázek 2: Jarek Rossignac.

zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

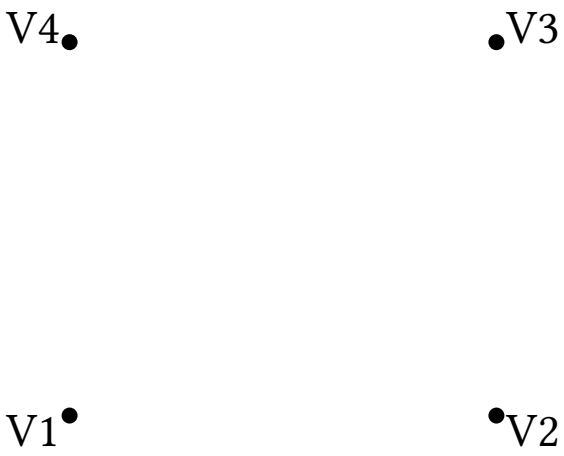
disk vs RAM

2. Definice problému



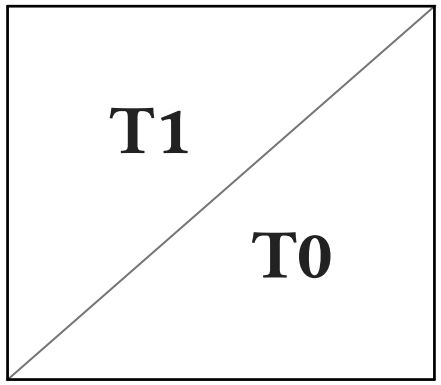
Obrázek 3: Trojúhelníková síť (mesh).

2.1 Mesh: geometrie + konektivita



Obrázek 3: Trojúhelníková síť (mesh).

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Obrázek 3: Trojúhelníková síť (mesh).

2.2 Jak schovat informaci?

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pořadí vertexů

[V1 V2 V3]

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[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

2.2 Jak schovat informaci?

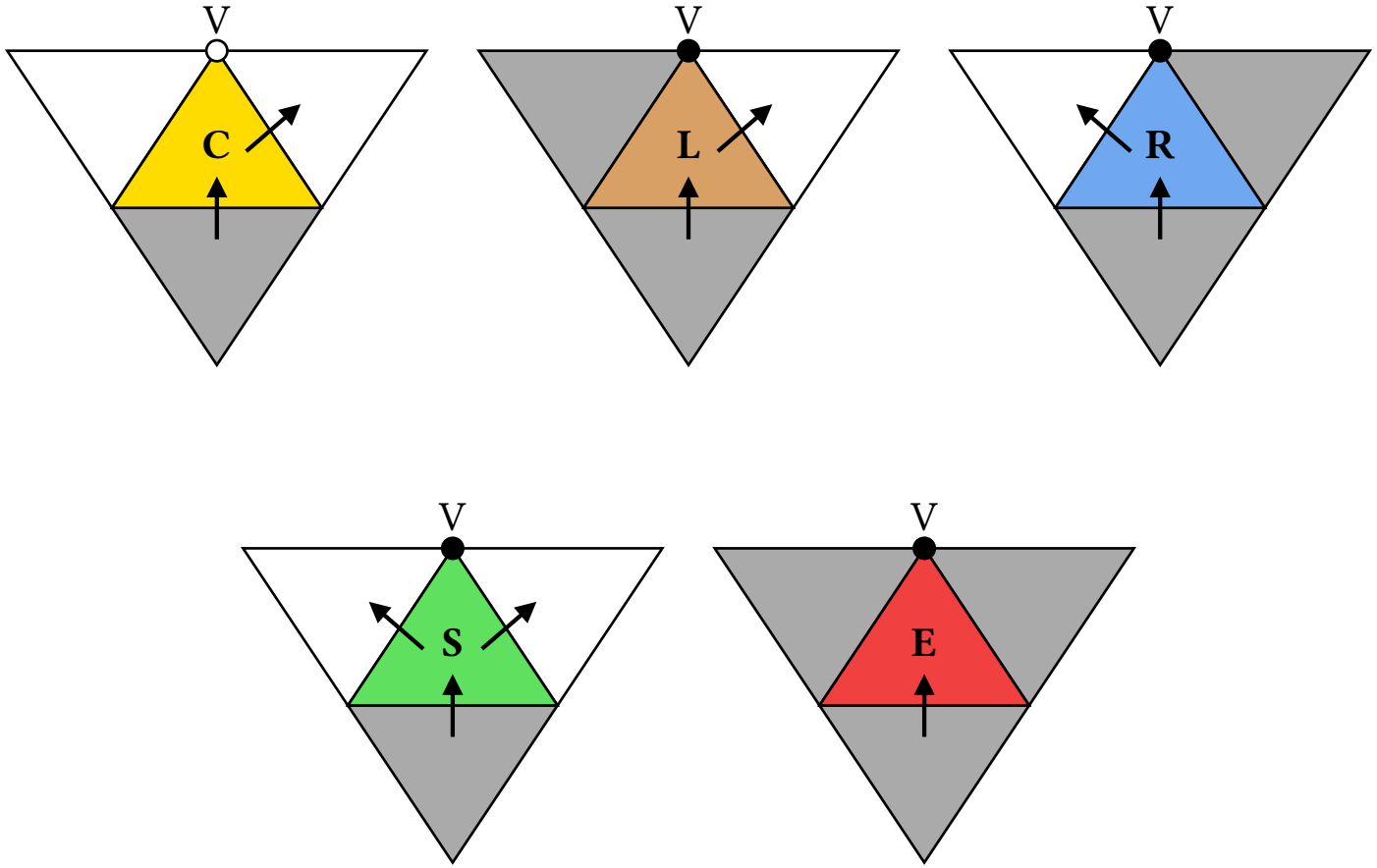
pořadí vertexů

[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

CLERS řetězec

2.2 Jak schovat informaci?

pořadí vertexů	CLERS řetězec
[V1 V2 V3]	CLRSLRCLEE
[V1 V3 V2]	
[V2 V1 V3]	
[V3 V1 V2]	
[V2 V3 V1]	
[V3 V2 V1]	



Obrázek 4: Základní CLERS operace.

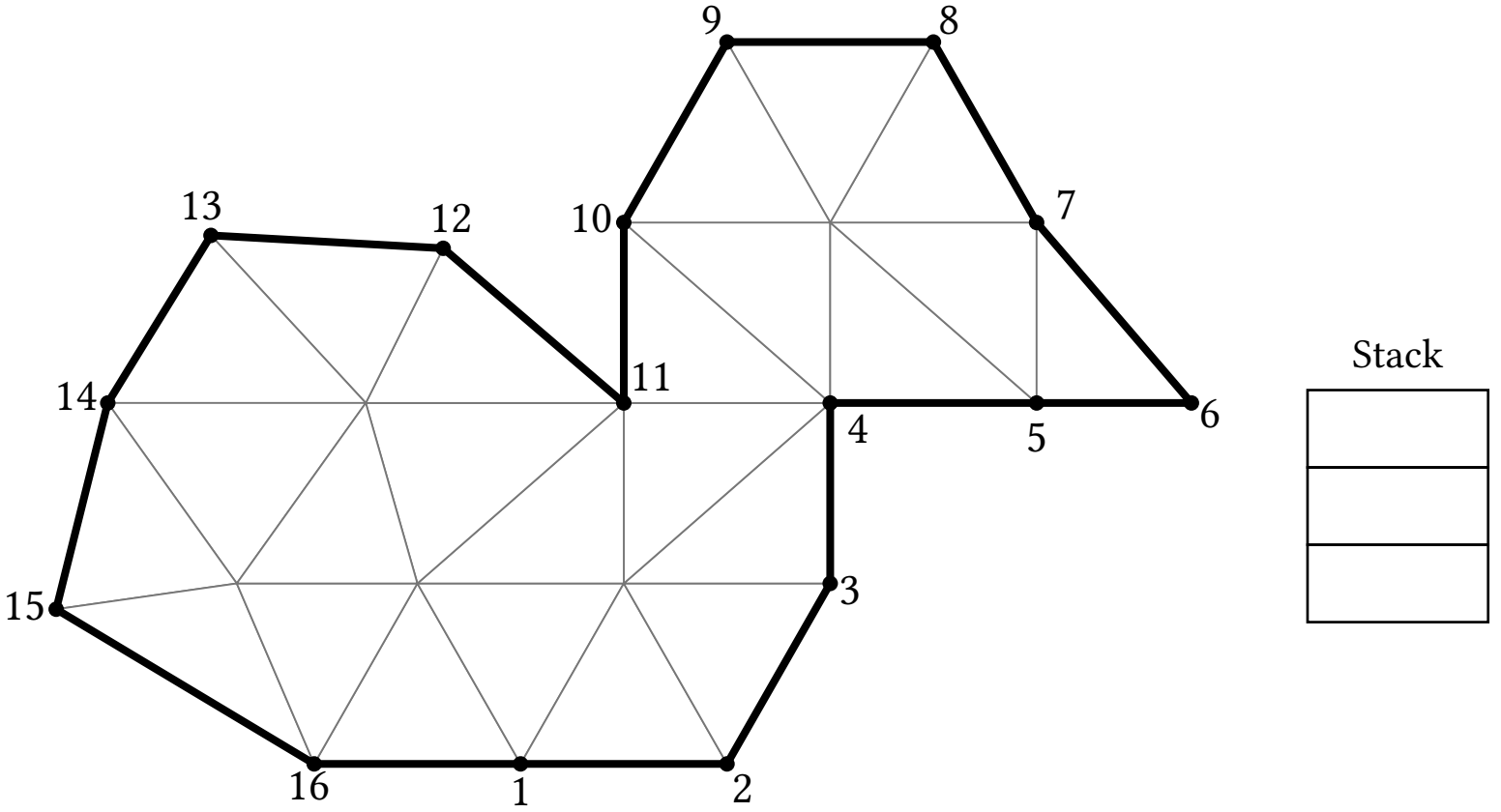
3. Komprese

3.1 Omezení

Mesh musí být **orientovatelný manifold** homeomorfní s koulí/kruhem.

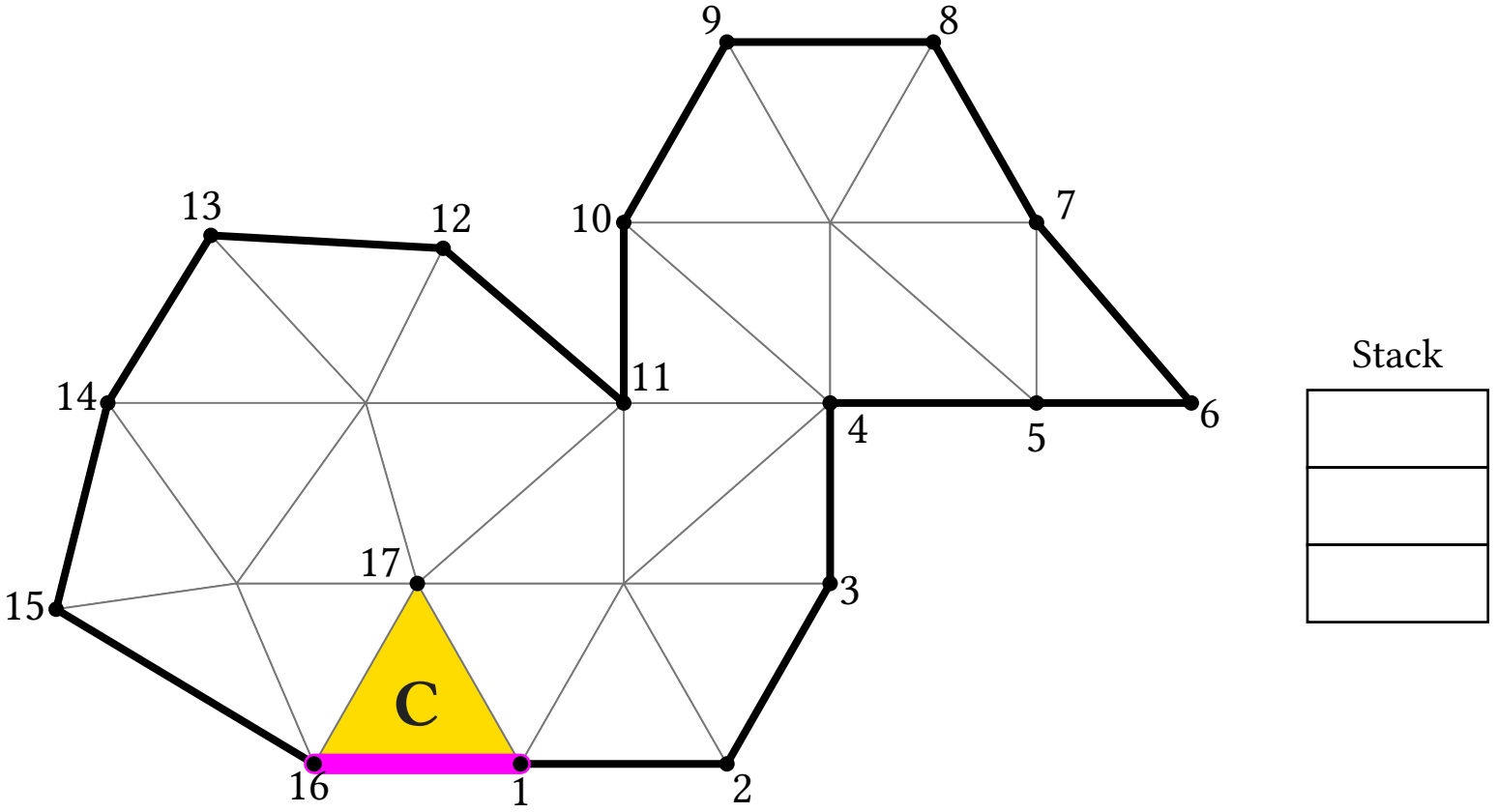
Lehce upravená verze algoritmu zvládne i *díry* a *handly*, obecně meshe s vyšším genusem.

Historie =



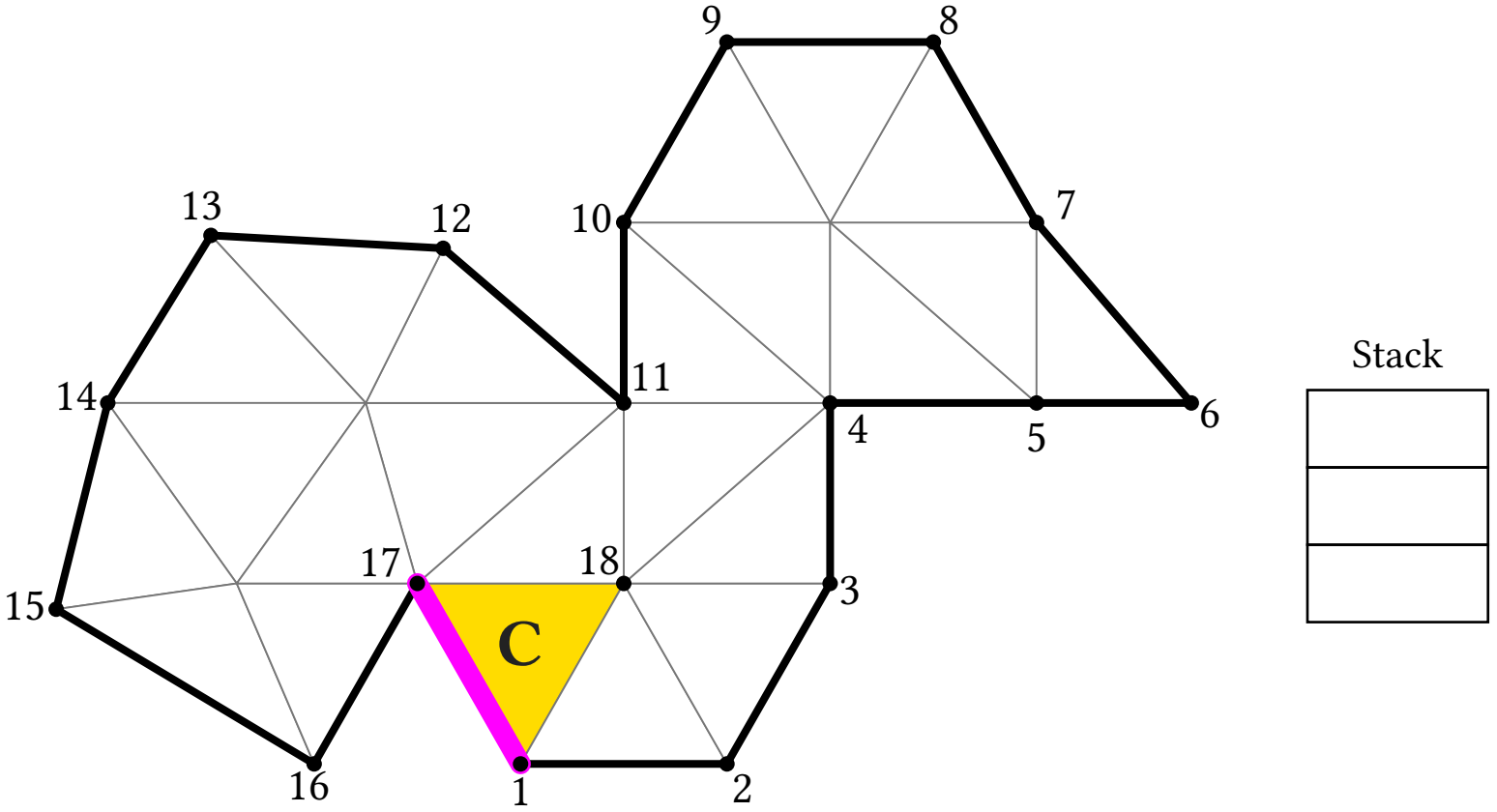
Obrázek 5: Jednoduchá komprese.

Historie = C



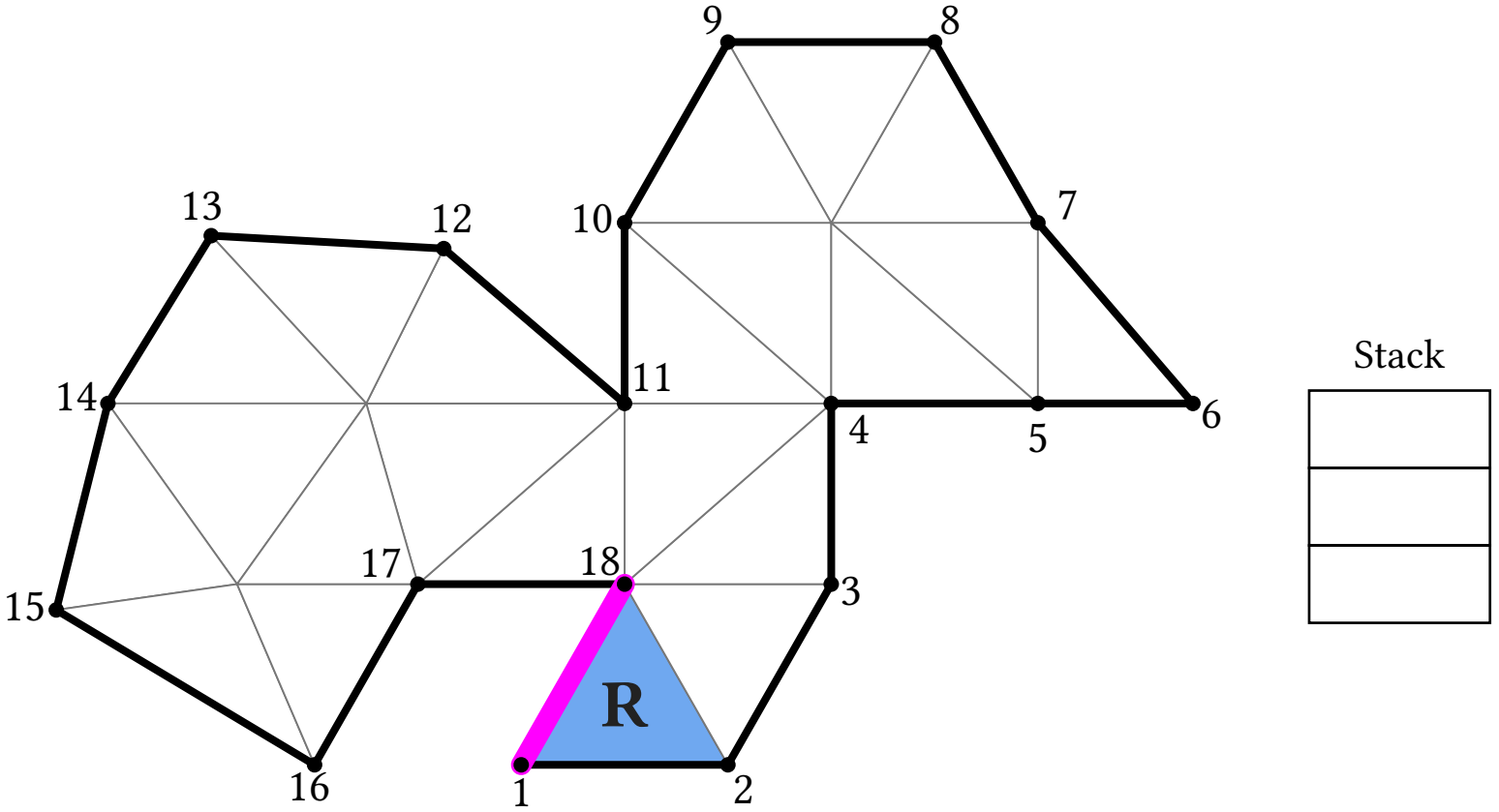
Obrázek 5: Jednoduchá komprese.

Historie = CC



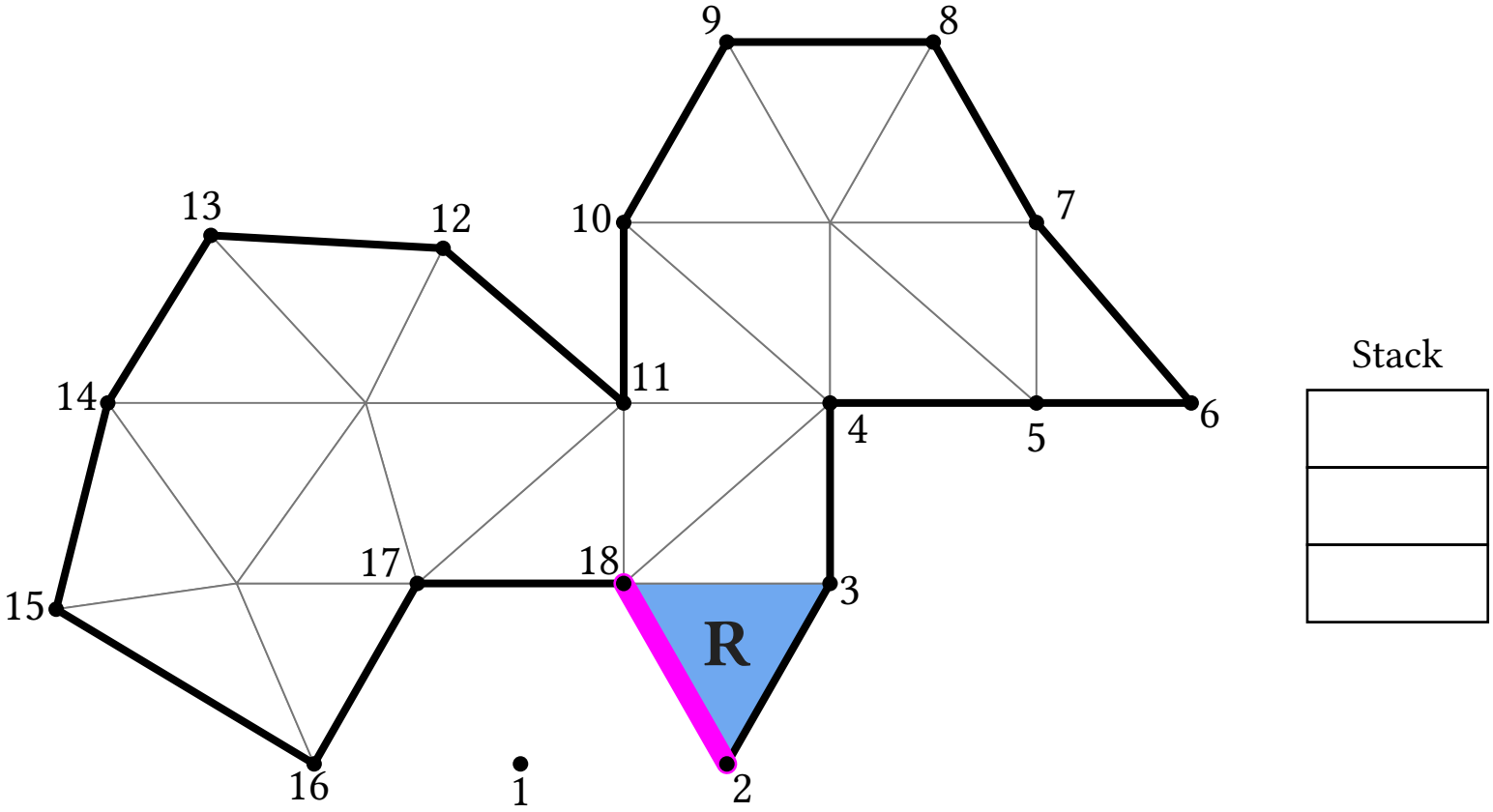
Obrázek 5: Jednoduchá komprese.

Historie = CCR



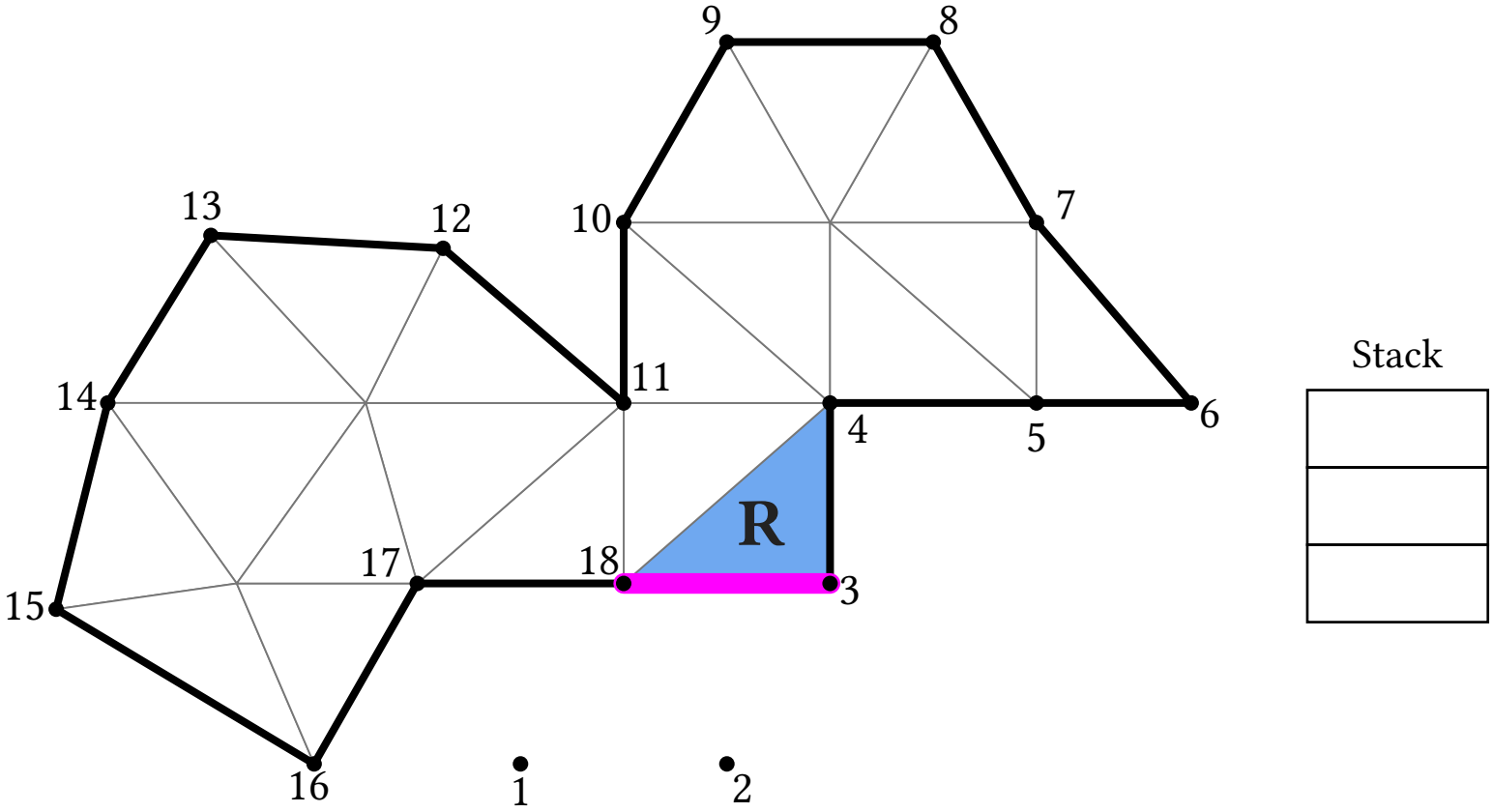
Obrázek 5: Jednoduchá komprese.

Historie = CCRR



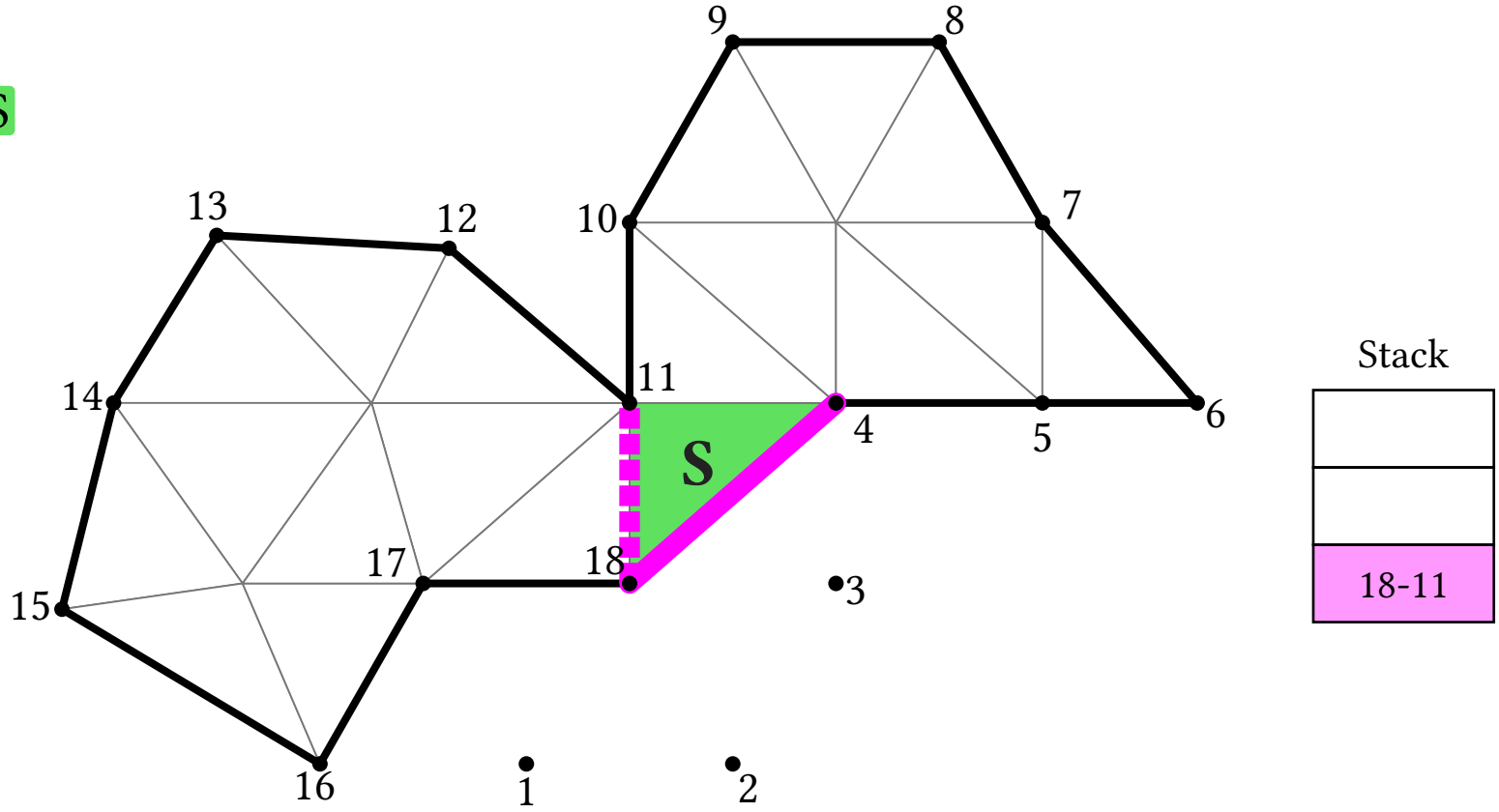
Obrázek 5: Jednoduchá komprese.

Historie = CCRRR



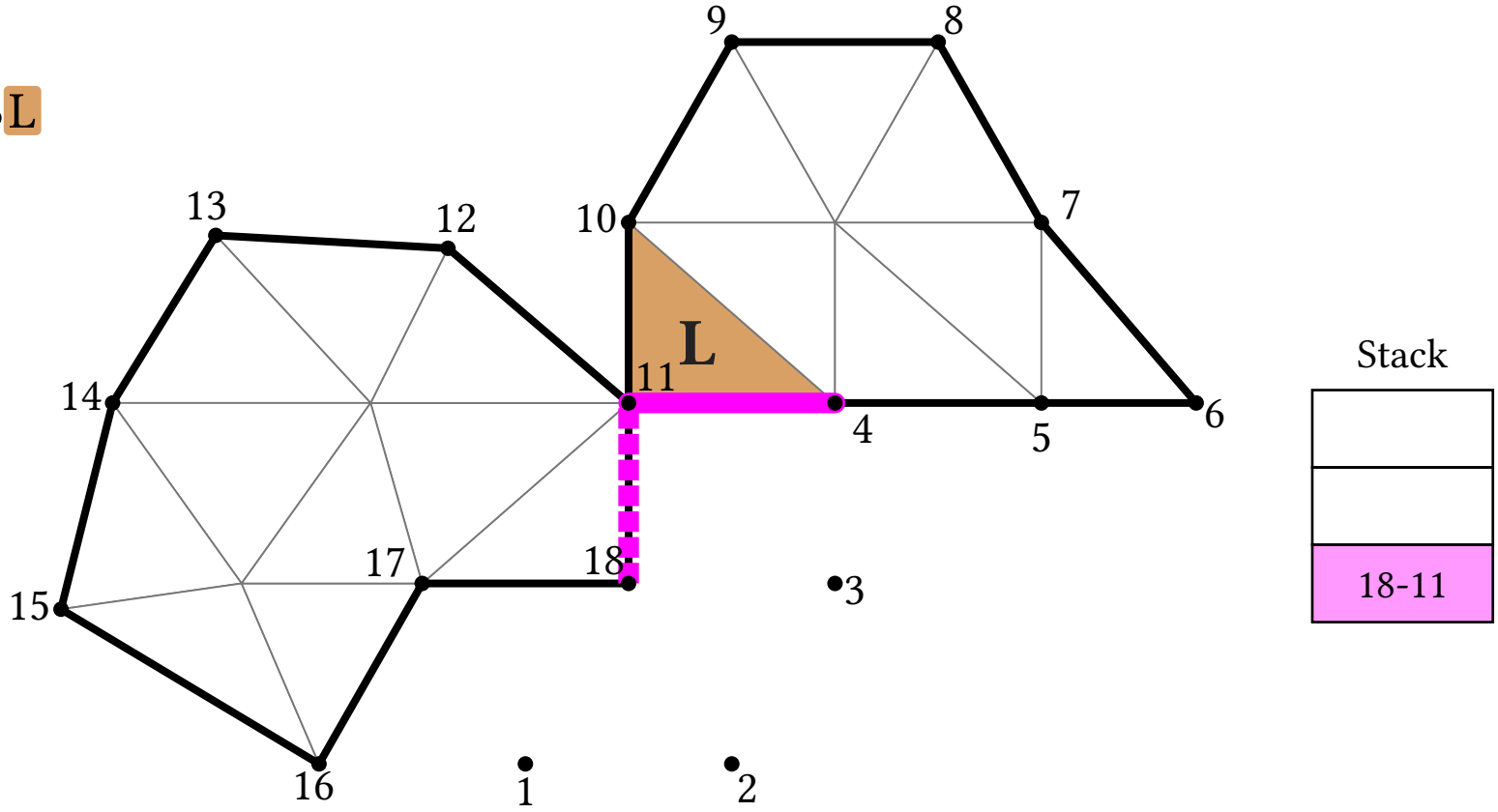
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRS



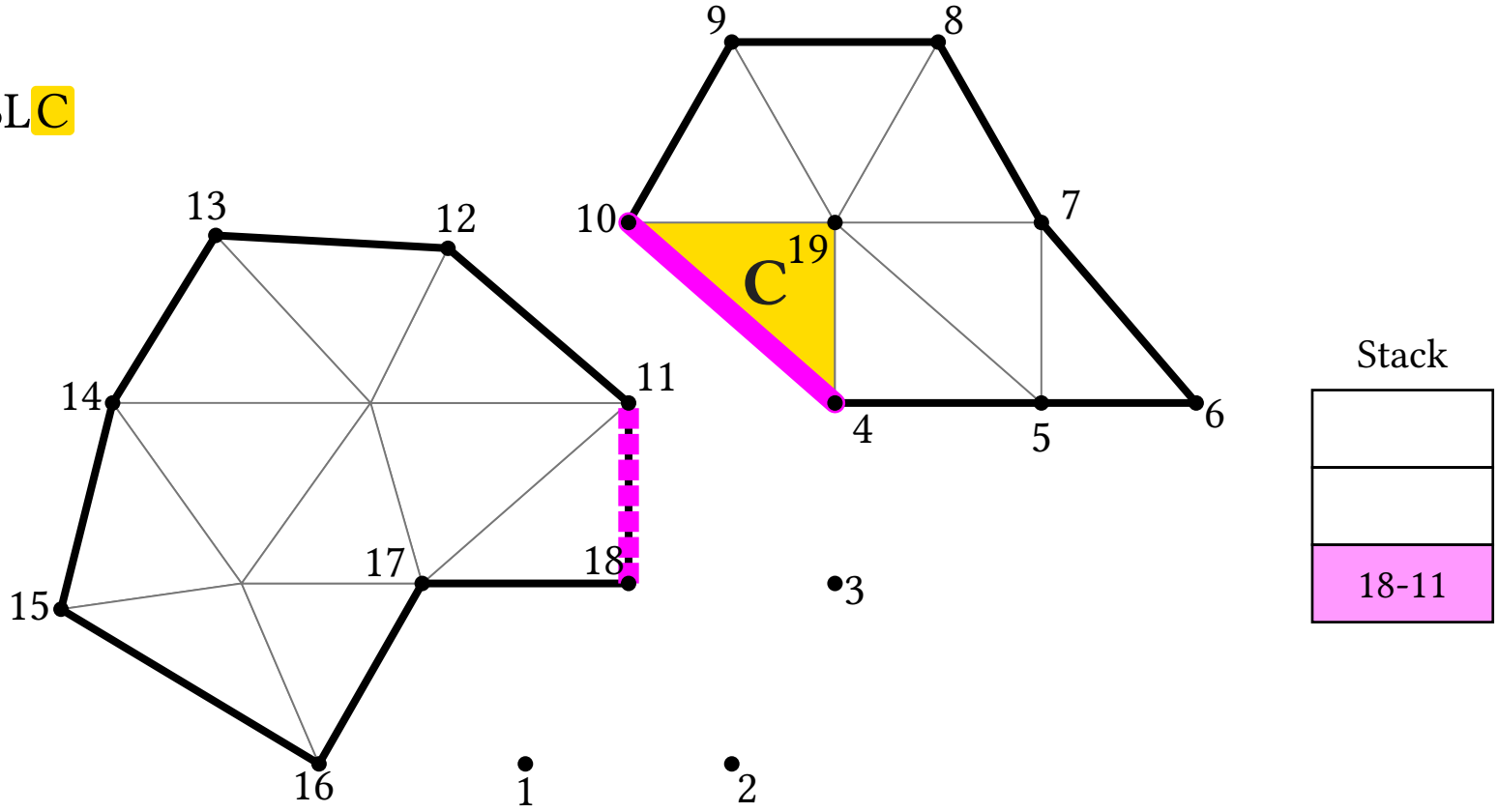
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSL



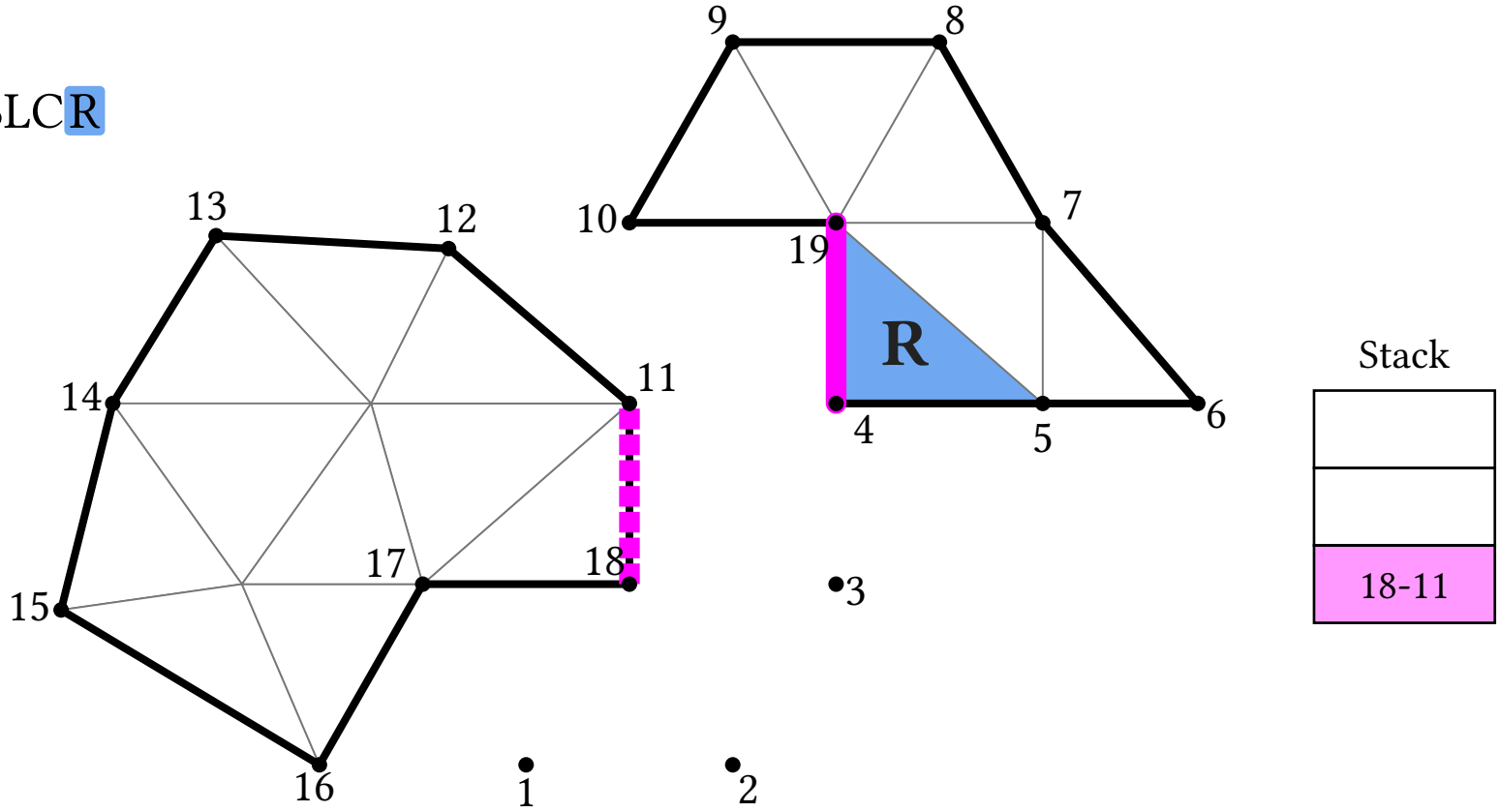
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLC



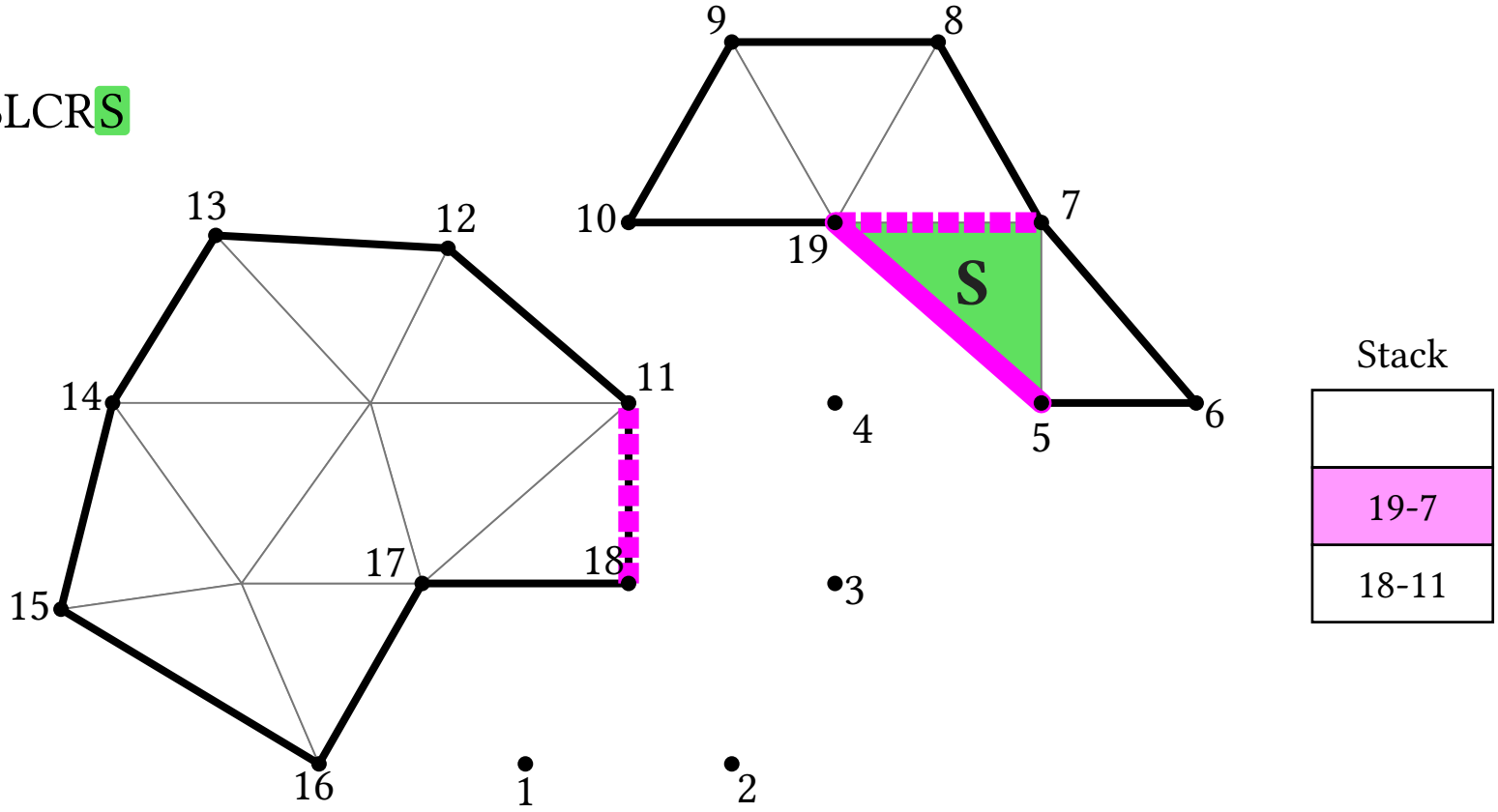
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCR



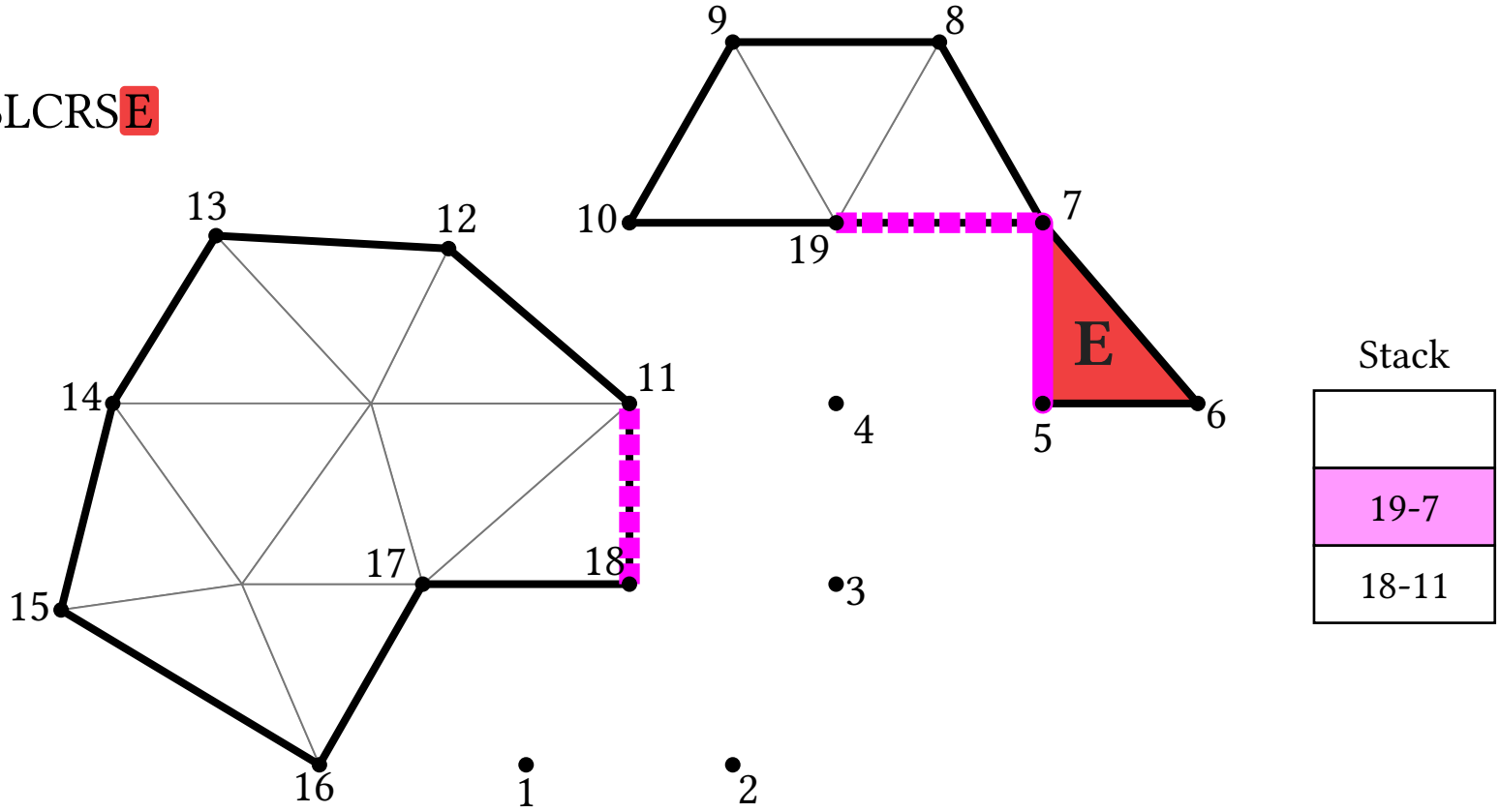
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRS



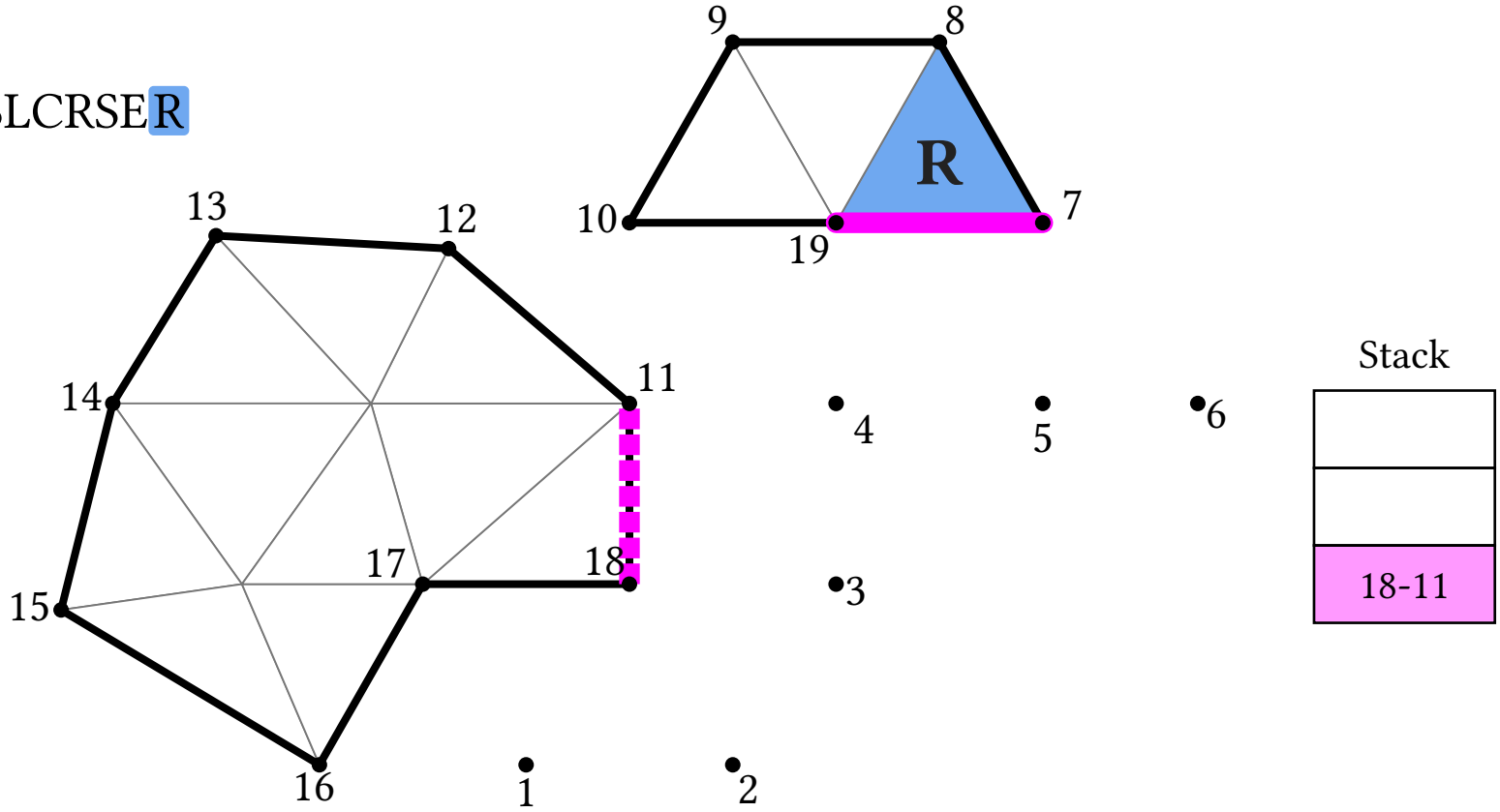
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSE



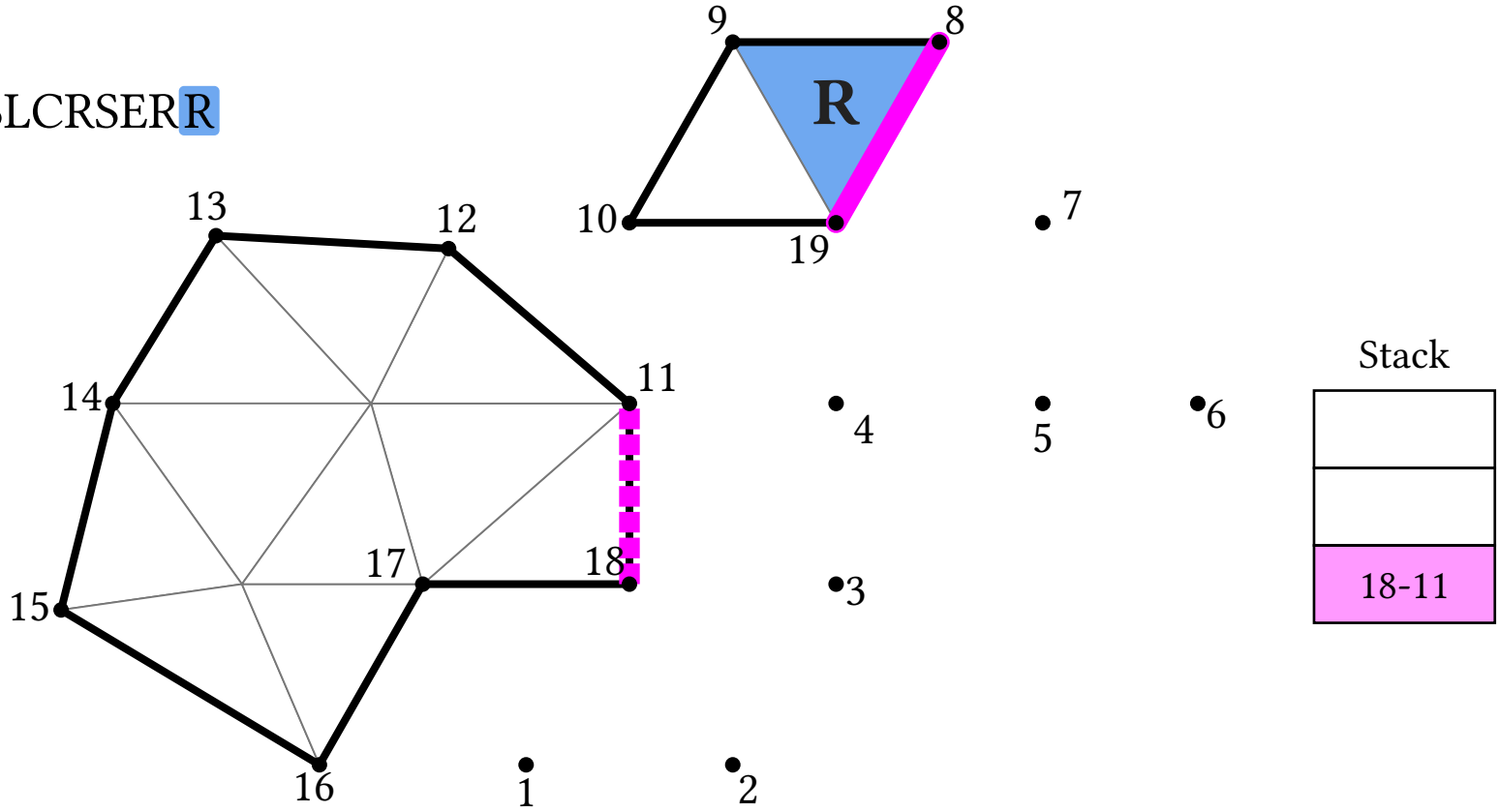
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSER



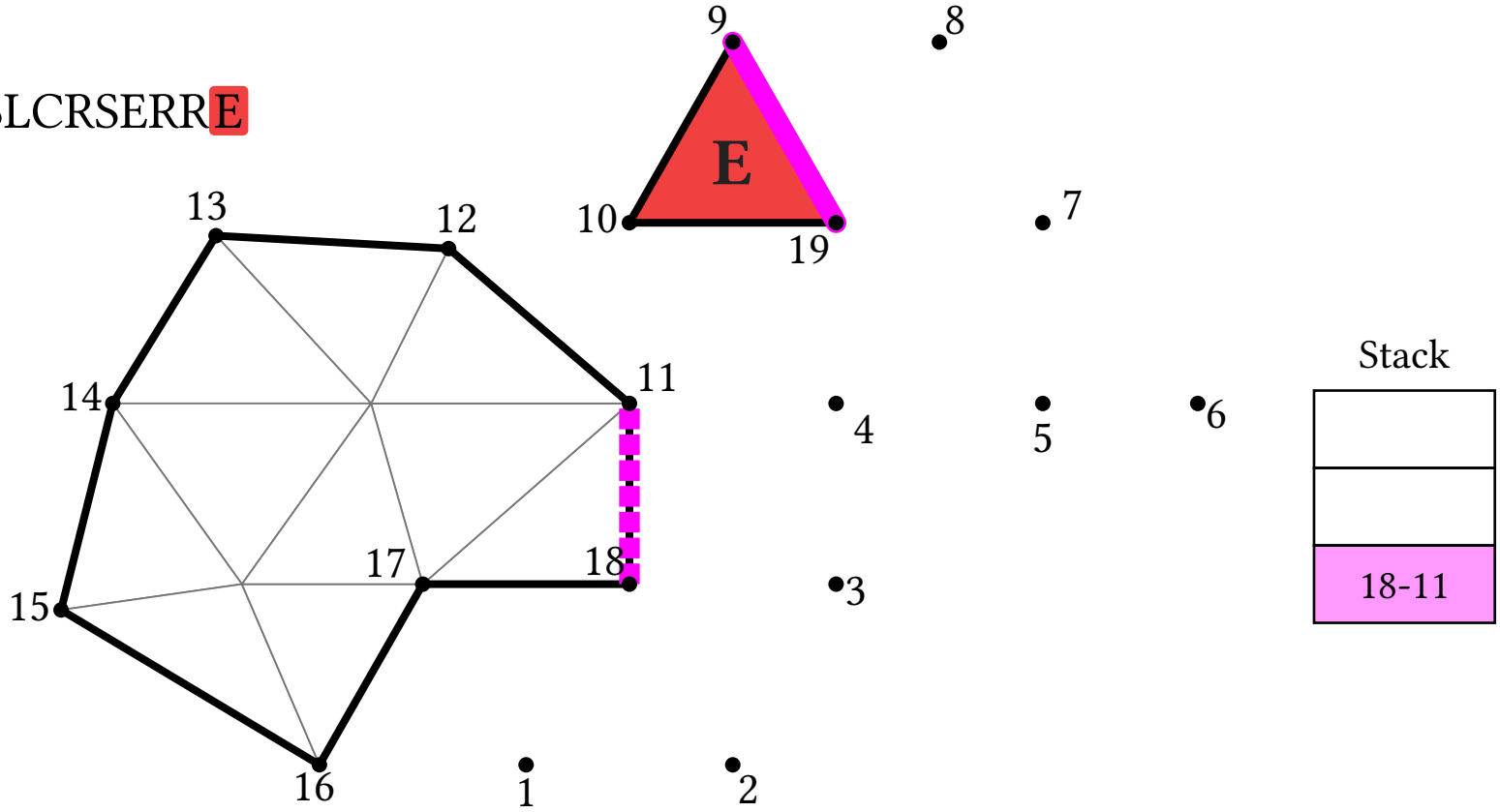
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERR



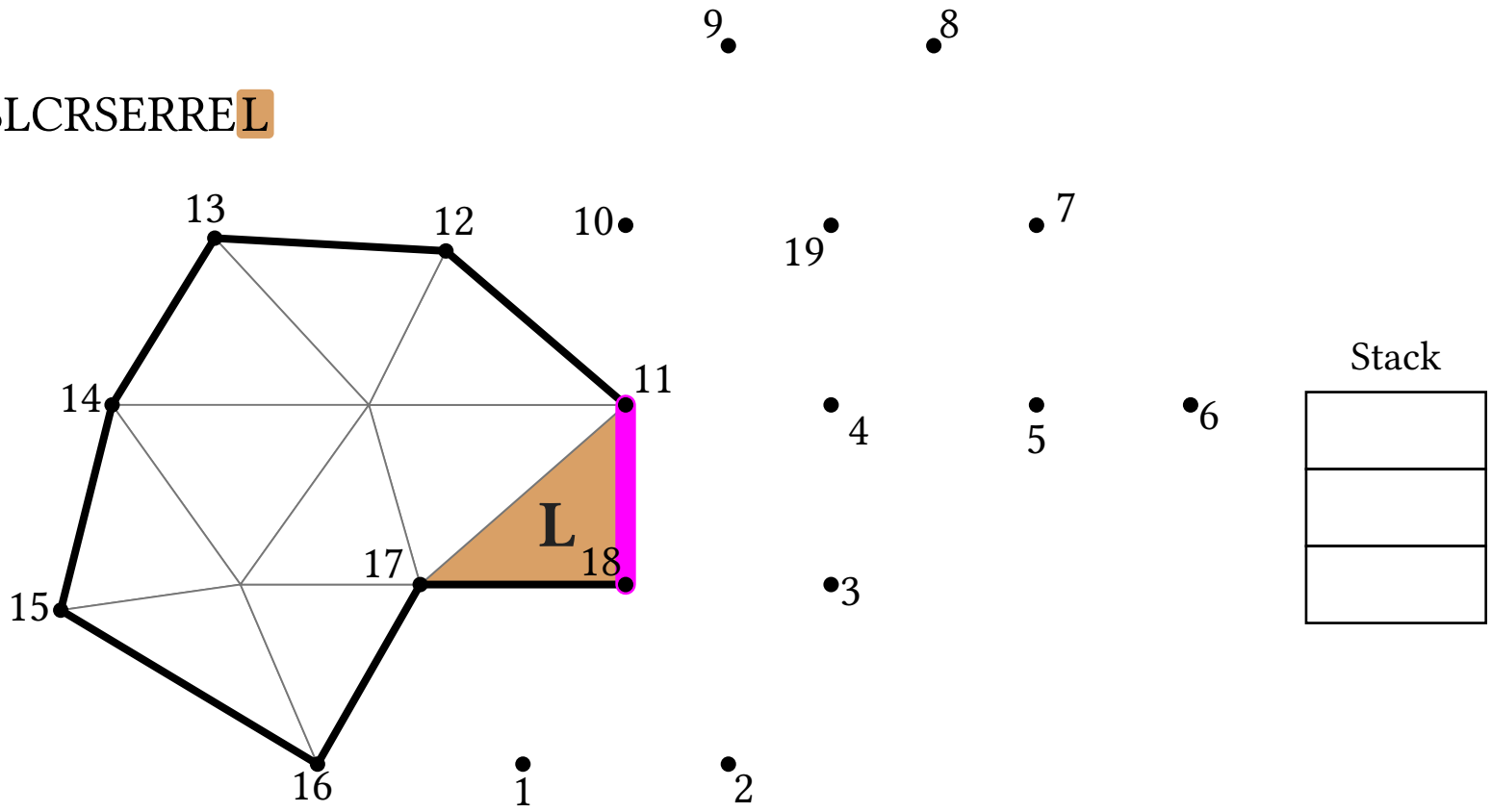
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRE



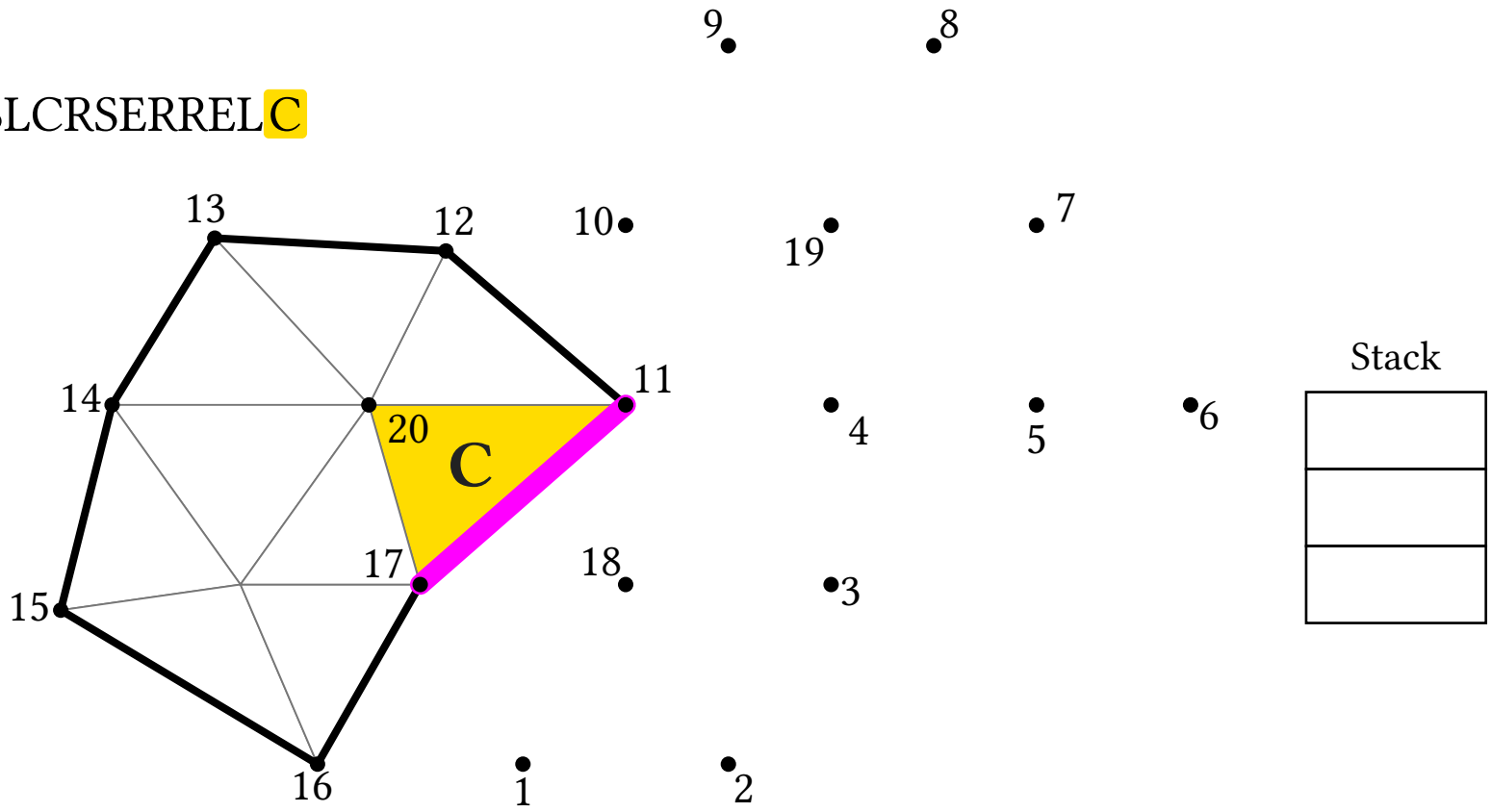
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERREL



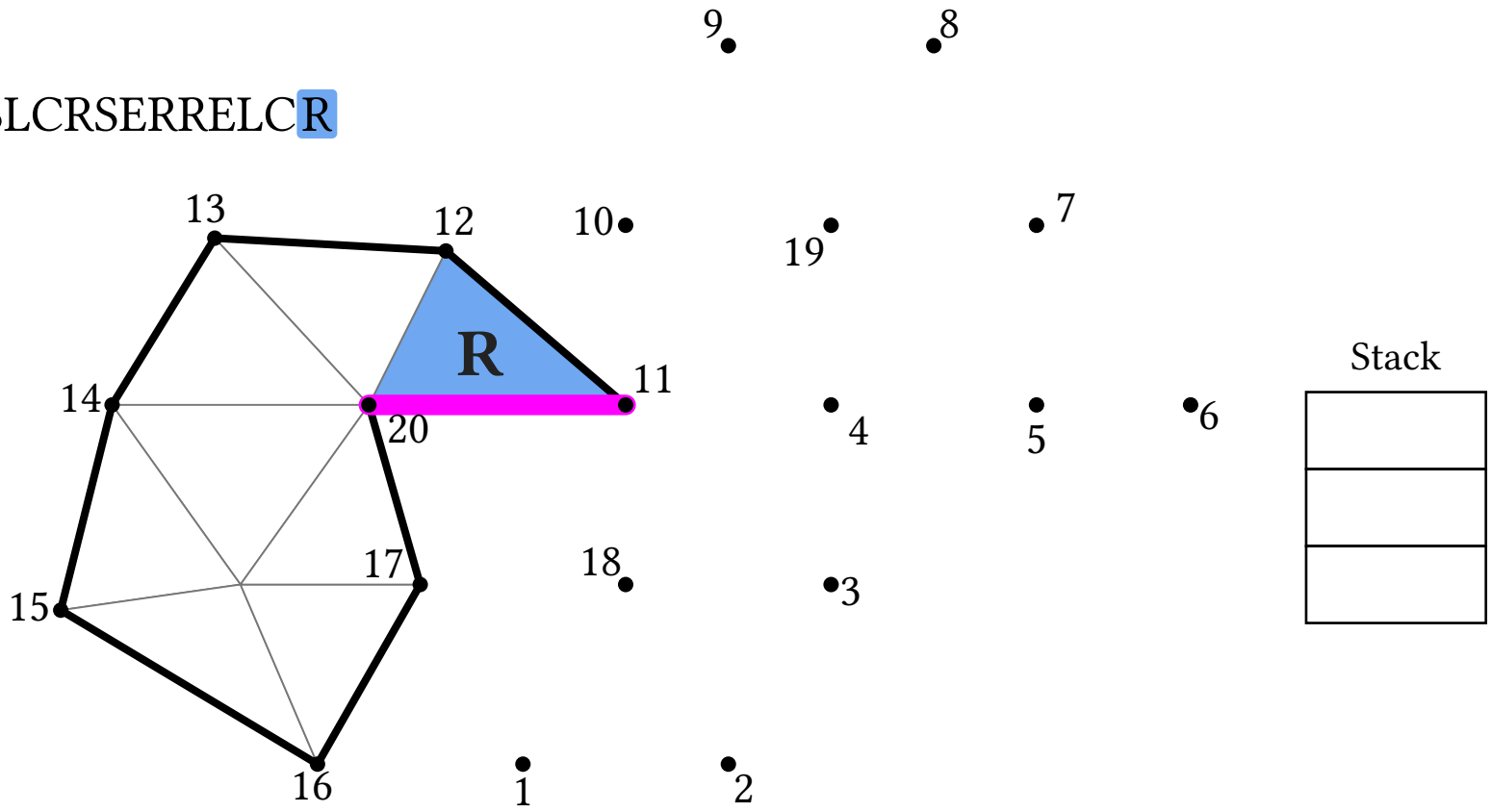
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERREL**C**



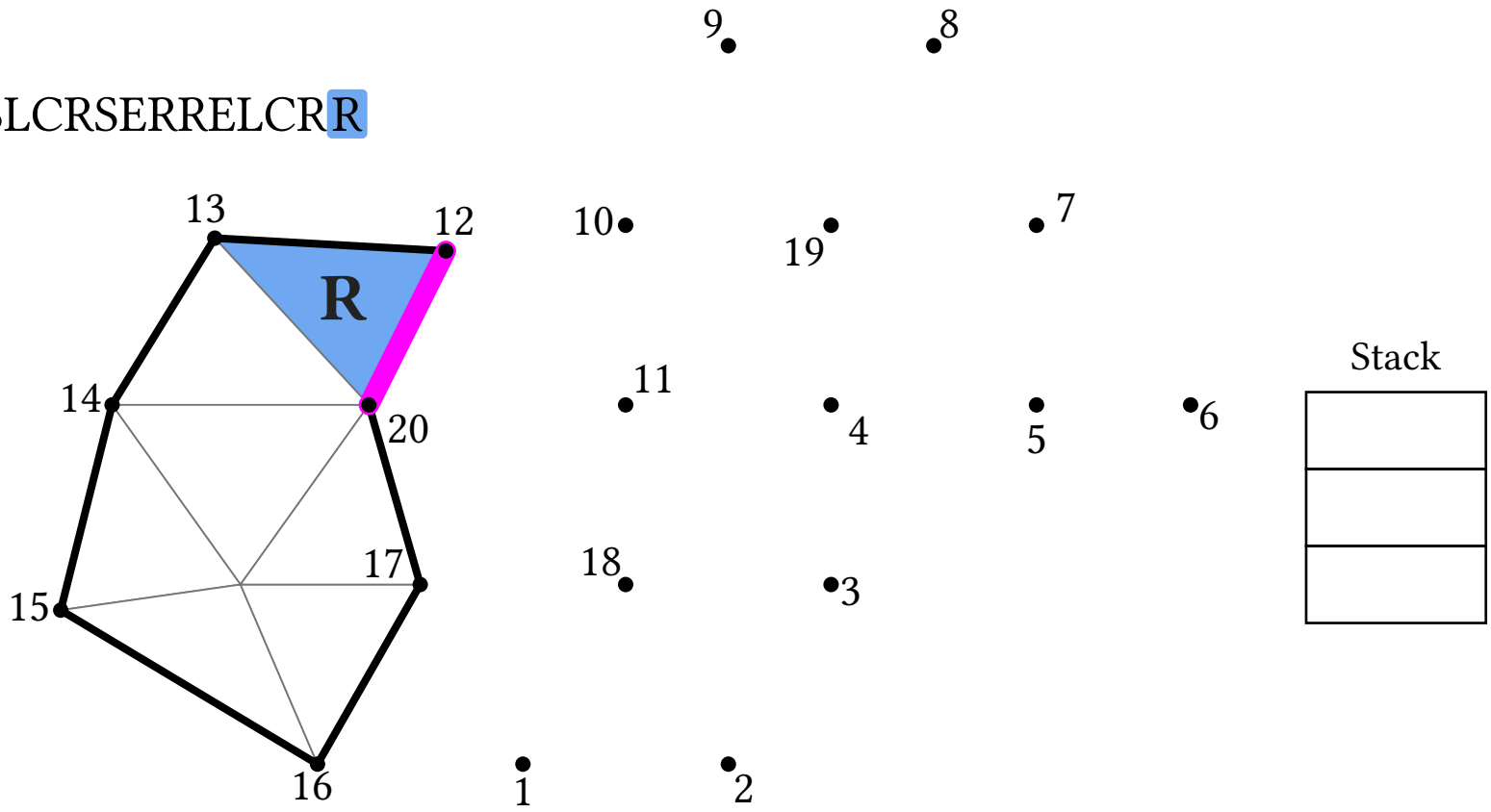
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCR



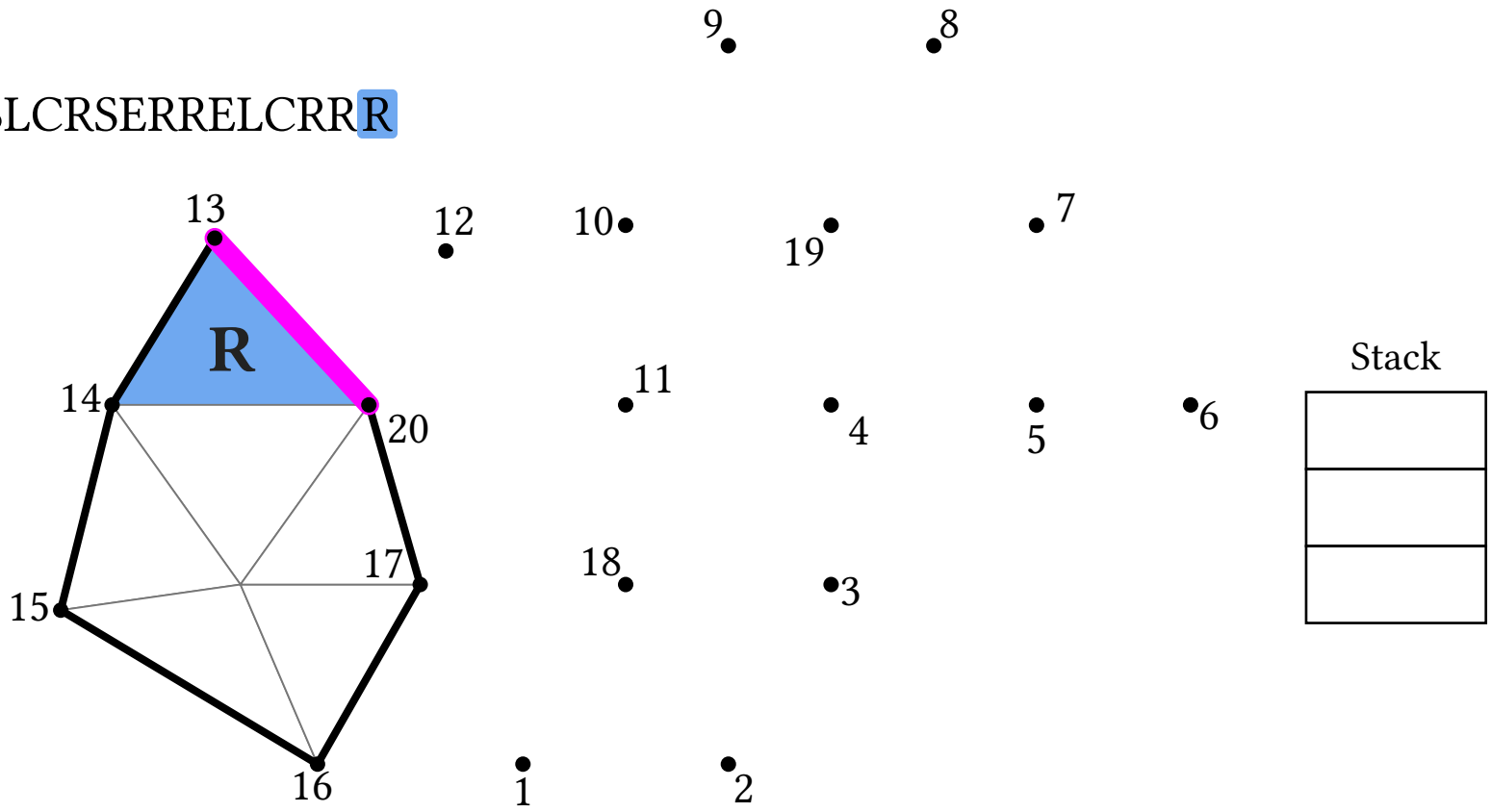
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRR



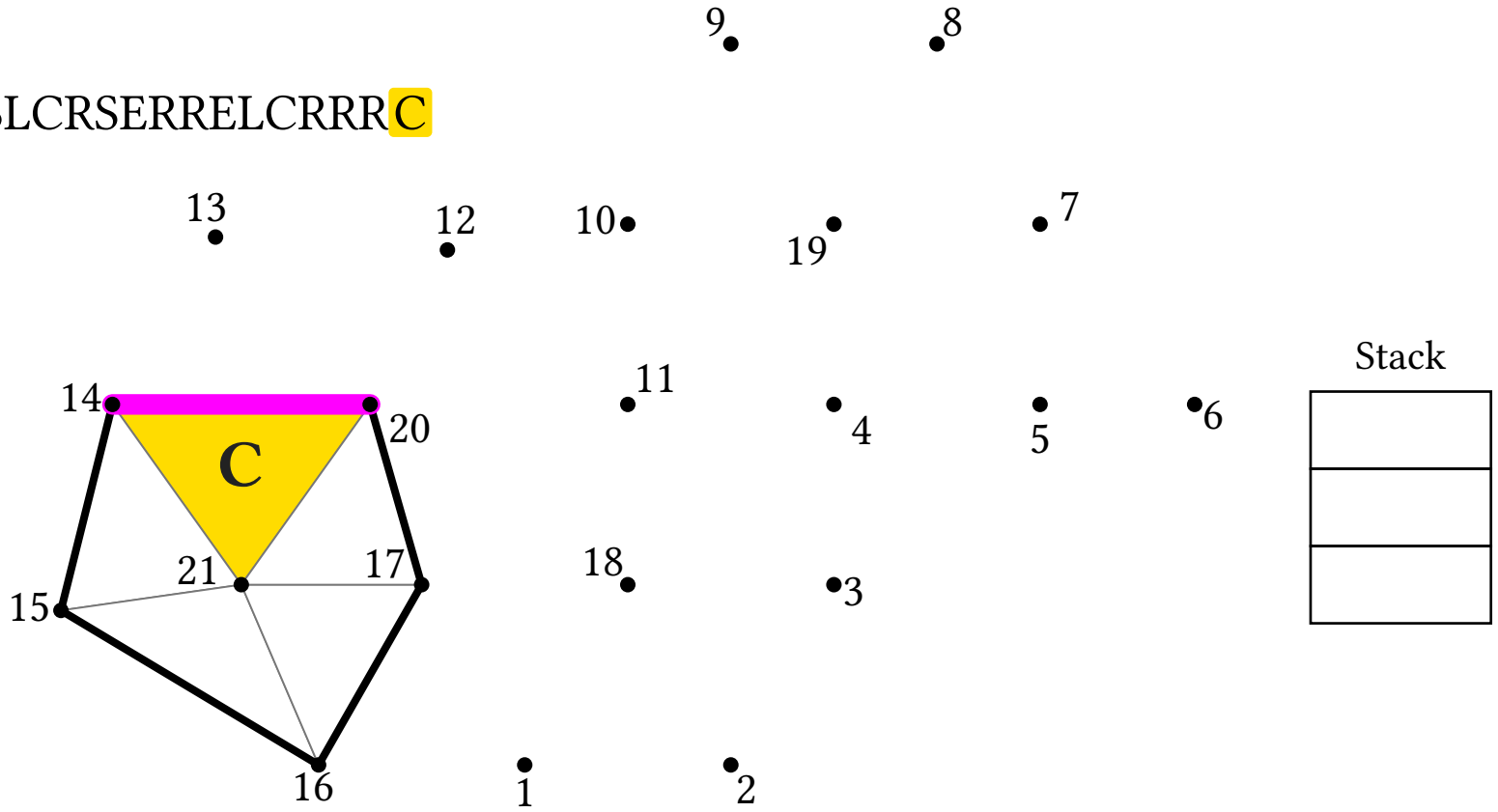
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRR



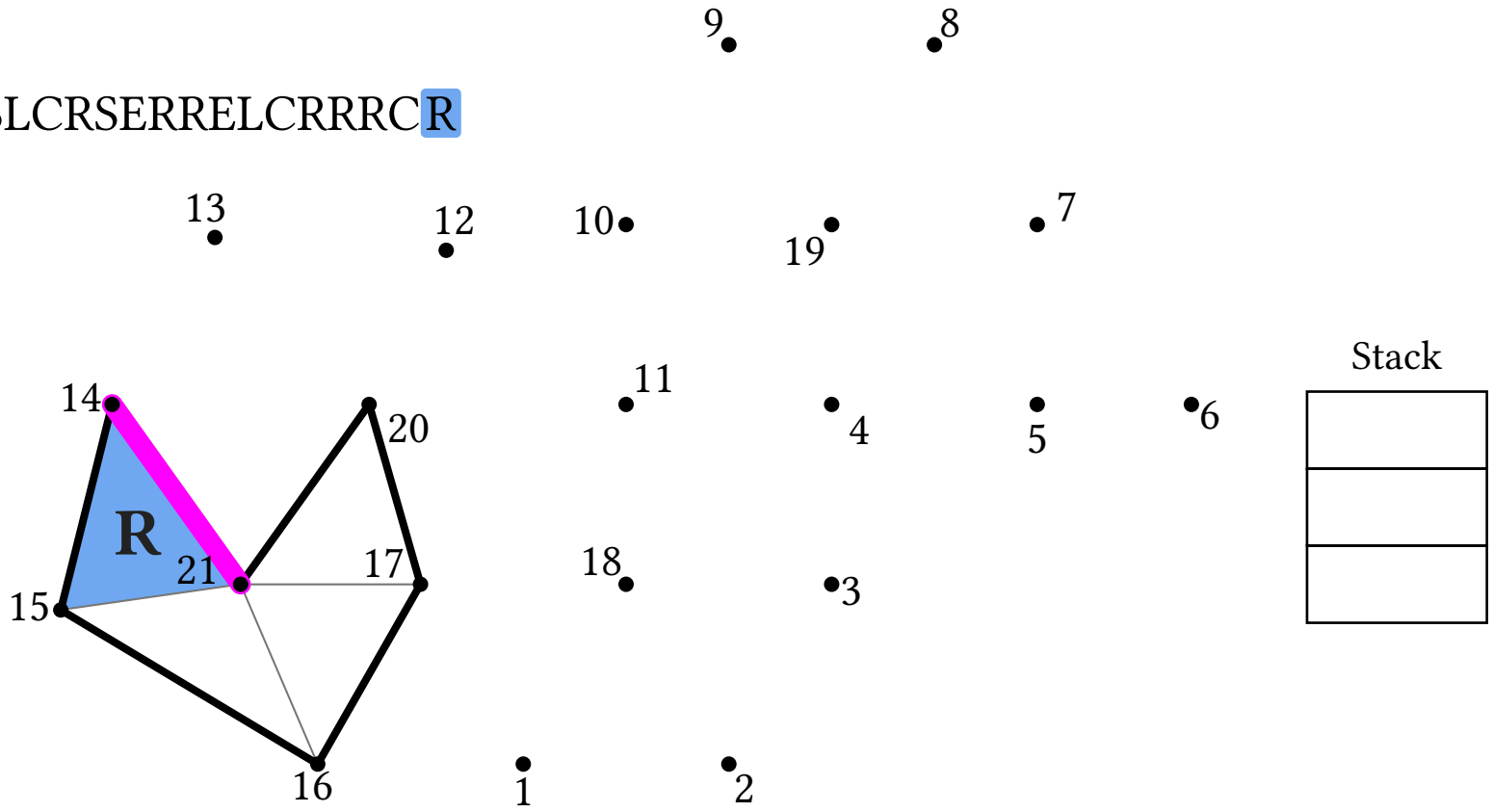
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRC



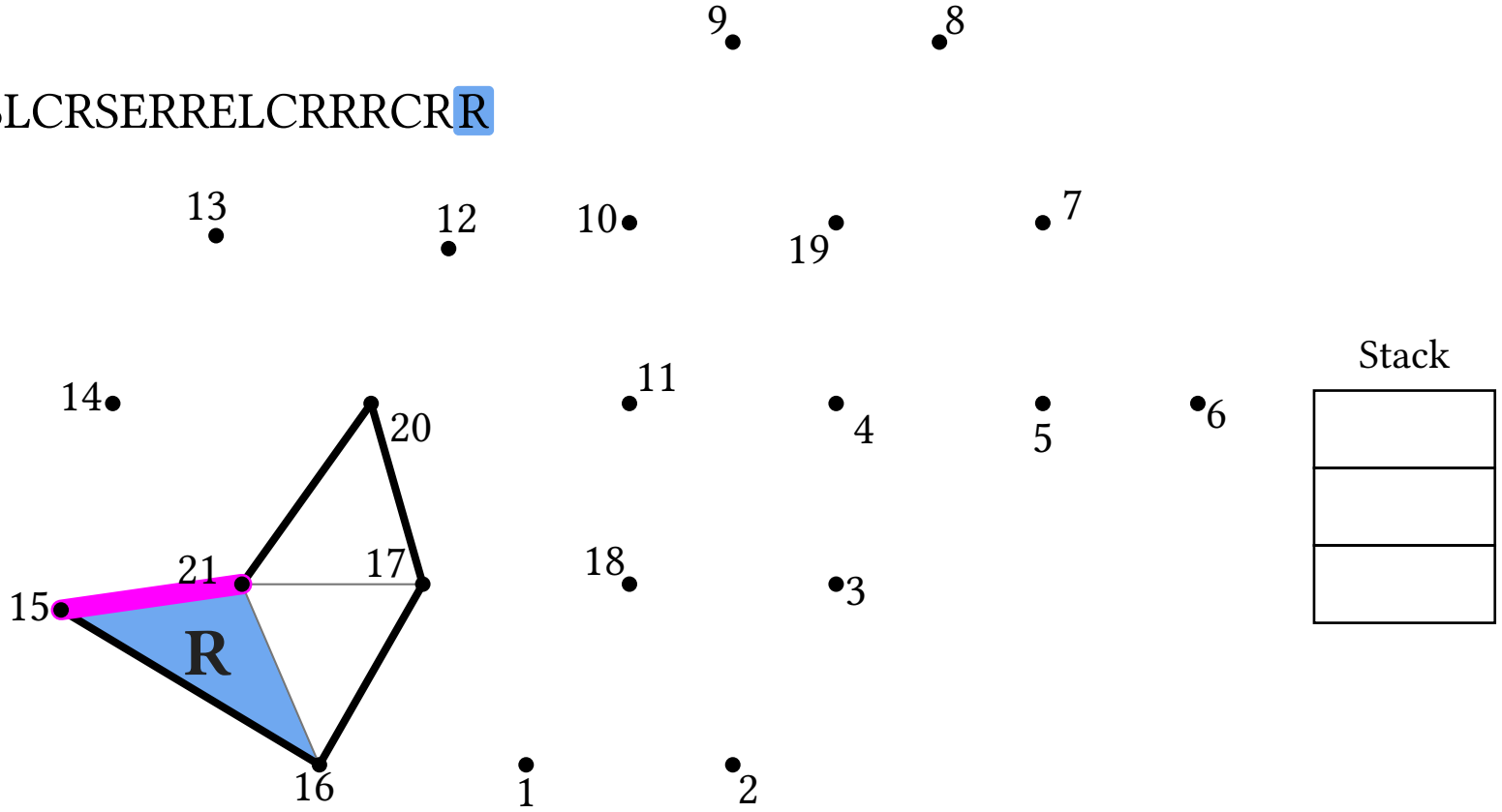
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Historie = CCRRRSLCRSERRELCRRRCR



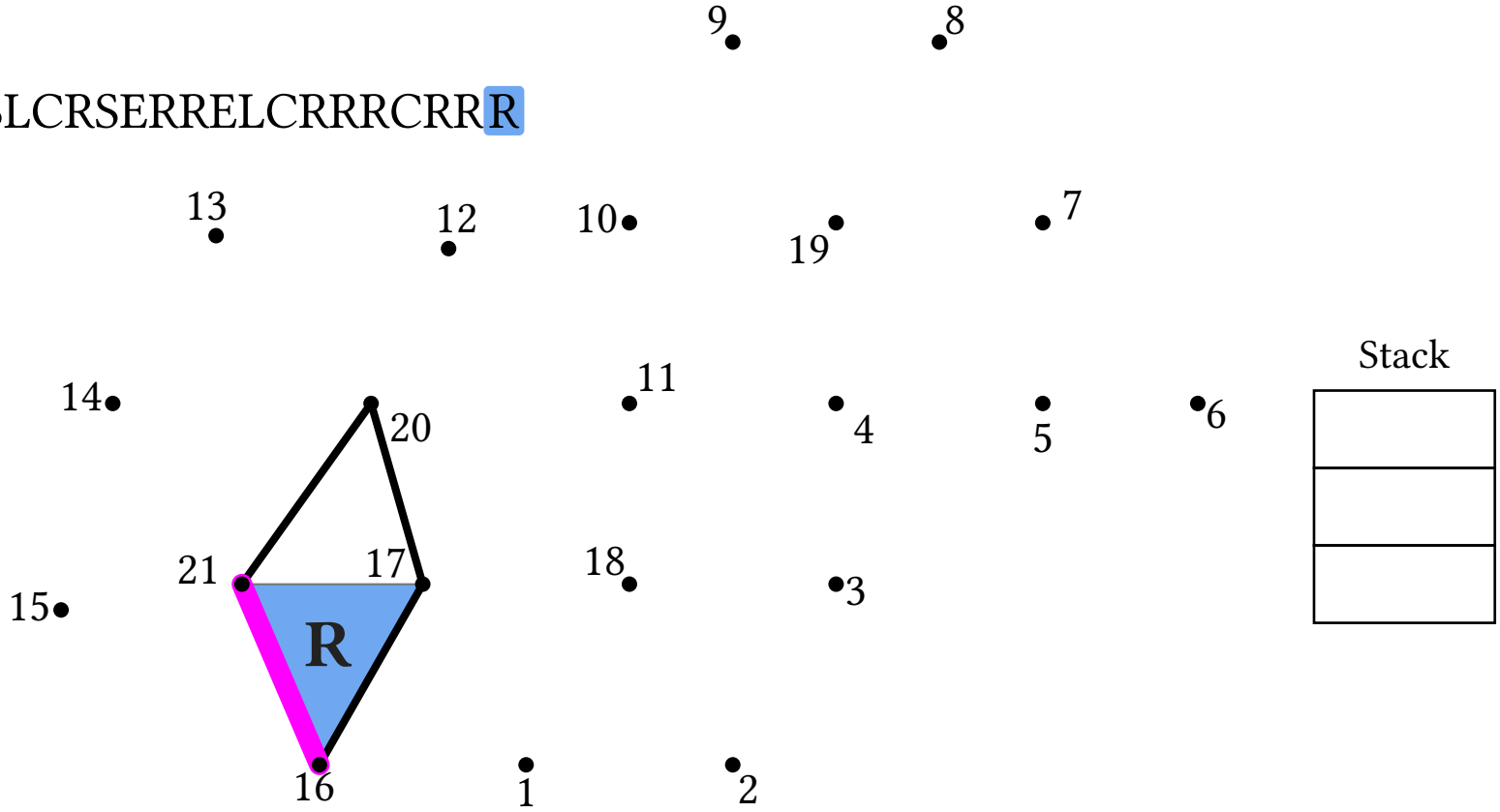
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRR



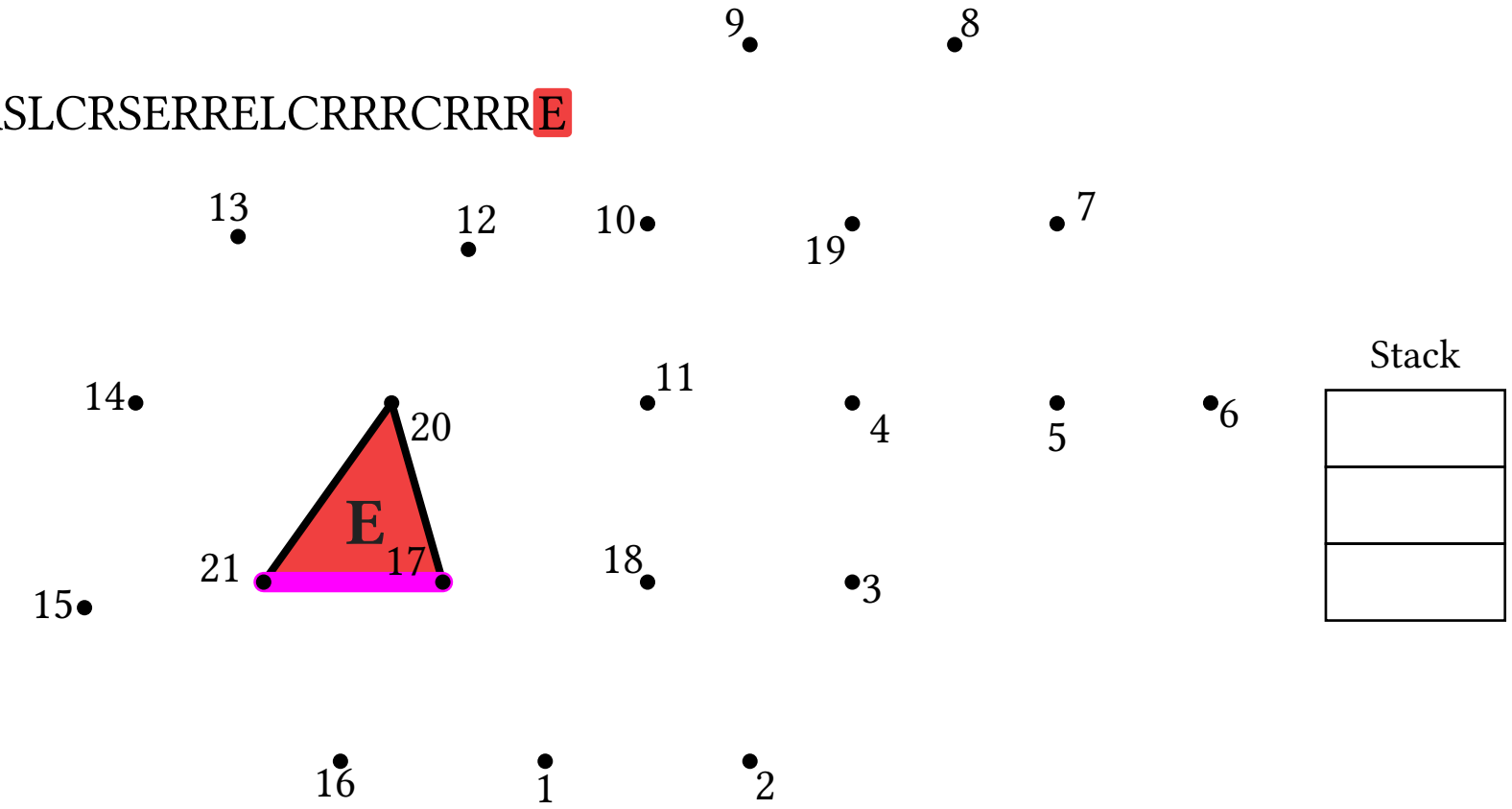
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRRR



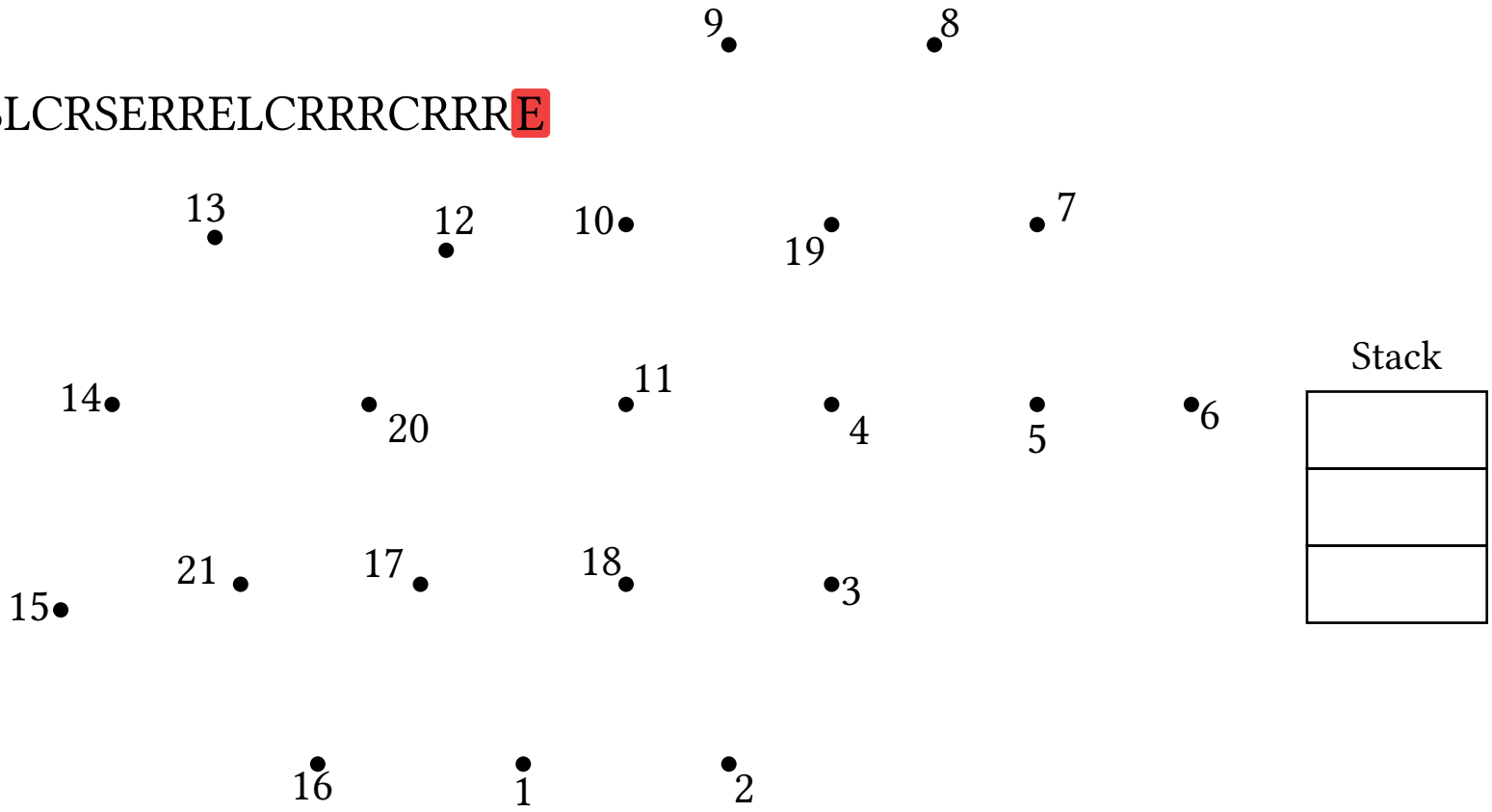
Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE



Obrázek 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE



Obrázek 5: Jednoduchá komprese.

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

Kódování: C = 0, S = 100, R = 101, L = 110, E = 111

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

Kódování: C = 0, S = 100, R = 101, L = 110, E = 111

(CLERS) 0110111101100 bitové pole:

00101101 10110011 00101100 11110110 11111100 10110110 10101101 101111

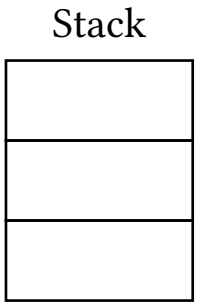
4. Dekomprese

4.1 Možnosti

Existuje víc algoritmů, implementuji spirale-reversi a wrap&zip.

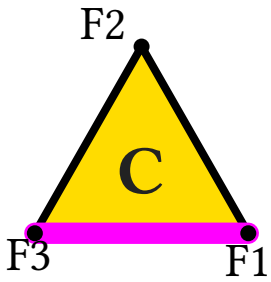
Demonstrováno na wrap&zip, je intuitivnější.

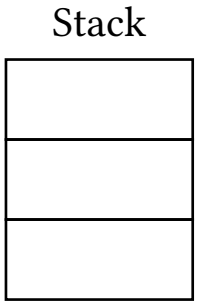
Historie = CCRRRSLCRSERRELCRRRCRRRE



Obrázek 6: Dekomprese.

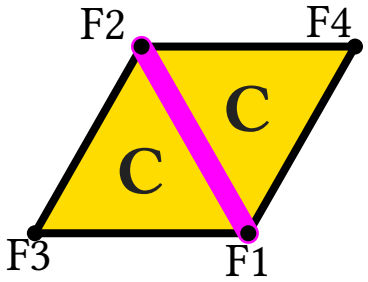
Historie = CCRRRSLCRSERRELCRRRCRRRE

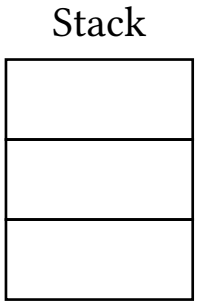




Obrázek 6: Dekomprese.

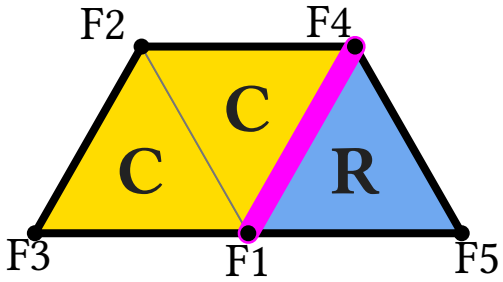
Historie = CCRRRSLCRSERRELCRRRCRRRE





Obrázek 6: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

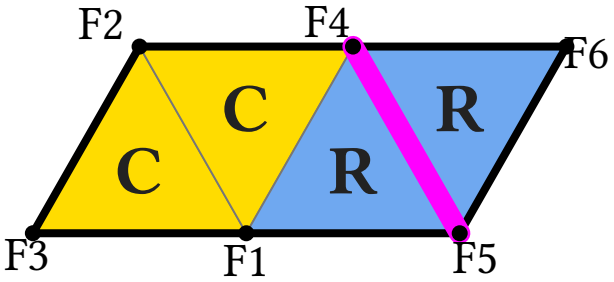


Stack



Obrázek 6: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

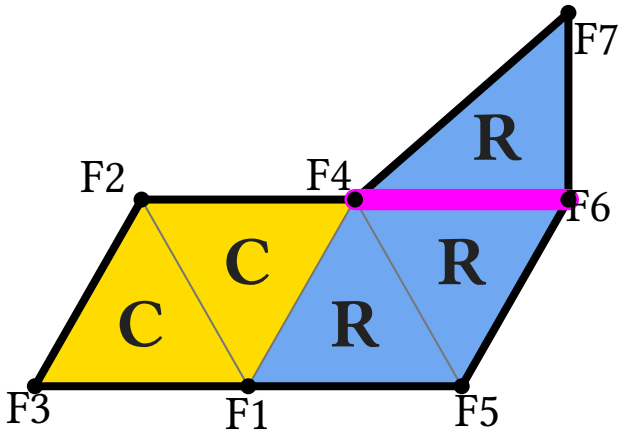


Obrázek 6: Dekomprese.

Stack



Historie = CCRRRSLCRSERRELCRRRCRRRE

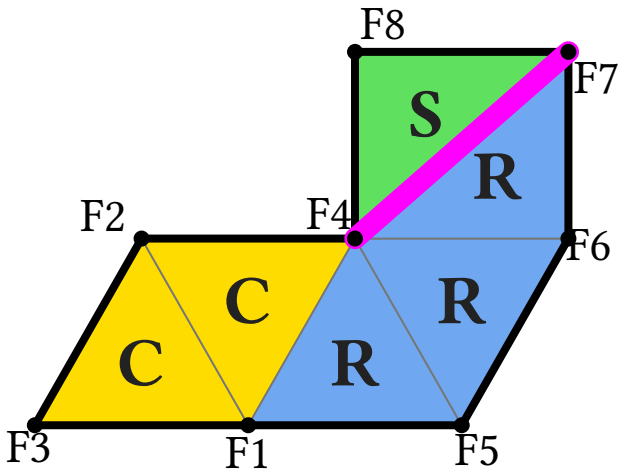


Stack



Obrázek 6: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

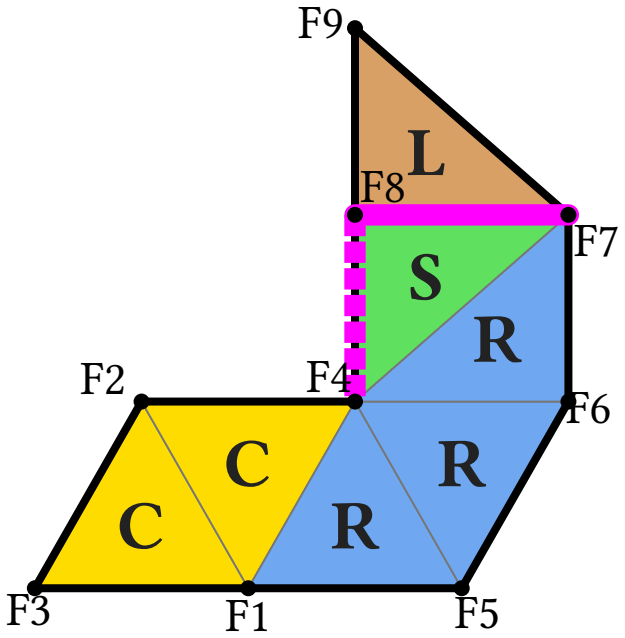


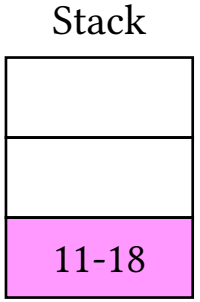
Stack



Obrázek 6: Dekomprese.

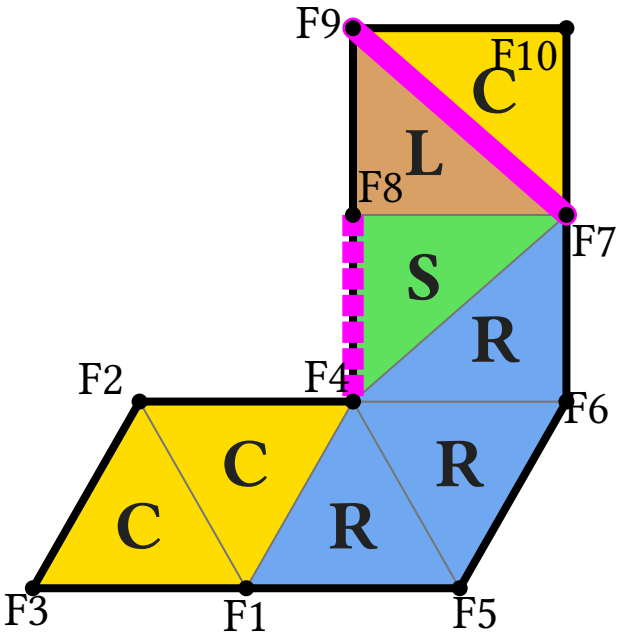
Historie = CCRRRSLCRSERRELCRRRCRRRE

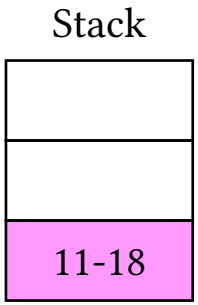




Obrázek 6: Dekomprese.

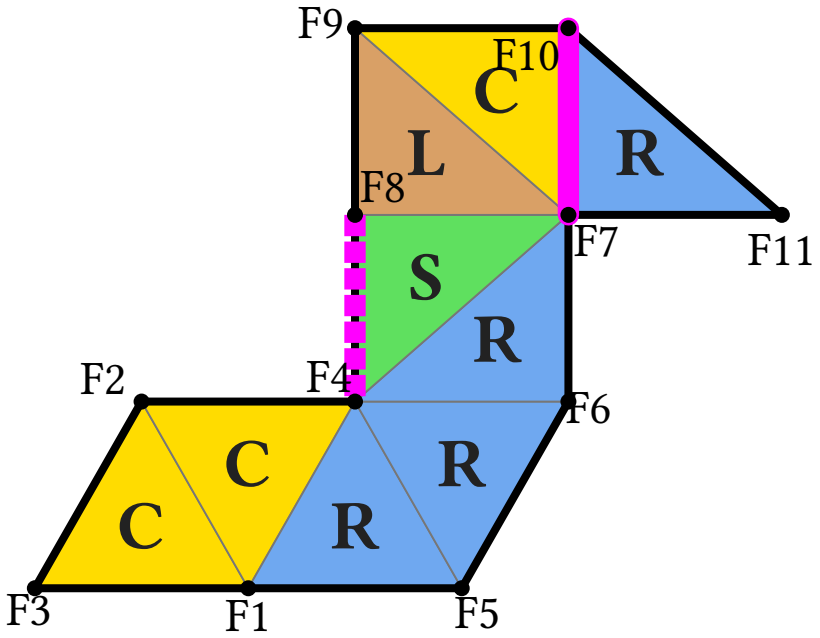
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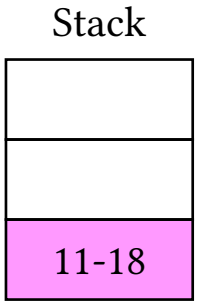




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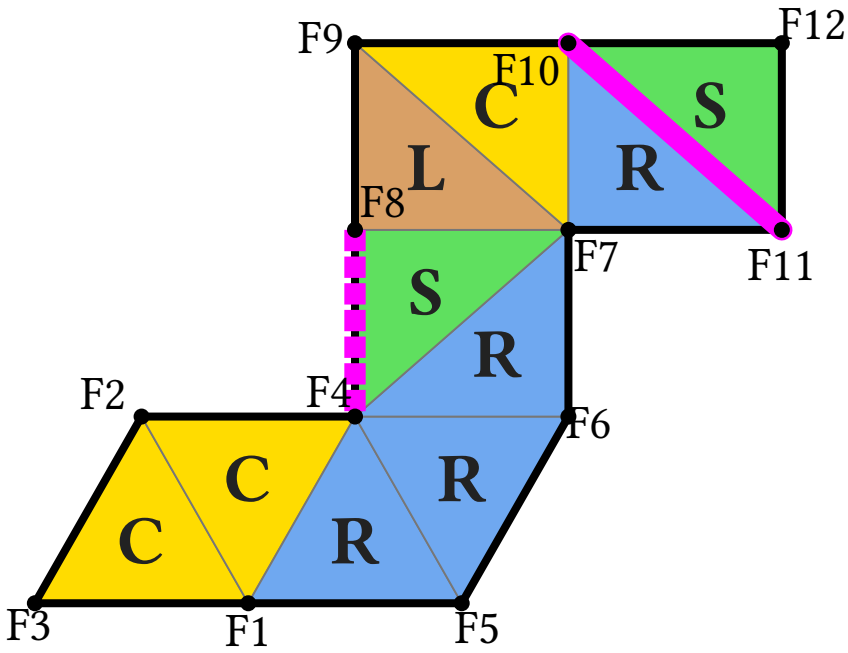
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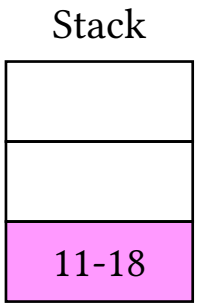




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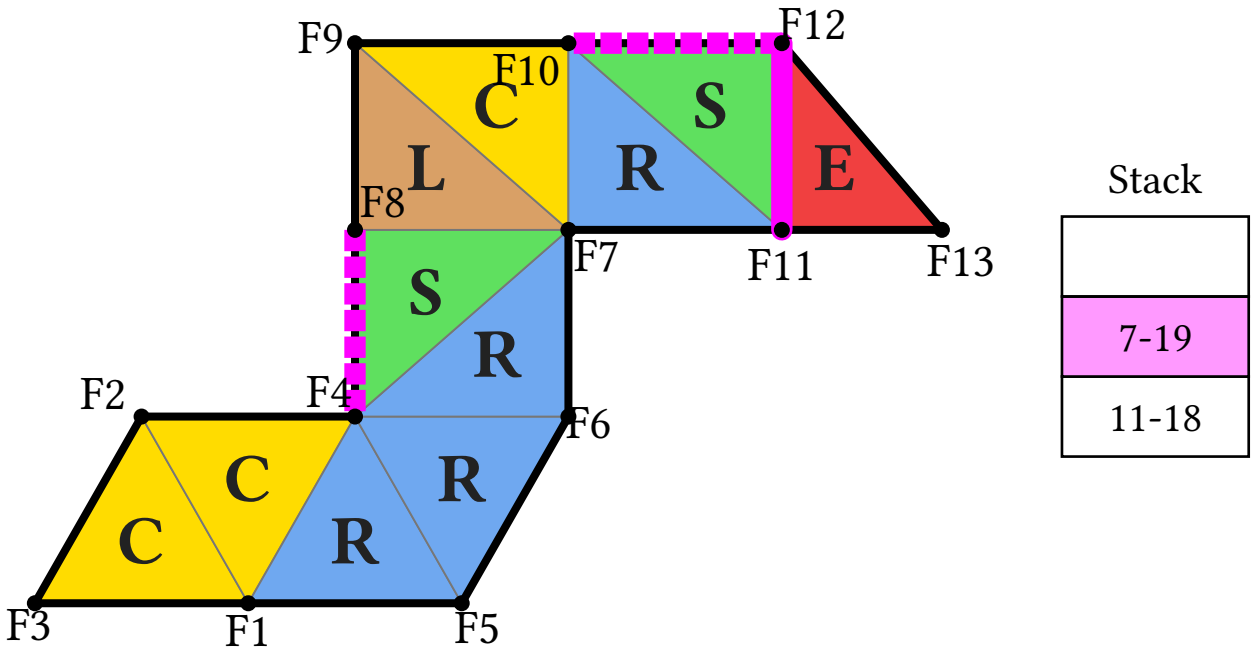
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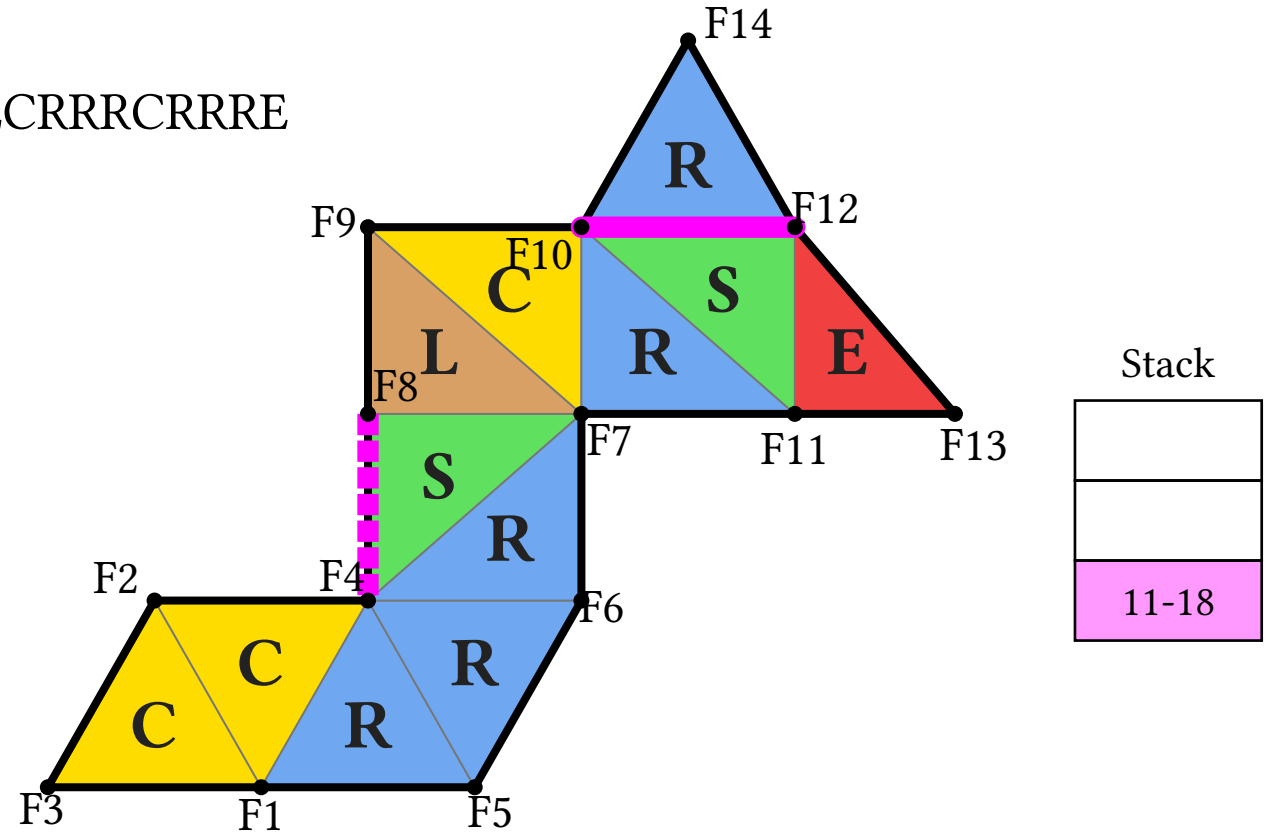
Obrázek 6: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE



Obrázek 6: Dekomprese.

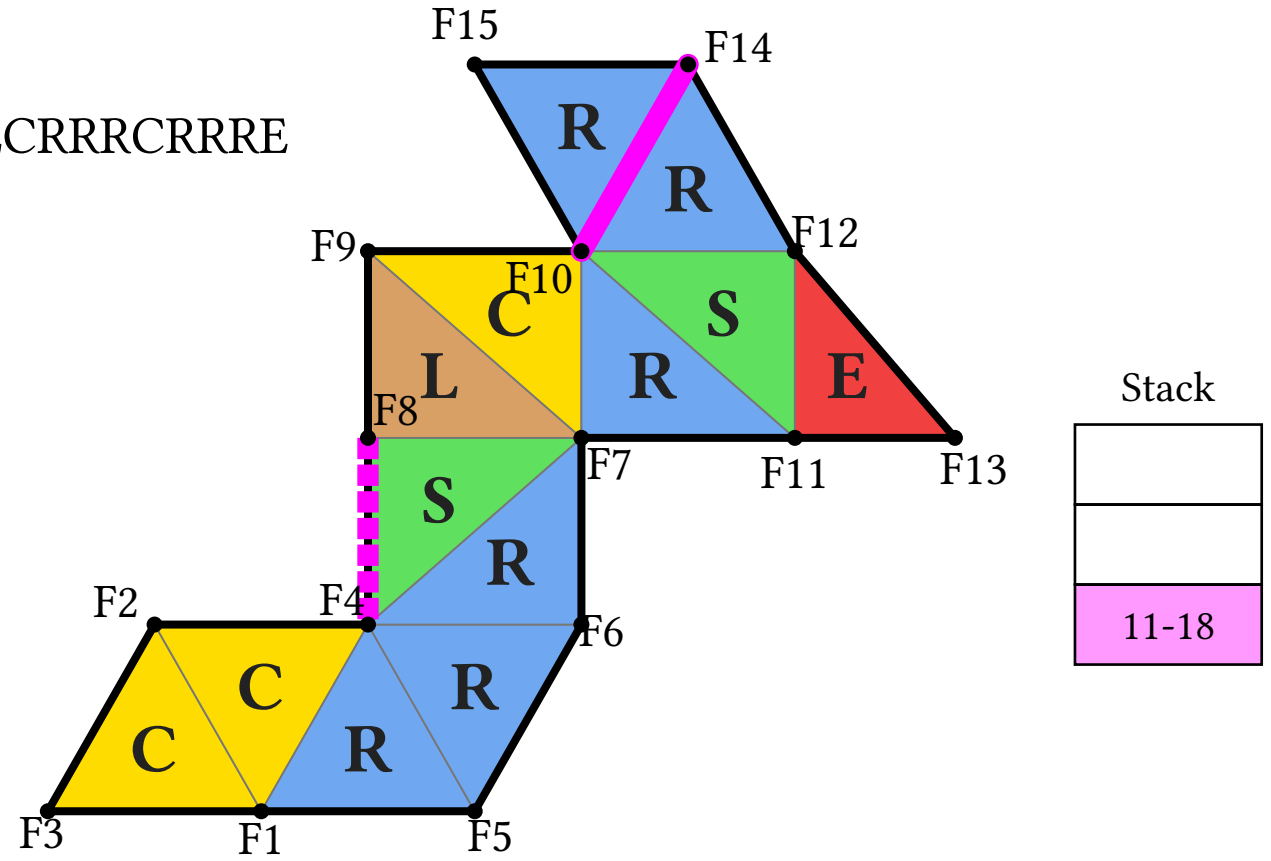
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Obrázek 6: Dekomprese.

4.2 Animace

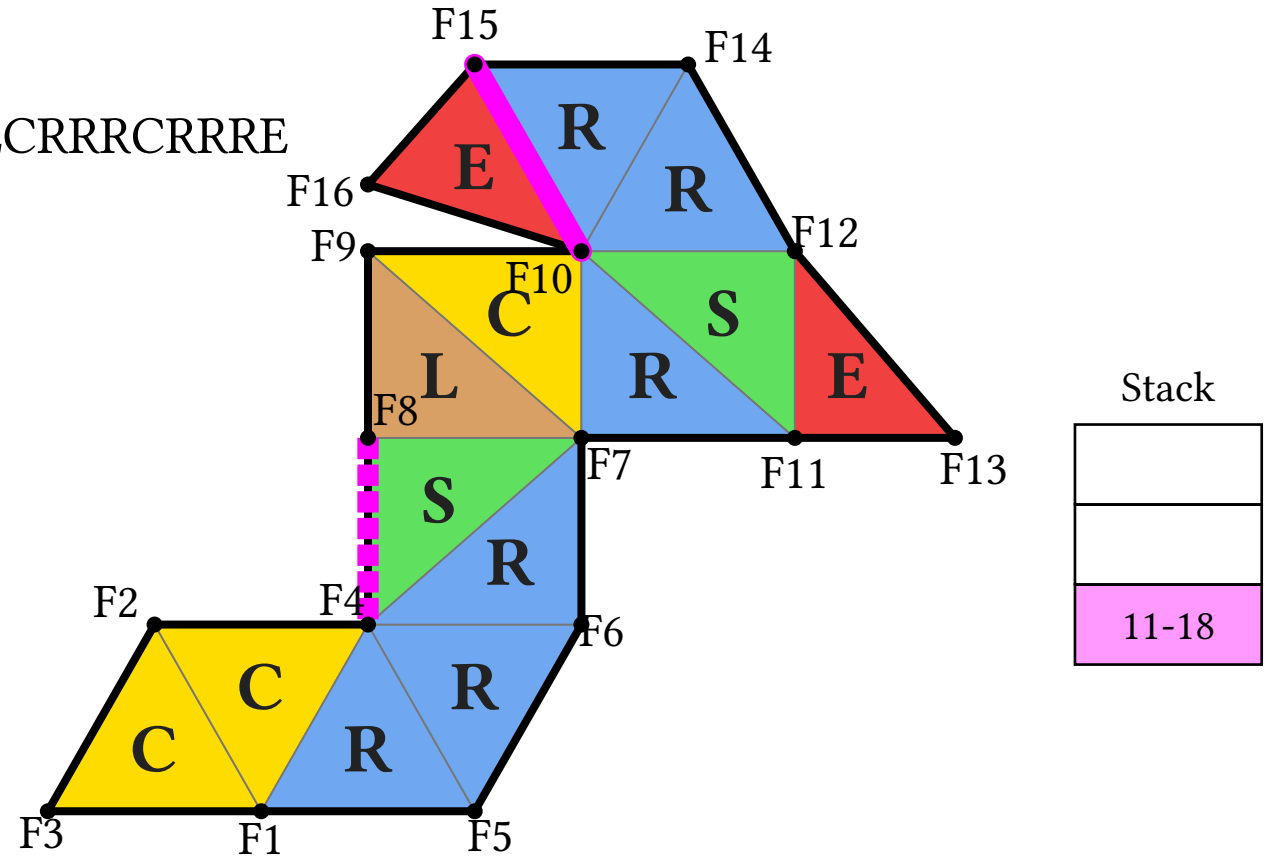
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Obrázek 6: Dekomprese.

4.2 Animace

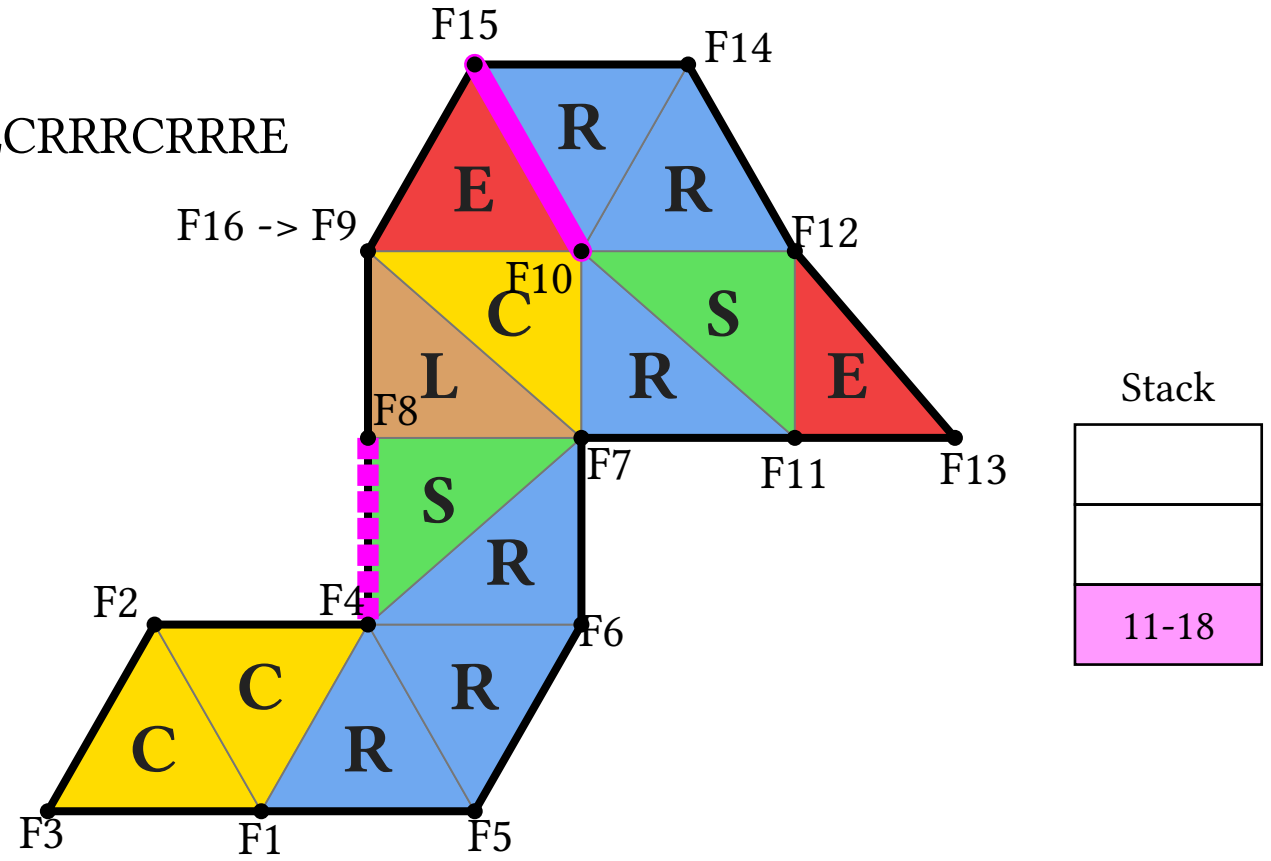
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Obrázek 6: Dekomprese.

4.2 Animace

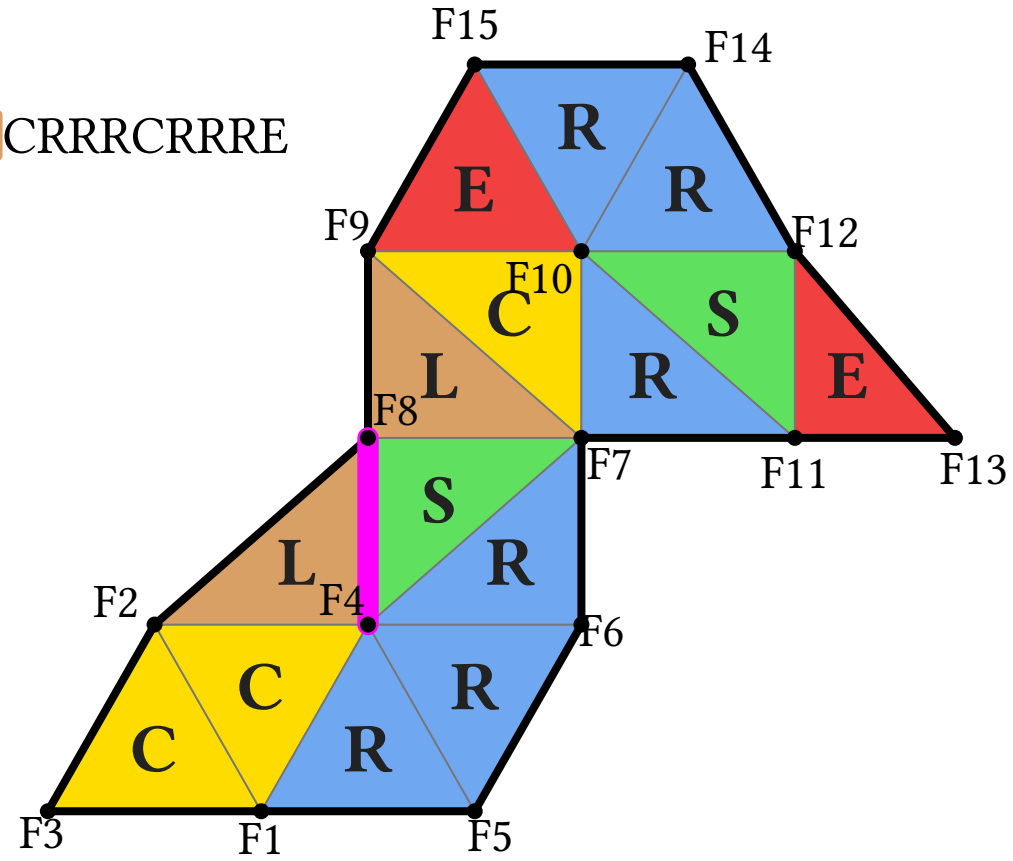
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Obrázek 6: Dekomprese.

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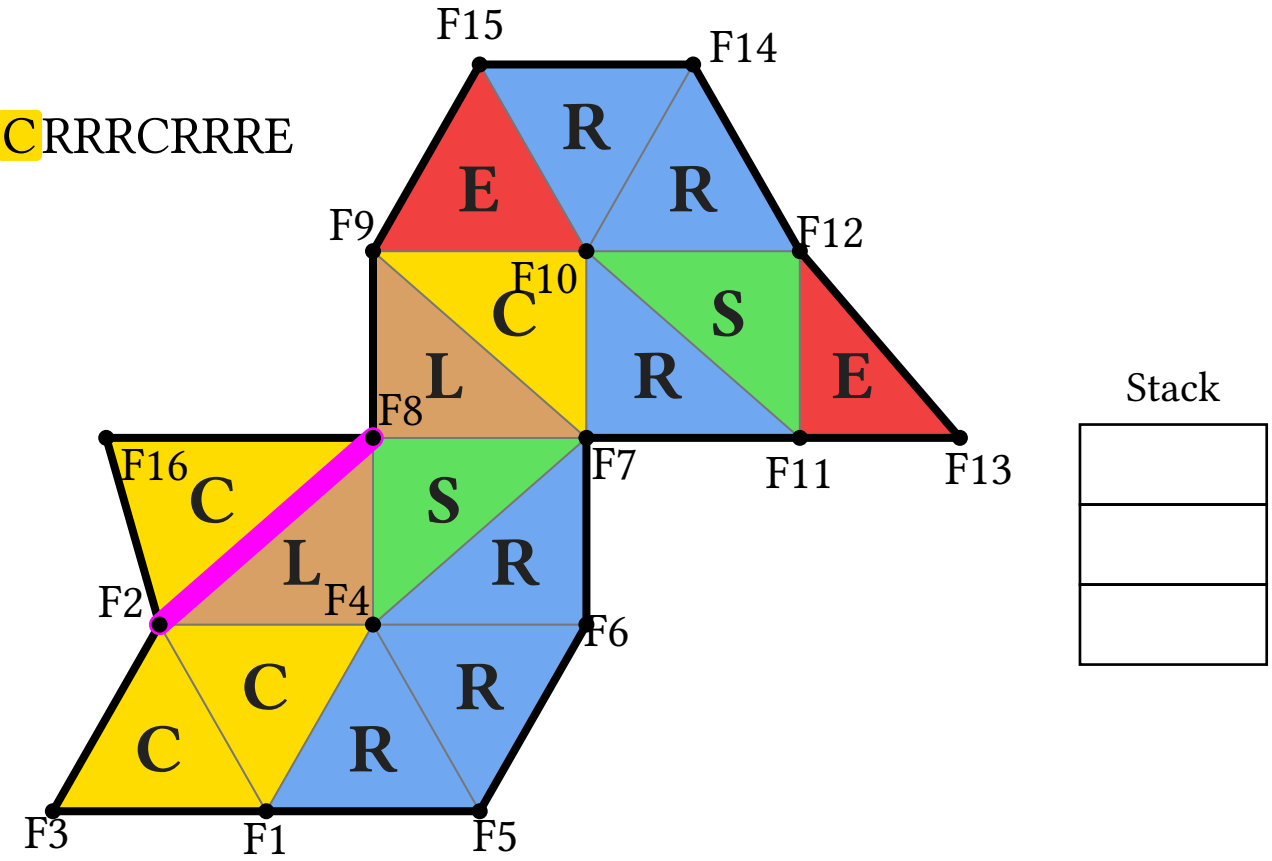
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Obrázek 6: Dekomprese.

4.2 Animace

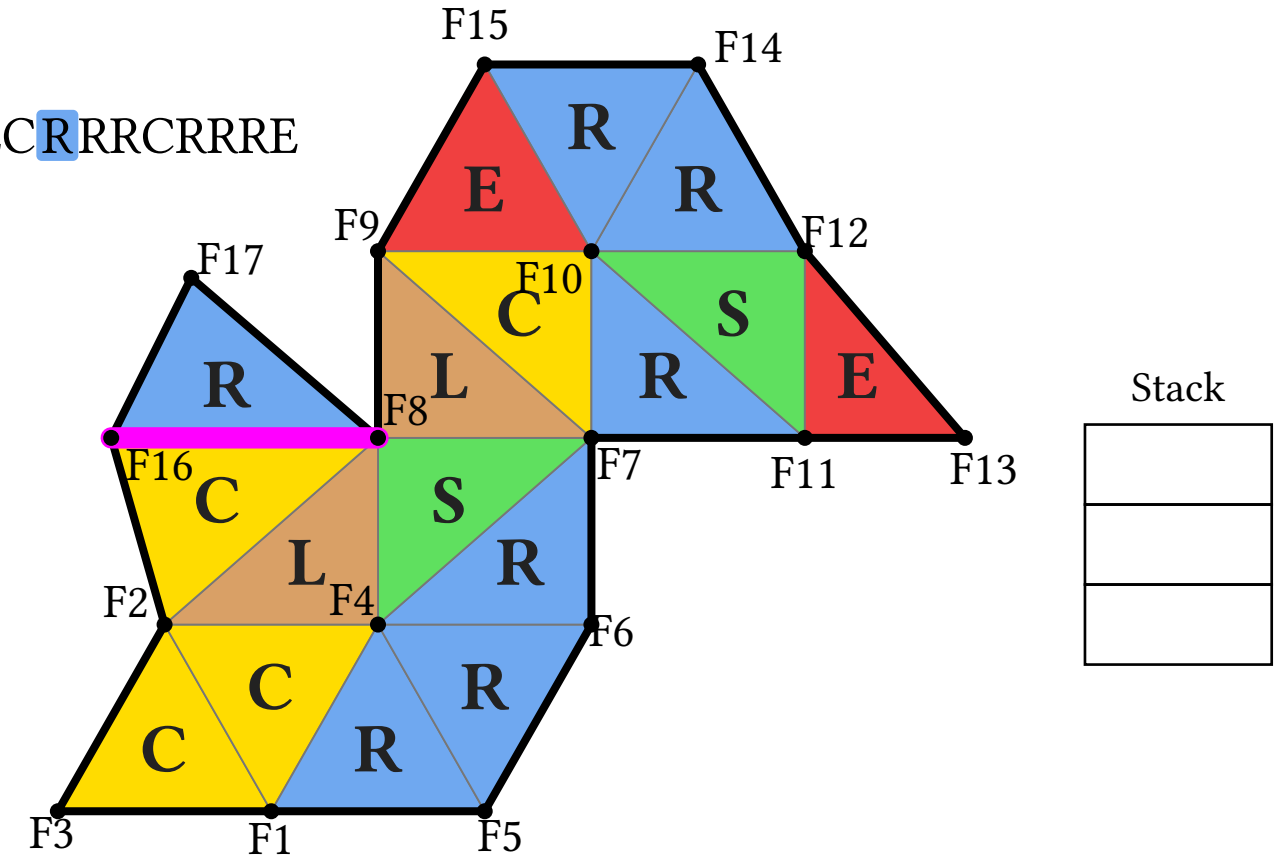
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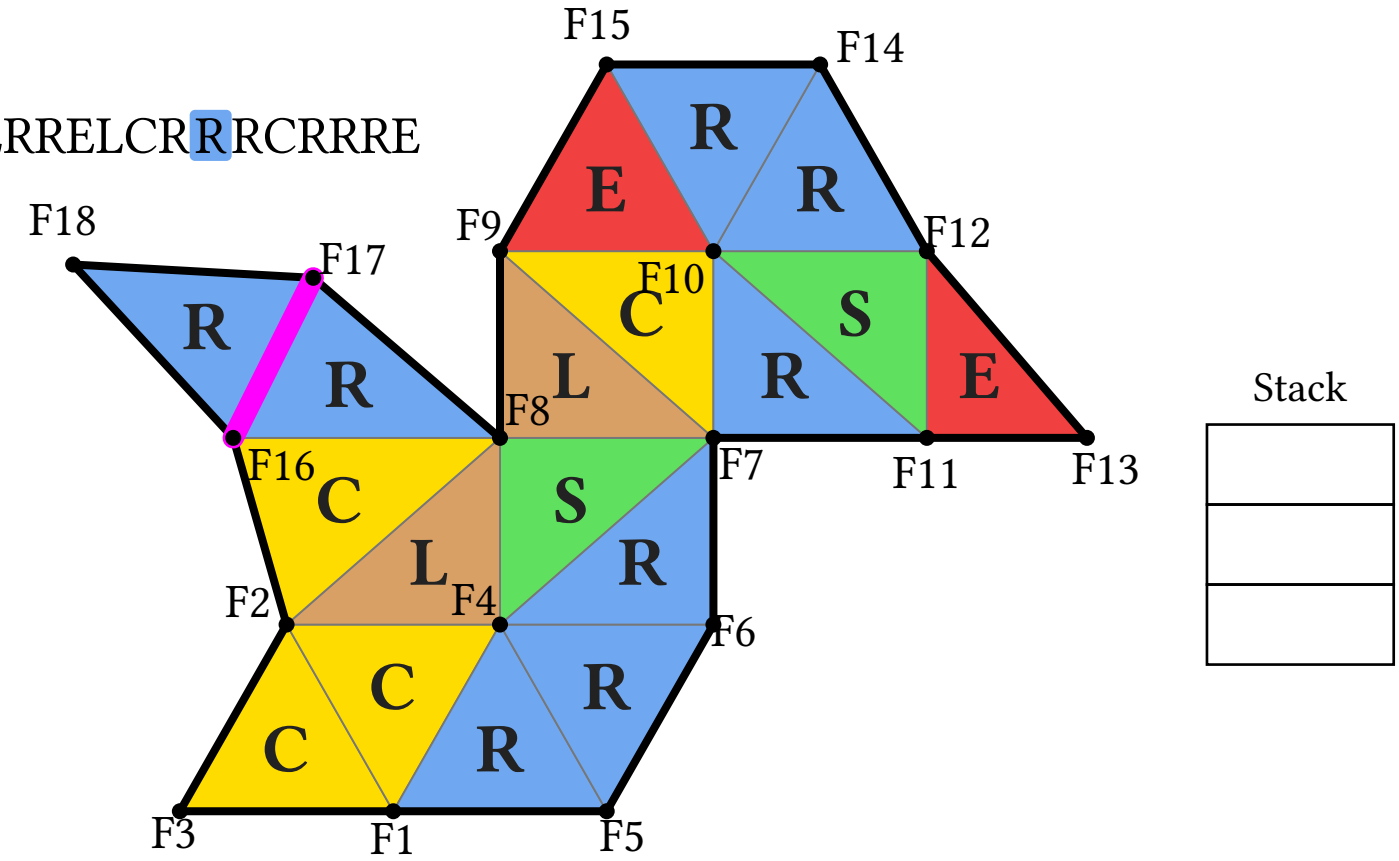
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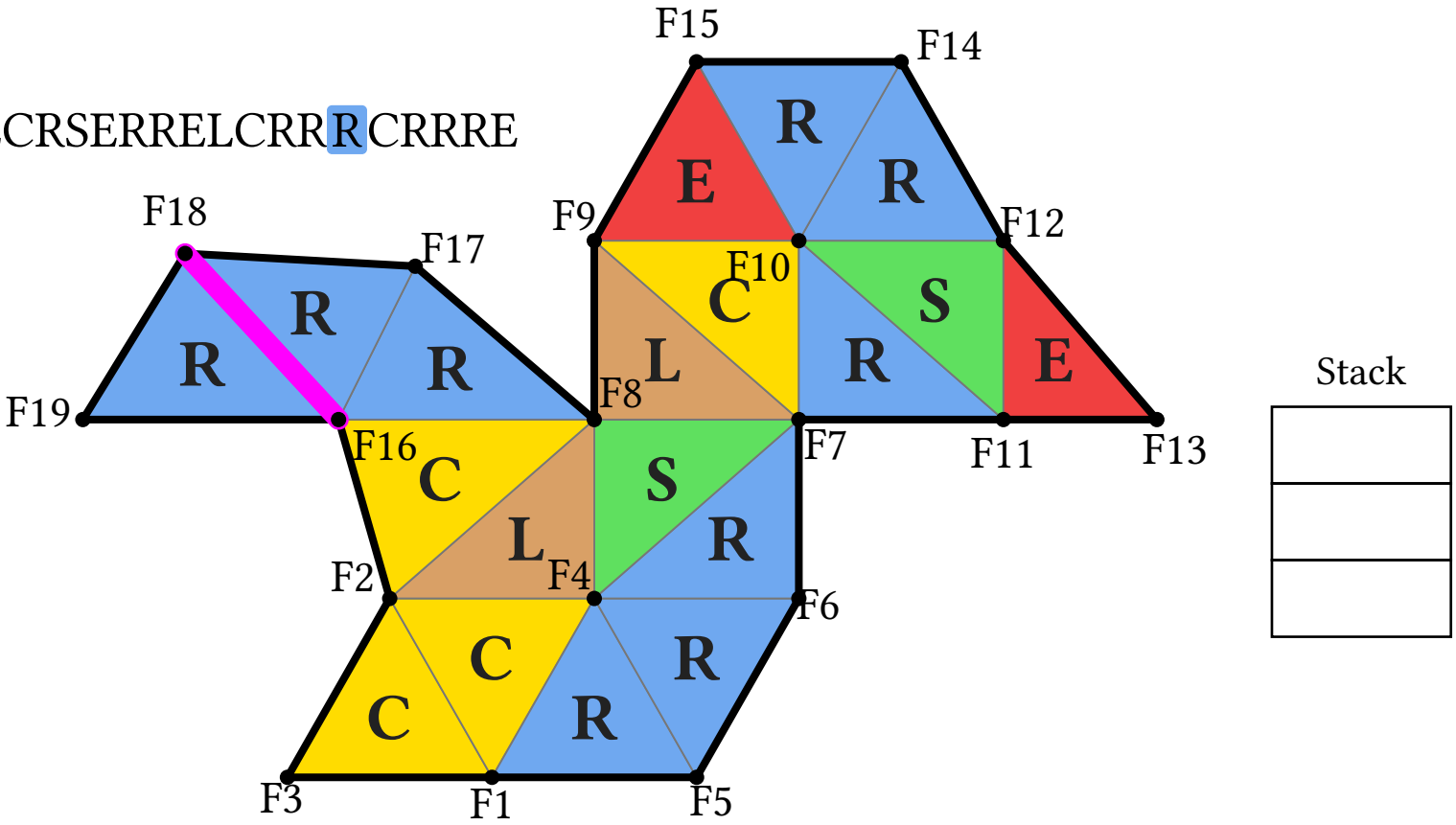
Obrázek 6: Dekomprese.

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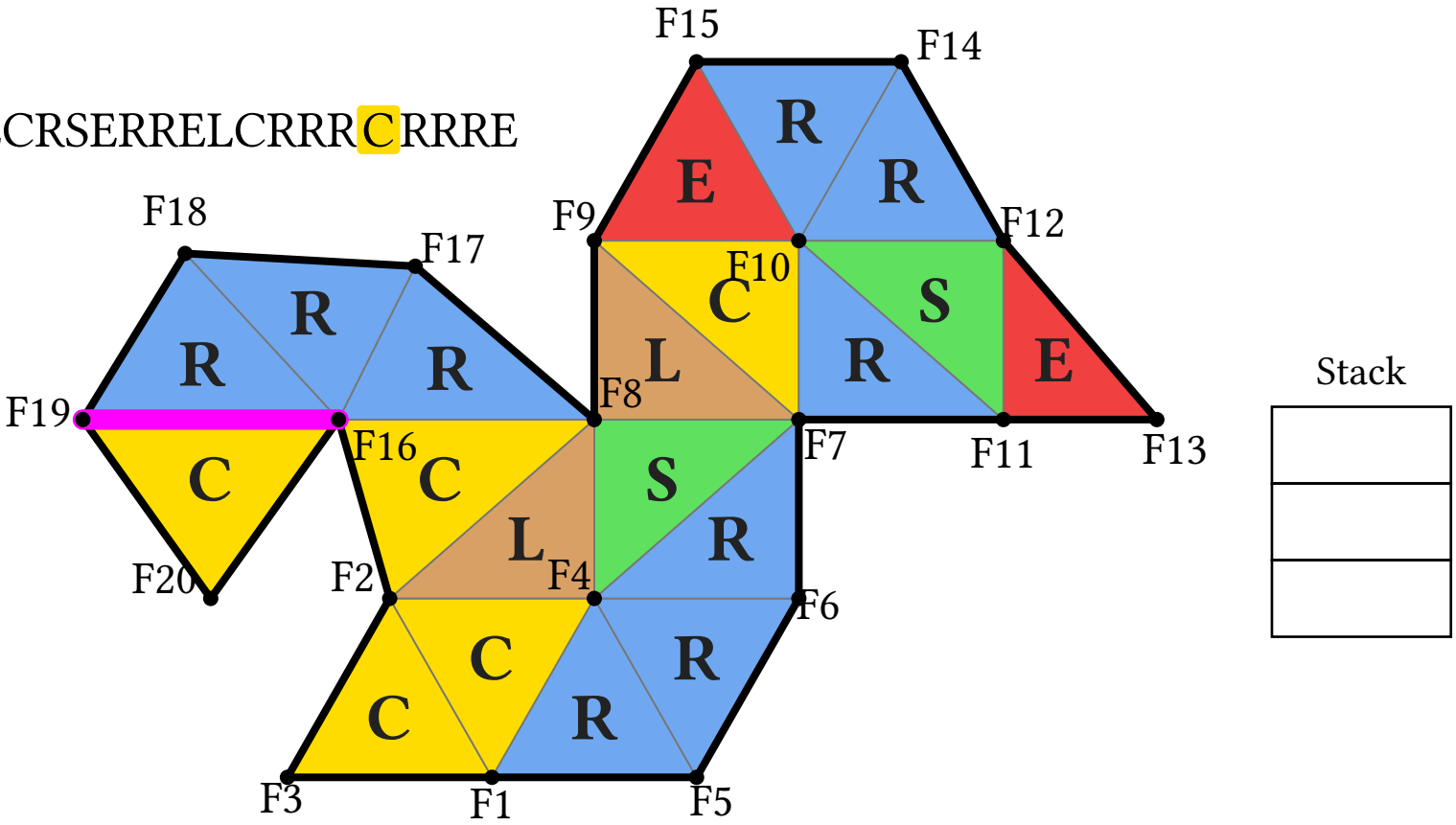
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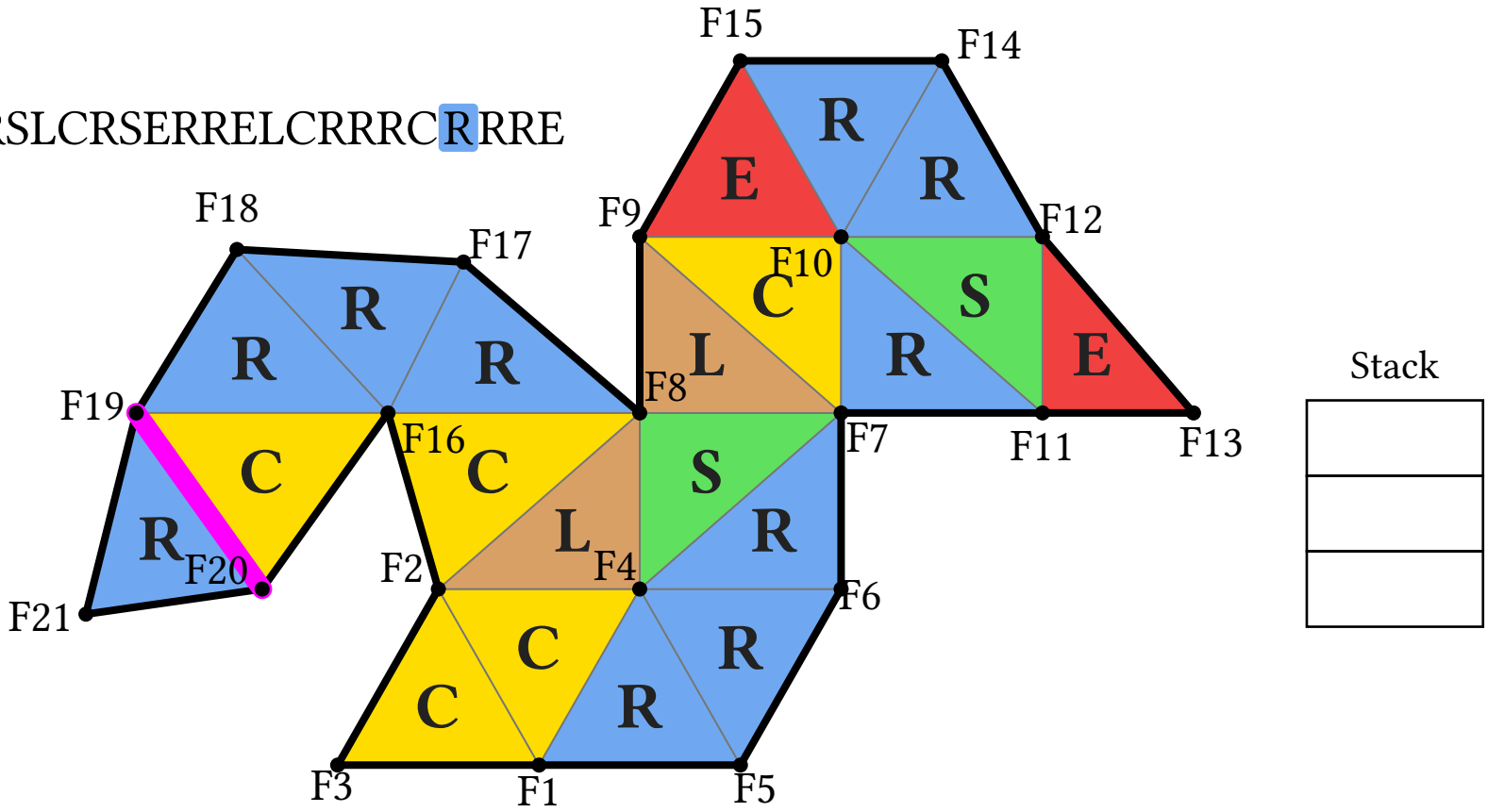
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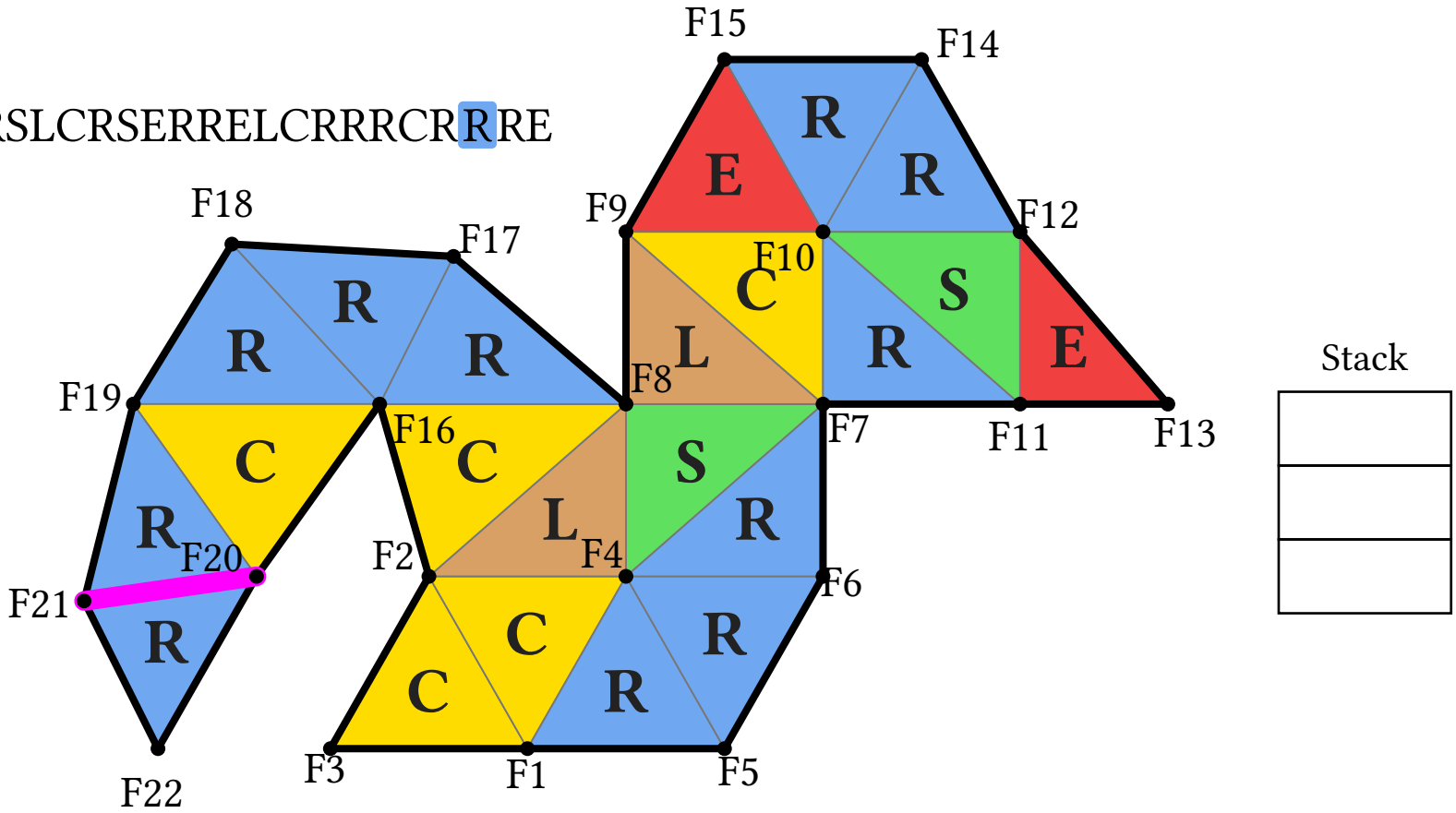
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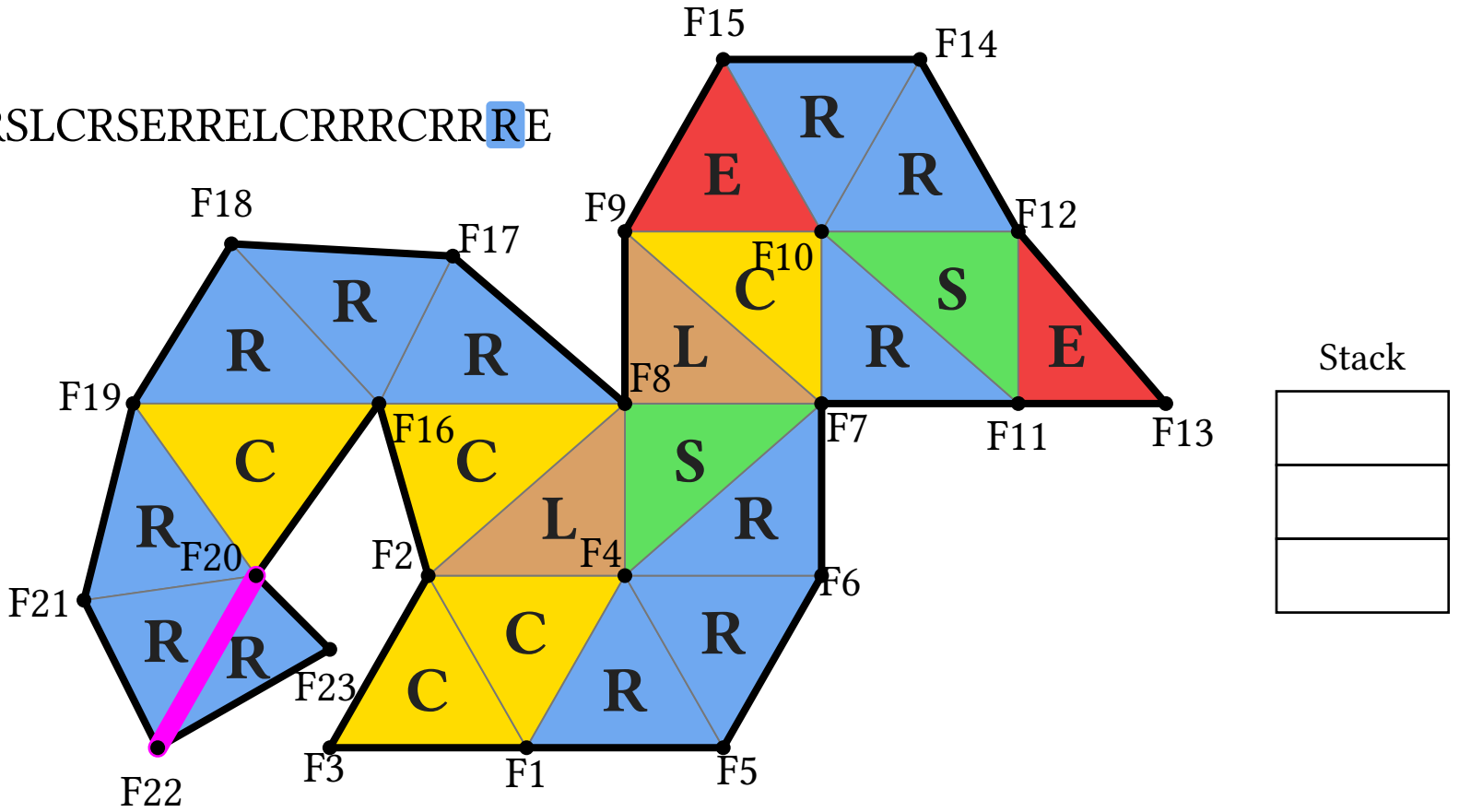
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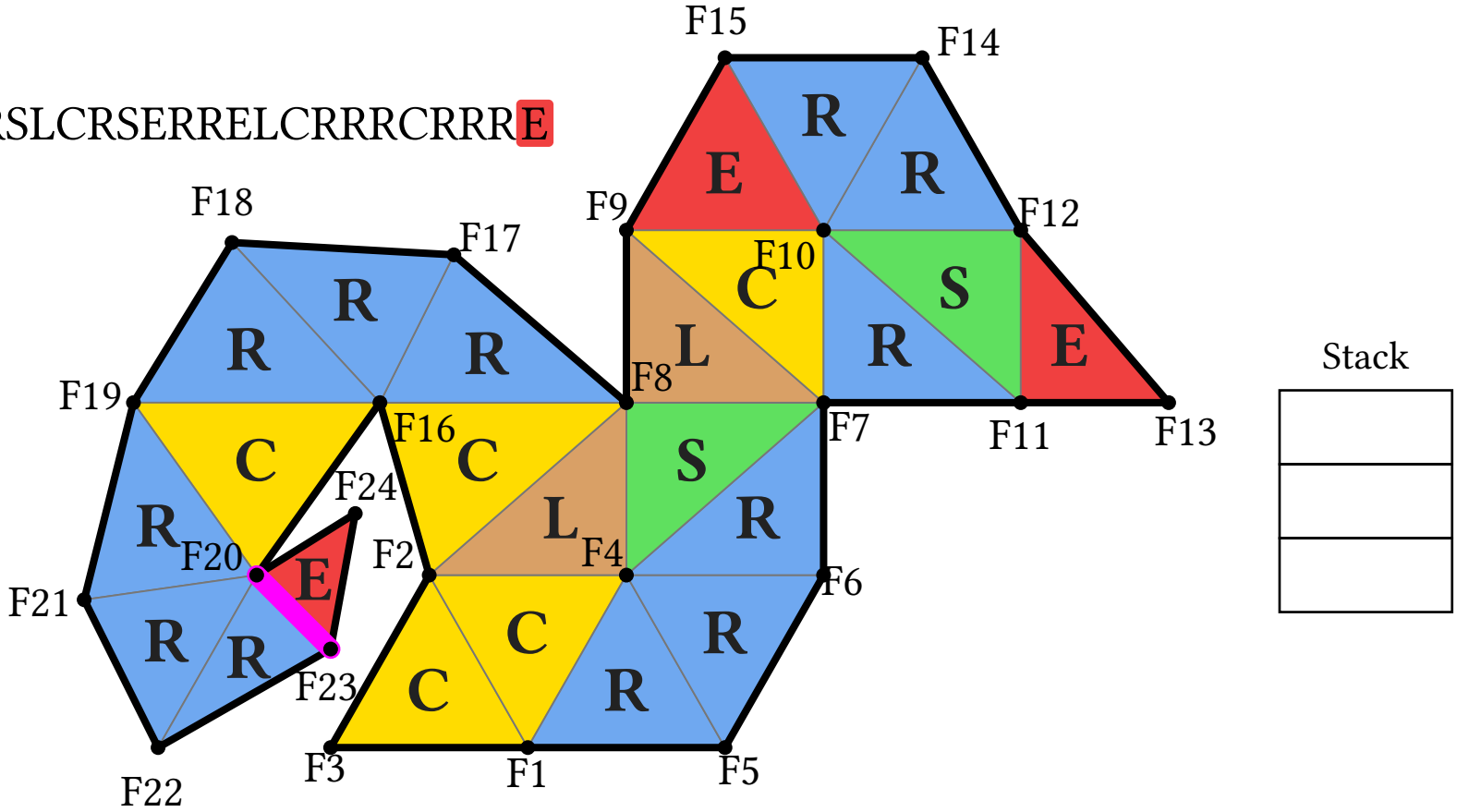
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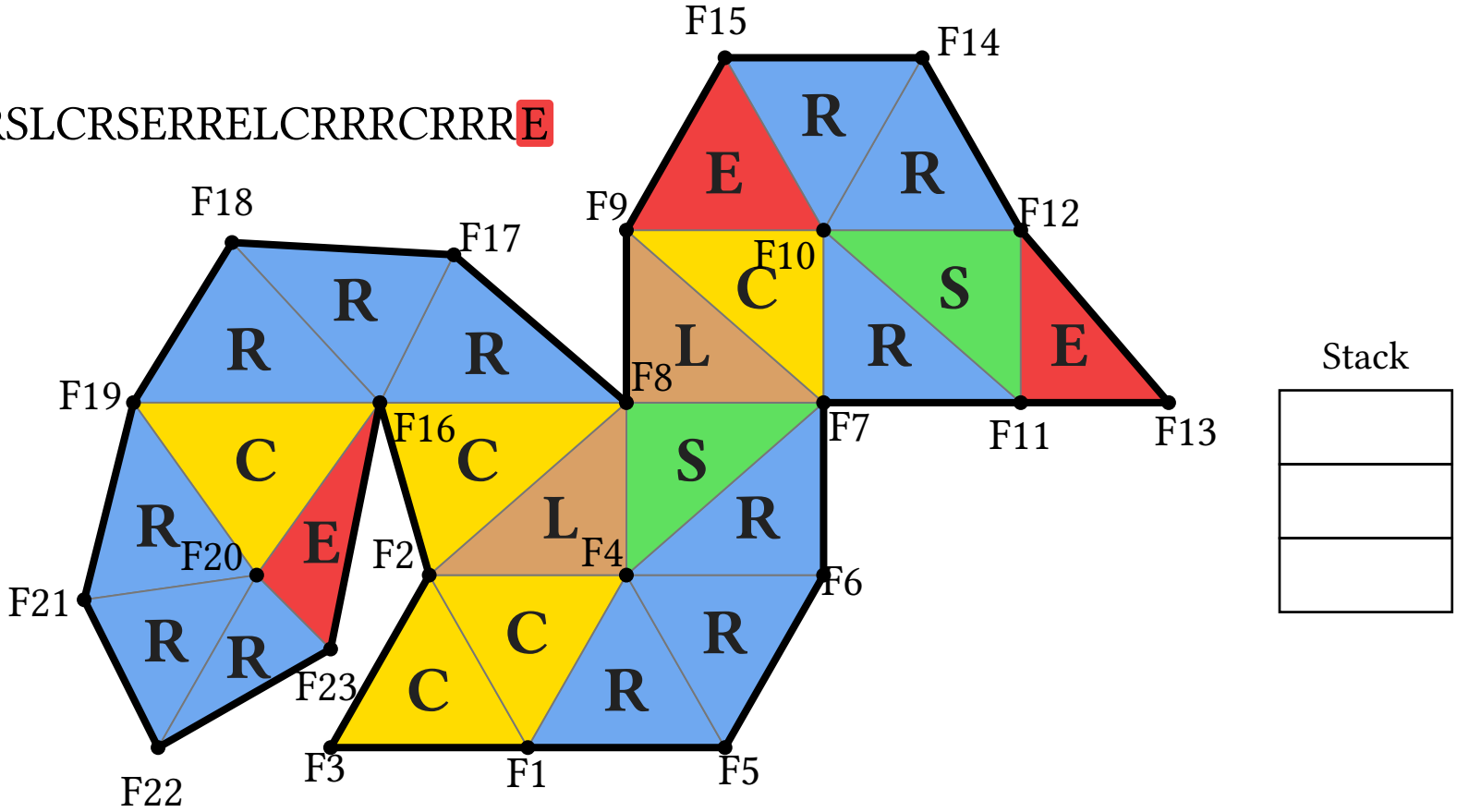
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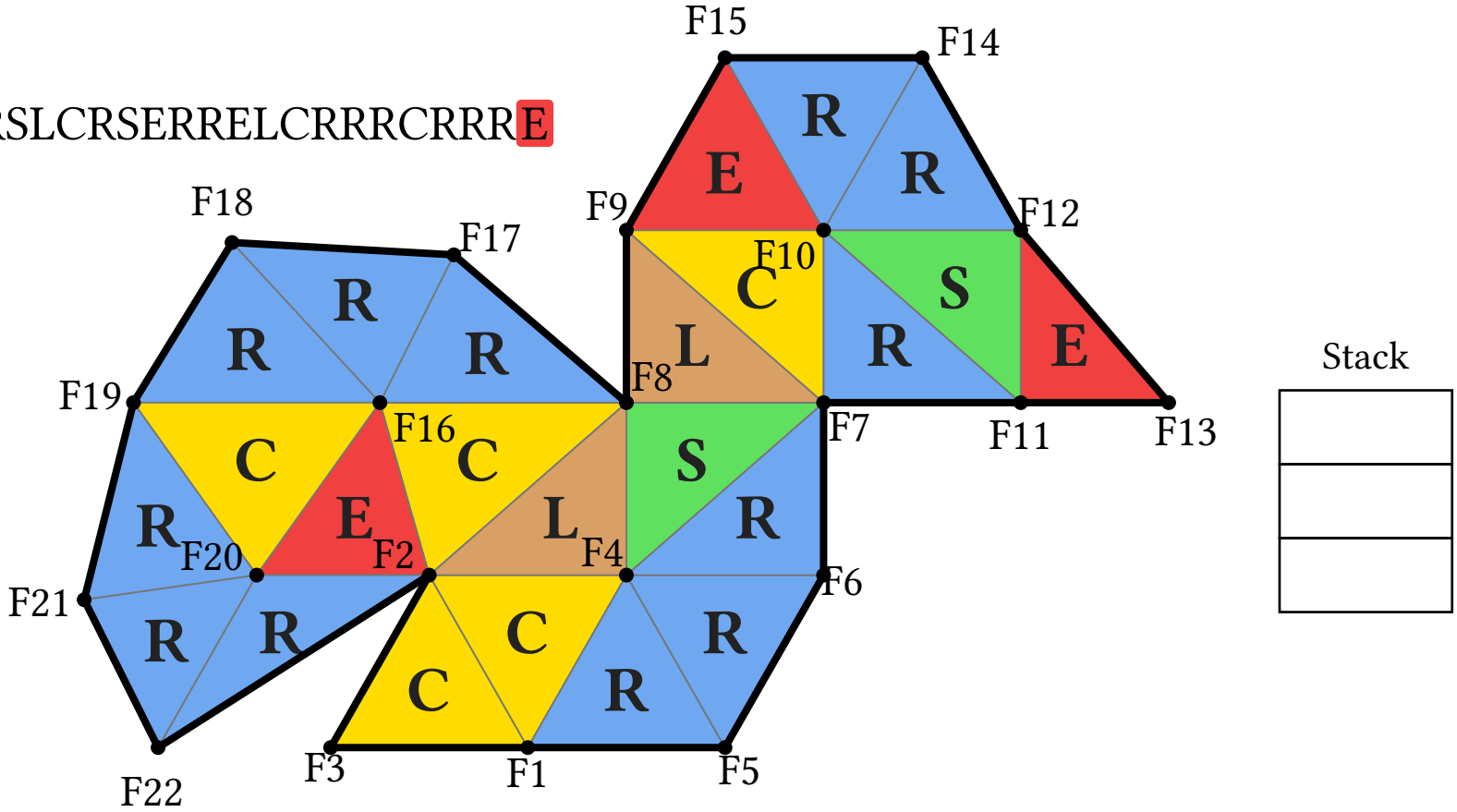
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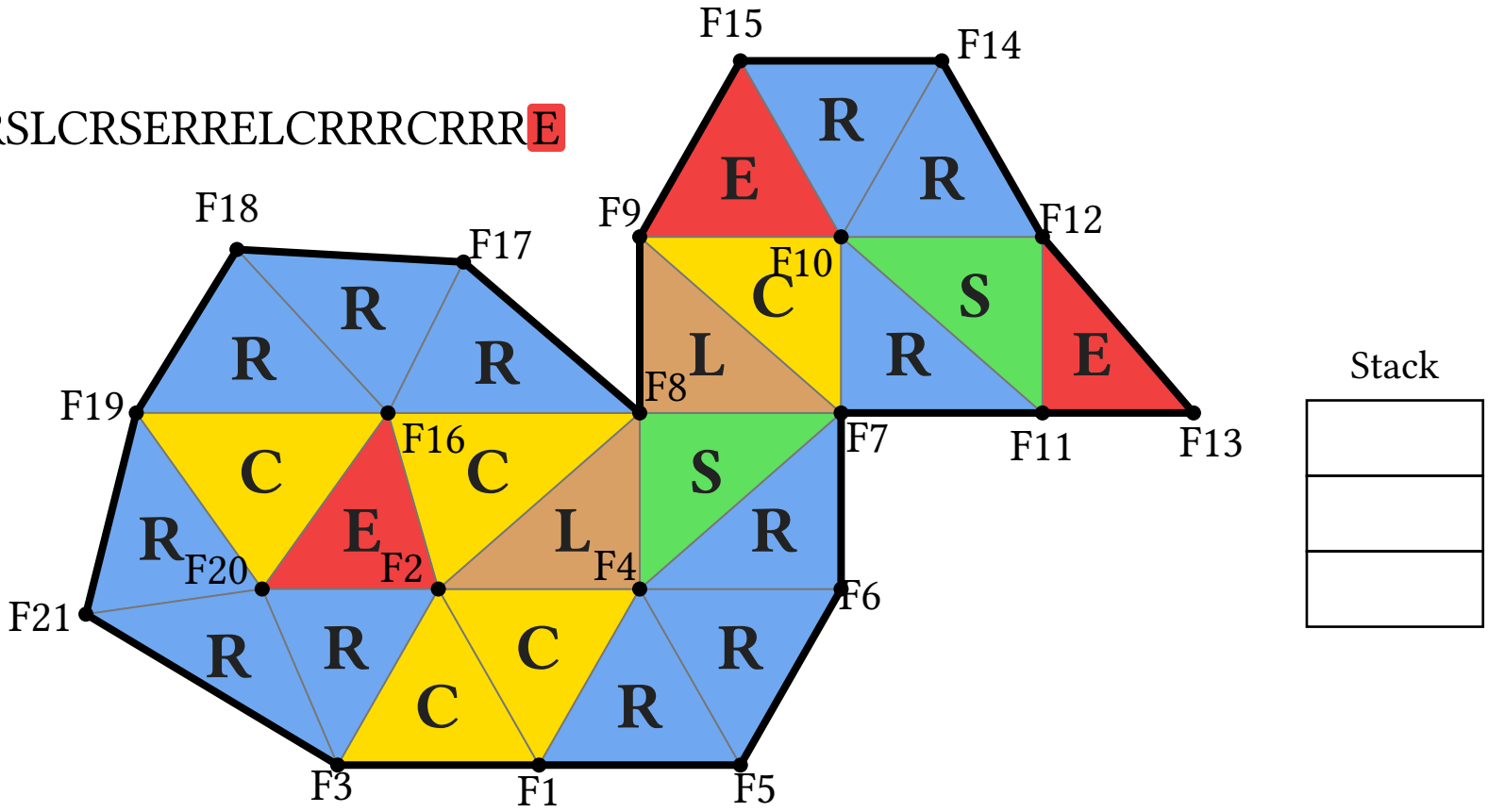
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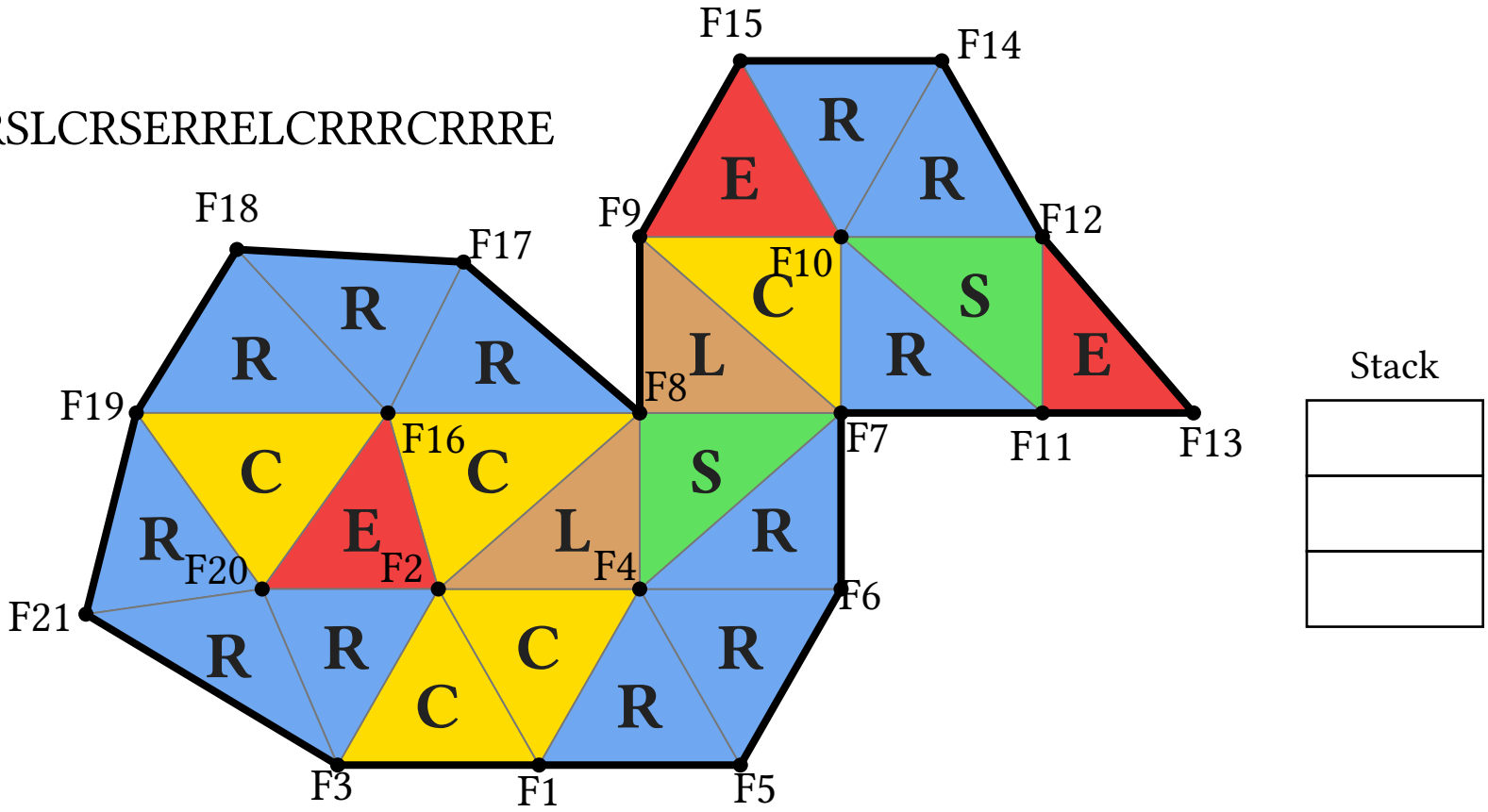
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Obrázek 6: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE



Obrázek 6: Dekomprese.

5. Vyšší genus

5.1 Řešení

Zavádíme obecnější operaci S.

Navíc je třeba ukládání topologie.

5.1 Řešení

Zavádíme obecnější operaci S.

Navíc je třeba ukládání topologie.

5.1.1 Díry

Objevení díry => push topologie s délkou díry.

5.1 Řešení

Zavádíme obecnější operaci S.

Navíc je třeba ukládání topologie.

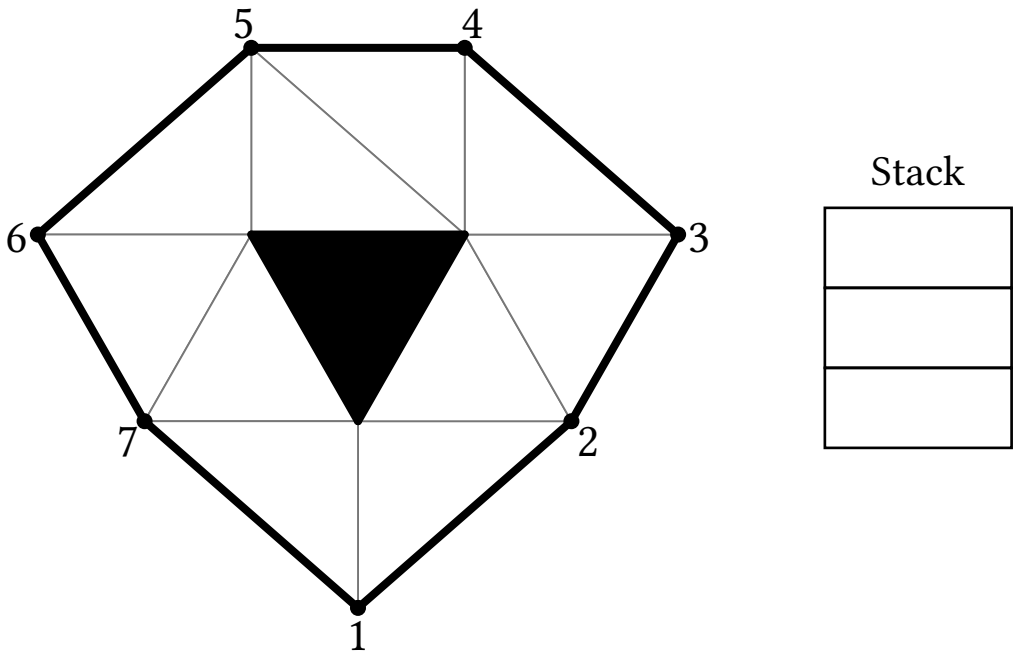
5.1.1 Díry

Objevení díry => push topologie s délkou díry.

5.1.2 Handly

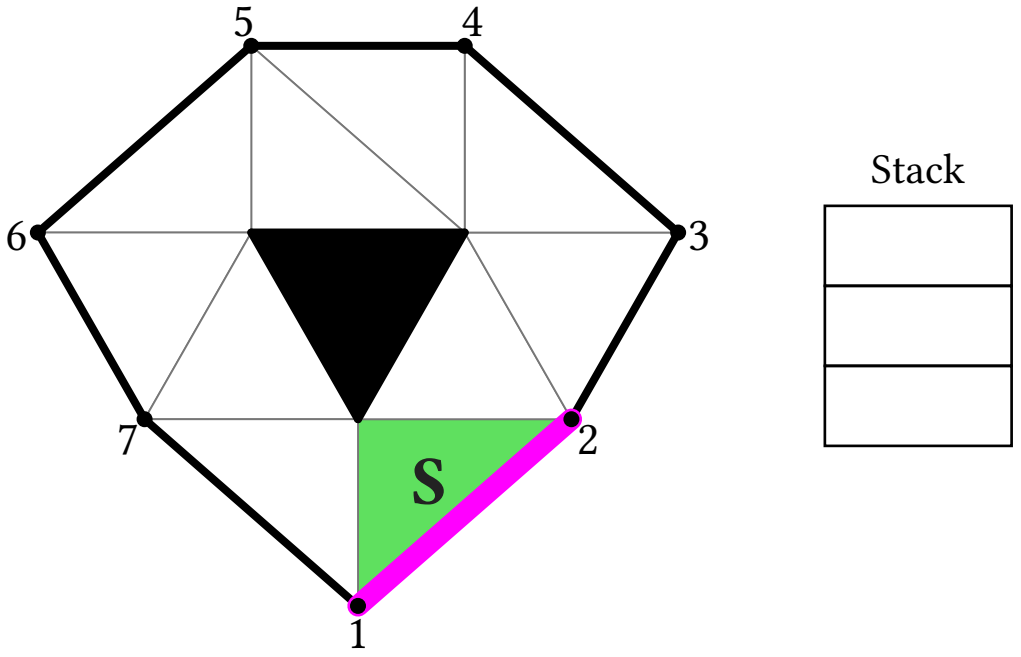
Následuje S => push topologie se “zipem”.

Historie =



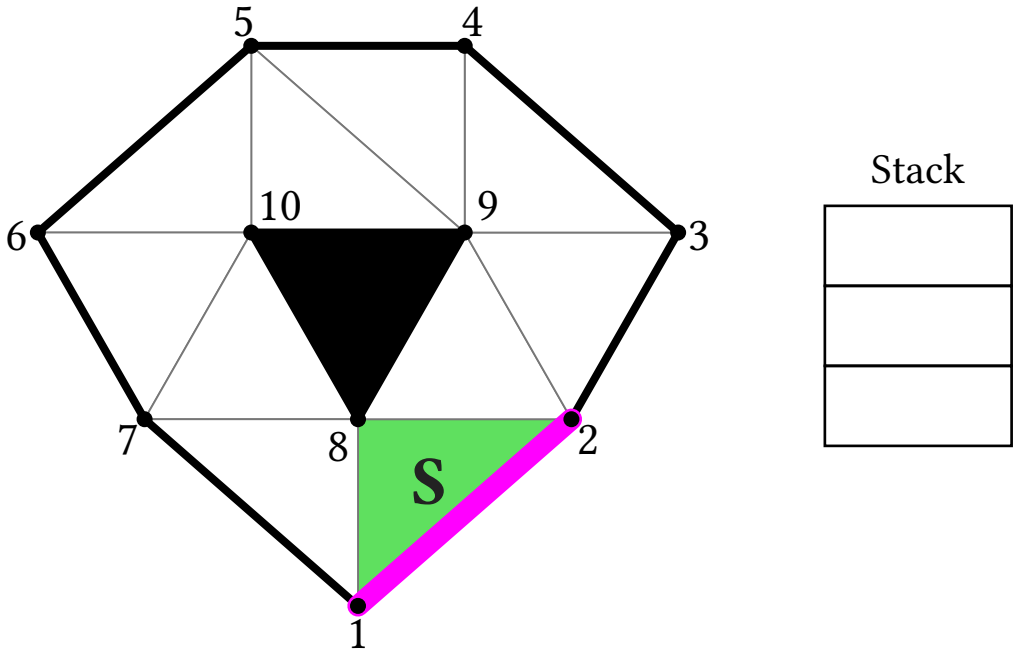
Obrázek 7: Složitější komprese.

Historie = S



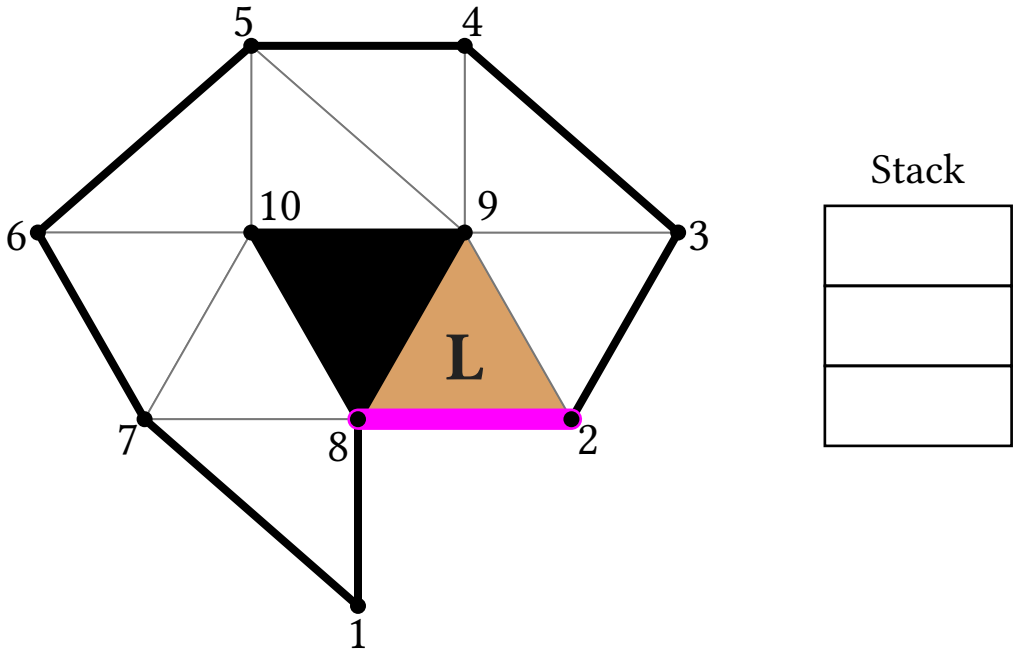
Obrázek 7: Složitější komprese.

Historie = S



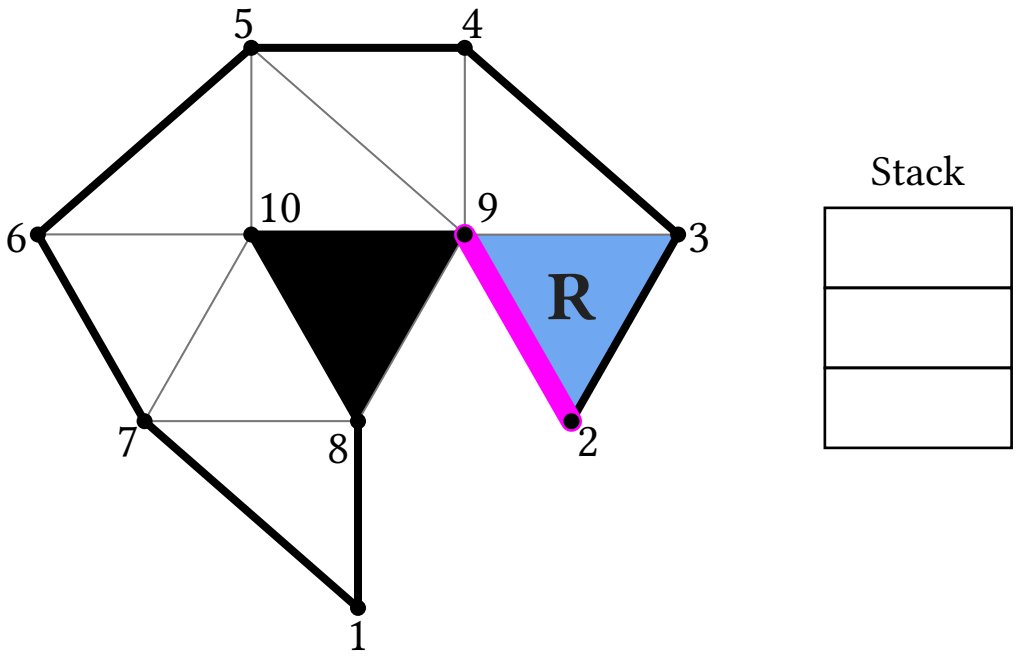
Obrázek 7: Složitější komprese.

Historie = SL



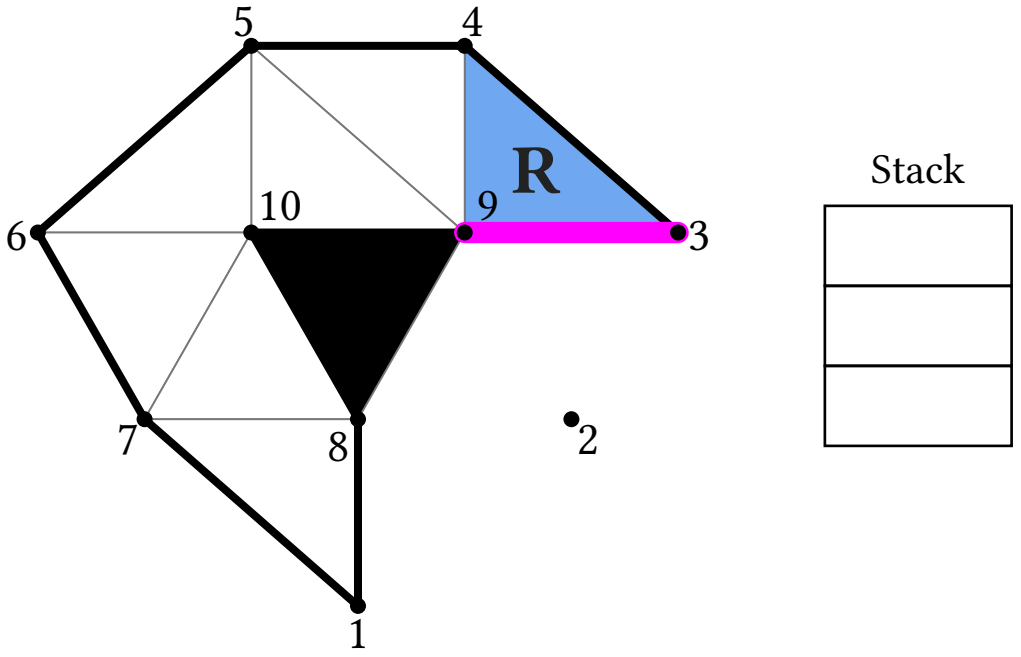
Obrázek 7: Složitější komprese.

Historie = SLR



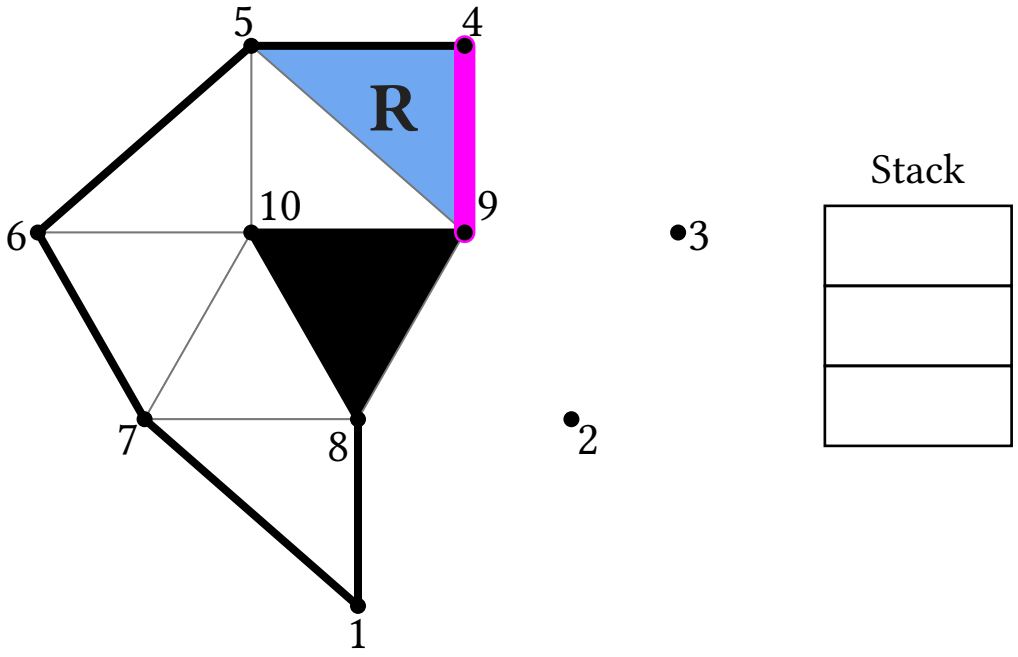
Obrázek 7: Složitější komprese.

Historie = SLRR



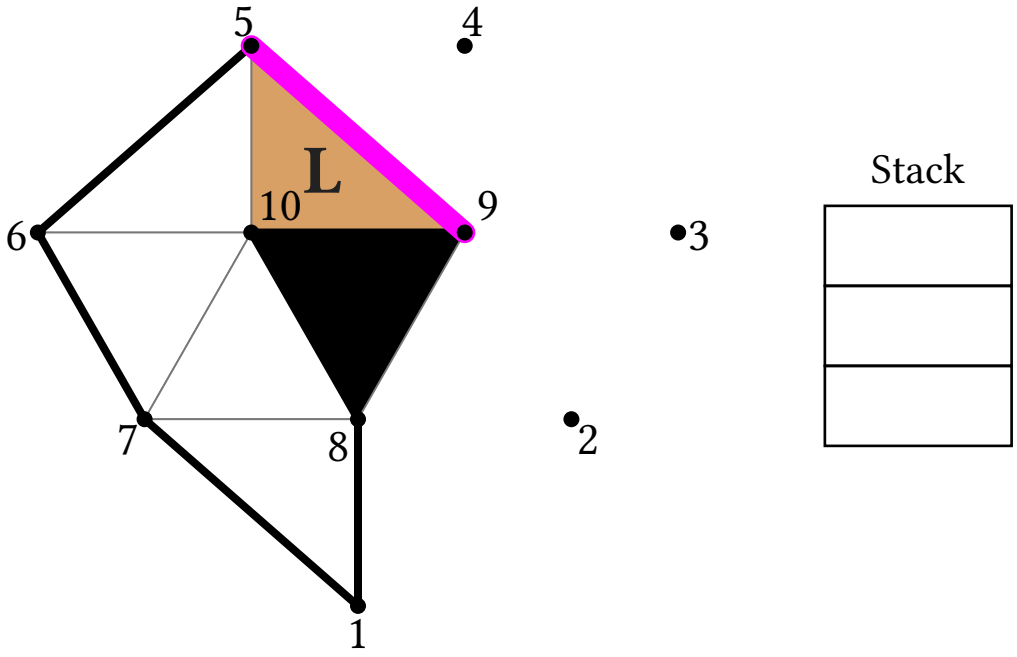
Obrázek 7: Složitější komprese.

Historie = SLRRR

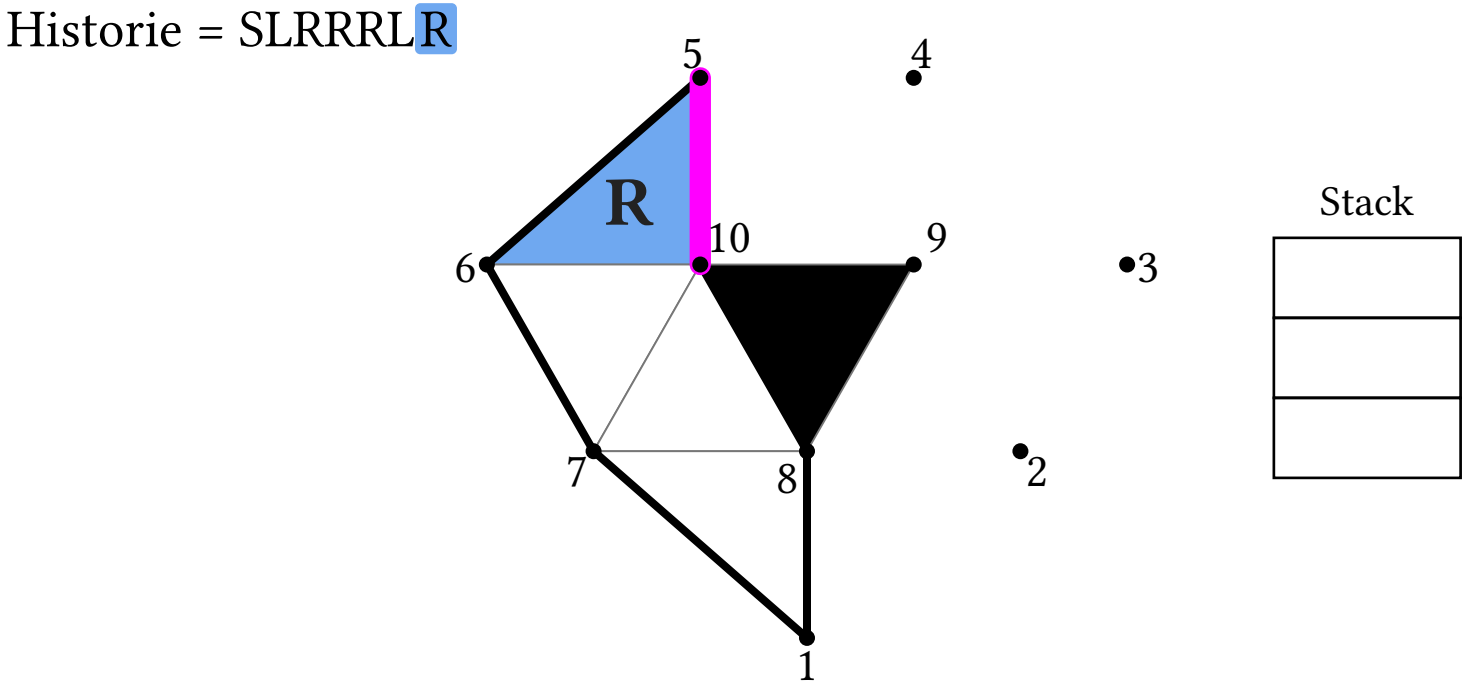


Obrázek 7: Složitější komprese.

Historie = SLRRRL

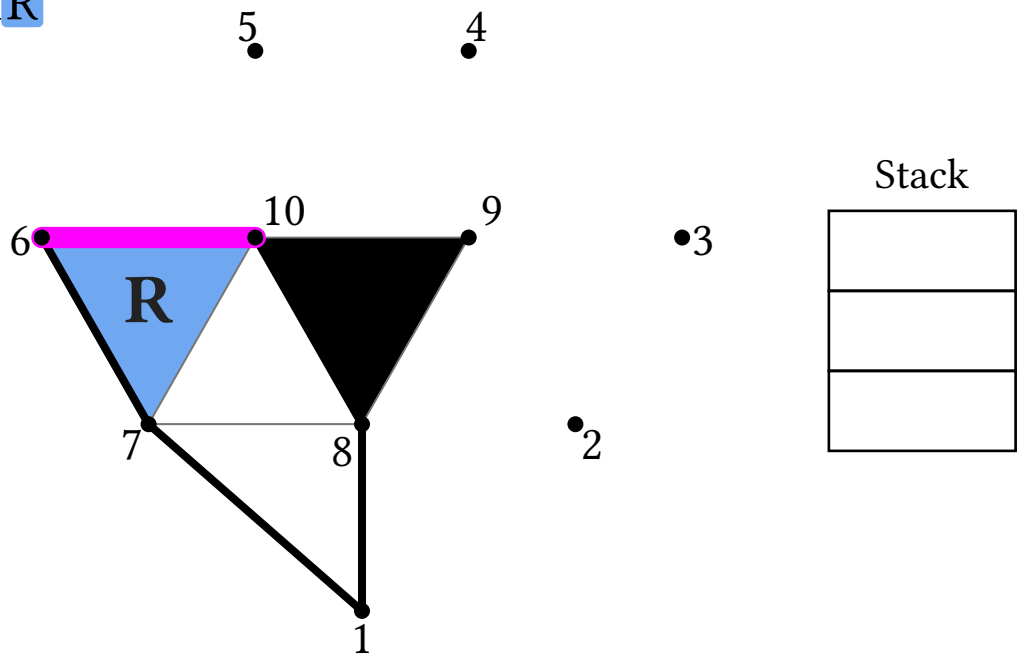


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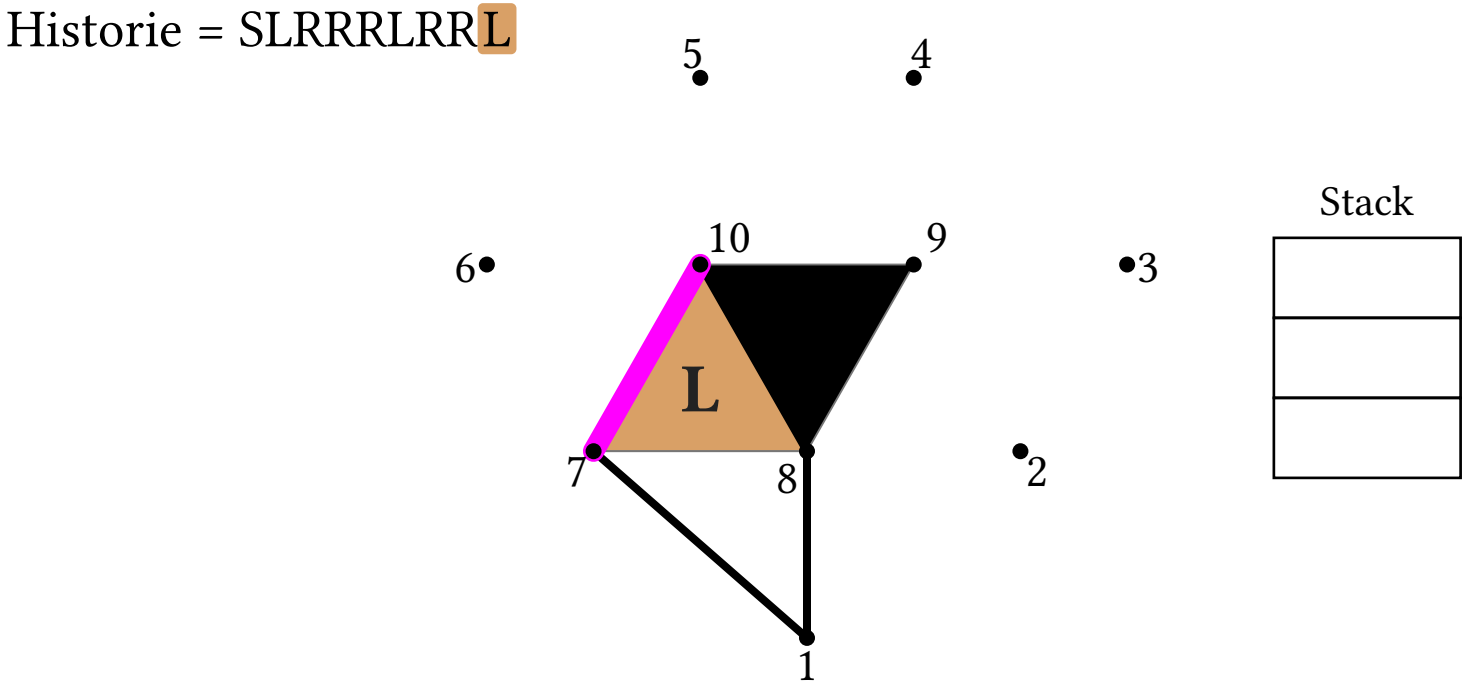


Obrázek 7: Složitější komprese.

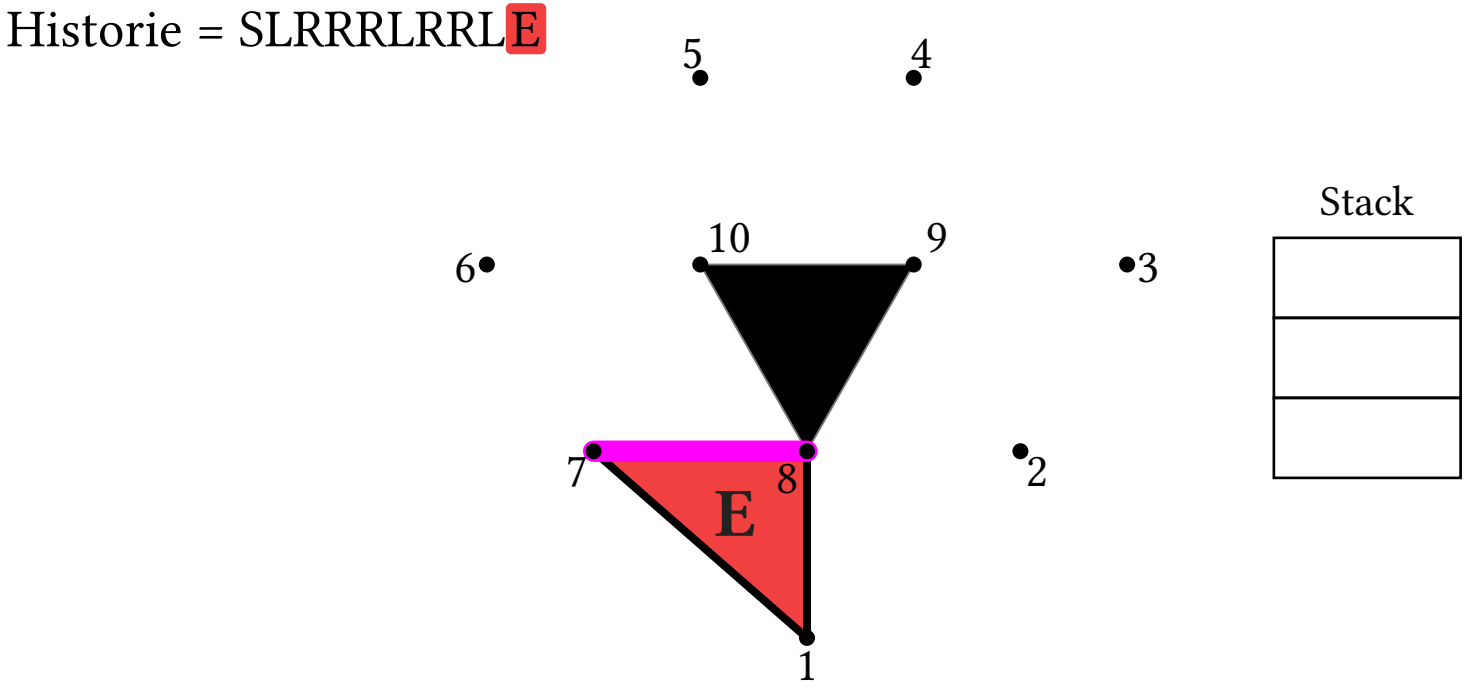
Historie = SLRRRLRR



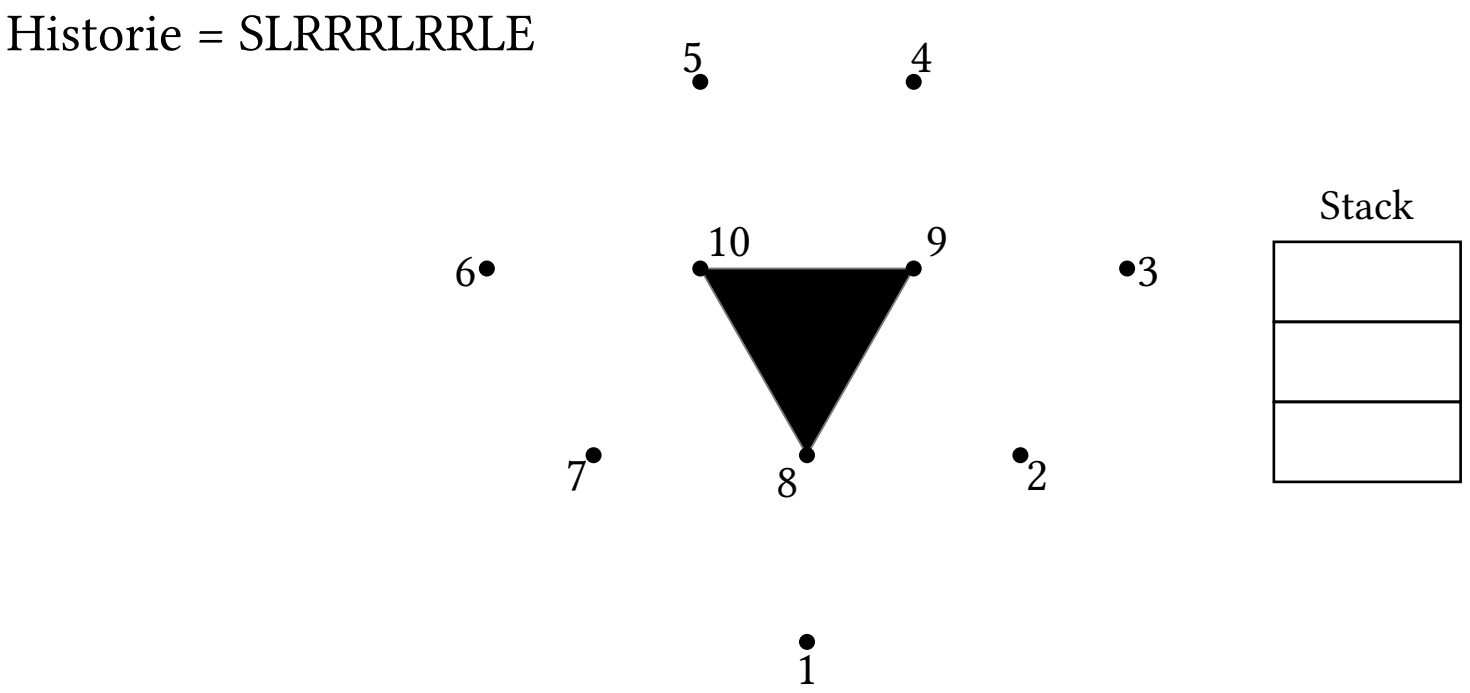
Obrázek 7: Složitější komprese.



Obrázek 7: Složitější komprese.



Obrázek 7: Složitější komprese.



Obrázek 7: Složitější komprese.

6. Zdroje

6.1 Seznam obrázků

Obrázek 1: Dobový kontext vzniku algoritmu Edgebreaker. 2
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Obrázek 3: Trojúhelníková síť (mesh). 5
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6.2 Obrázky (dobový kontext)

1	**Nagano** (2/1998)	Wikipedia, *fair use*
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3	**Windows 98** (5/1998)	SoundCloud (artwork)
4	**MrBeast** (5/1998)	YouTube (avatar kanálu)
5	**Google** (9/1998)	logodix.com
6	**Half-Life** (11/1998)	Valve, valvearchive.com Wikipedia *fair use*
7	**NATO** (3/1999)	nato.int
8	**Futurama** (3/1999)	ČSFD (plakát)
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10	**Napster** (6/1999)	Jusore, GPLv3, Wikimedia Commons
11	**Putin** (12/1999)	Kremlin.ru, CC BY 4.0, Wikimedia Commons

6.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

[1] J. Rossignac, “Edgebreaker: Connectivity Compression for Triangle Meshes,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 5, no. 1, pp. 47–61, Jan. 1999, doi: 10.1109/2945.764870.

[2] J. Rossignac, A. Safonova, and A. Szymczak, “Edgebreaker on a Corner Table: A Simple Technique for Representing and Compressing Triangulated Surfaces,” in *Hierarchical and Geometrical Methods in Scientific Visualization*, G. Farin, B. Hamann, and H. Hagen, Eds., Berlin, Heidelberg: Springer, 2003, pp. 41–50. doi: 10.1007/978-3-662-07443-5_3.

[3] T. Lewiner, H. Lopes, J. Rossignac, and A. W. Vieira, “Efficient Edgebreaker for Surfaces of Arbitrary Topology,” in *Proceedings of SIBGRAPI 2004 (XVII Brazilian Symposium on Computer Graphics and Image Processing)*, Curitiba, Brazil, 2004, pp. 218–225. doi: 10.1109/SIBGRA.2004.1352965.

[4] J. Rossignac and A. Szymczak, “Wrap&Zip: Linear Decoding of Planar Triangle Graphs,” technical report GIT-GVU-99-08, Feb. 1999.

[5] “Edgebreaker: Connectivity Compression for Triangle Meshes.” Accessed: Nov. 03, 2026. [Online]. Available: https://graphics.stanford.edu/courses/cs468-08-winter/slides/sid2.pdf